

Local rates, stalling, and finite certificates for iterated optimization-based bound tightening

Abstract

Iterated optimization-based bound tightening (OBBT) optimizes each coordinate over a convex relaxation with an objective cutoff and rebuilds it on the tightened box. For a valid relaxation family that is monotone under box restriction, three things limit further tightening: the hull of feasible points with objective at most the cutoff, boxes that the relaxation cannot shrink, and the rate of approach to the limit. Near a minimizer, the relaxation’s second-order tangent model defines an order-preserving map on box shapes. If its upper Collatz–Wielandt factor is strictly below one, it bounds the local linear rate, and the limit has width $O(\sqrt{\epsilon})$ and leaves a relaxation gap $O(\epsilon)$ for incumbent suboptimality ϵ . Points of negative tangent-model value on every face of a box shape certify stalling at every sufficiently small scale. The model is exact for quadratic objectives with McCormick relaxations, giving closed-form rates and stall conditions, including stalls of strongly convex objectives under that relaxation; for factorable functions with C^2 univariate factors, we prove a second-order expansion of composite McCormick relaxations. Finite point pools feasible in the relaxation rebuilt on their own hull certify boxes that no later round removes, with exact cutoff thresholds; residual and sensitivity bounds limit the remaining movement. In 120 prospective runs, an adaptive SCIP policy using only current-round screening solved 20.9% fewer auxiliary LPs than a fixed policy but spent more propagator time; neither produced an additional solve or demonstrated speedup over native SCIP. These certificates bound a specified operator’s movement, not search time.

Keywords. Optimization-based bound tightening; global optimization; McCormick relaxations; fixed-point certificates; mixed-integer nonlinear programming.

1 Introduction

Optimization-based bound tightening (OBBT) improves the bounds of a variable by minimizing and maximizing it over a convex relaxation of the feasible set, usually together with an objective cutoff given by the incumbent value. Tighter bounds strengthen every relaxation that depends on them, such as the McCormick envelopes of products, and can therefore shrink a branch-and-bound tree. A round costs up to $2n$ auxiliary optimization problems for n variables, and after a round the relaxation can be rebuilt on the smaller box and the round repeated. Solvers ration this work: they run OBBT mainly at the root, skip directions that a known relaxation point already shows cannot improve, order the problems to reuse warm starts, impose work limits, and derive reusable bound inequalities from the dual solutions [15, 40]. Learned variants choose which variables to tighten in each round [11] or which domain-reduction configuration to use on an instance [16].

Whenever OBBT is available, three decisions remain: whether to start it at a node, when to stop repeating it, and when a change of the search state, such as a better incumbent or a smaller node box, justifies running it again. Each decision needs some knowledge of what further tightening can still achieve. This paper studies that question for a specified relaxation family. We ask what

limits further OBBT, how the limit and the speed at which it is approached depend on the cutoff and on the relaxation, and which of these limits can be certified by finite computations.

The starting point is the order structure of the operator T_U that maps a box to the smallest box containing all of its points whose relaxed objective is at most the cutoff U . For a valid relaxation family that is monotone under box restriction, there are three limits on further tightening (Section 3):

- (i) the box hull of the original feasible points with objective at most U , which no valid tightening removes and which shrinks with a better cutoff when the problem domain is held fixed;
- (ii) the fixed boxes of T_U , which no number of further rounds of the same relaxation family removes and only a smaller cutoff, a stronger relaxation, or branching can shrink;
- (iii) the rate at which the iterates approach their limit, which determines how many rounds are needed to reach a given movement tolerance.

The first is a property of the problem, the second of the relaxation, and the third of the iteration. The paper develops each of them.

Local rates and stalling. Near a minimizer x^* we rescale the relaxation to boxes $x^* + tD(d)$ of a fixed shape d and let $t \rightarrow 0$ (Section 4). The limit is a second-order tangent model $Q(d, \xi)$ that depends on the shape of the box and on the position ξ in it. It defines an order-preserving, positively homogeneous map Φ on box shapes whose upper Collatz–Wielandt factor r^* bounds the local linear rate: if $r^* < 1$, then for every $\lambda \in (r^*, 1)$ the iterates from a sufficiently small box around x^* contract by the factor λ per round, in a suitable gauge, until their width is $O(\sqrt{\epsilon})$, where $\epsilon = U - f^*$ is the cutoff slack, and the relaxation gap that remains is $O(\epsilon)$. If points with negative tangent value attain all faces of a shape, boxes of that shape around x^* are fixed at every sufficiently small scale and every cutoff $U \geq f^*$, so OBBT stalls there. For quadratic objectives with McCormick relaxations the tangent model is exact. This gives closed forms: the rate $(\sqrt{2a^2 + 4|a|} - |a|)/2$ for $x^2 + y^2 + axy$ with $0 < |a| < 2$, a continuum of eigenvector boxes, the exact limit and remaining gap at a positive cutoff slack, row conditions for stalling and for contraction, and a strongly convex family that stalls exactly when $a(n-1)(n-2) \geq 2$. For composite McCormick relaxations of factorable functions with twice continuously differentiable univariate factors, we prove a second-order expansion that computes Q by a recursion over the factors. At minimizers on the boundary of the box with strict complementarity, the free coordinates obey the bounds of the interior theory, and the active widths of the limit are $O(\epsilon)$. These conditions are sufficient, not exhaustive: we give a valid monotone family that converges sublinearly and whose limit has width of order $\epsilon^{1/3}$.

Finite certificates. A finite set of relaxation points whose box hull is a box P , and which are feasible in the relaxation rebuilt on P , proves that P is fixed (Section 5). It therefore bounds the total movement of every endpoint under all later rounds and directional updates, at every cutoff not below the largest relaxation value of the points, as long as the box being tightened contains P . Established feasibility filtering, which checks the points only in the current relaxation, bounds a single frozen round. We give exact cutoff thresholds and the adaptation of cached witness pools to a new cutoff (Section 6), objective ceilings, the scope of the certificates under integer rounding, cuts, and branching, and the difference between checking a certificate and finding one. When the iterates converge only asymptotically, the residual of one exact round and a verified uniform sensitivity matrix bound the remaining movement (Section 7).

Constraints. Fixed-matrix LP duality and exact basis regions bound how the coordinate supports respond to a cutoff change across active-set changes; a complete cover by basis regions supplies the uniform sensitivity matrix that the residual certificates need; and a feasible-repair argument transfers relaxation and constraint errors into a contraction bound for nonlinear constraints, with explicit constants in a worked example (Section 8).

Algorithms, accounting, and evidence. An exact reference procedure commits a round only after verifying a primal–dual optimality proof for every endpoint, and reports a fixed box only when the finite certificate holds. A numerical policy inside SCIP applies dual bounds evaluated with directed rounding, screens directions with points of the current relaxation, stops unproductive pilots, and reconsiders tightening after the incumbent improves (Section 9). A cost ledger bounds optional work on the enhanced execution but cannot bound its run time relative to an unchanged baseline (Section 10). Section 11 reports the computational evidence.

The rate theory and the certificates answer different questions and complement each other. The rate theory identifies which limit dominates near a minimizer and how it scales with the cutoff, but its hypotheses involve x^* , f^* , and tangent data that a solver does not know. The certificates are computable once their witnesses or sensitivity data are available. A certificate for the current box proves exact fixedness; a certificate for an inner box bounds the remaining movement and can justify stopping at a specified tolerance. Neither requires positive width: a singleton is also a fixed box if its point belongs to its own cutoff relaxation. The rate theory describes which fixed boxes can occur locally. In the stall regime, positive-width fixed boxes exist at every sufficiently small scale and finitely many witnesses certify them. In the strictly contracting regime, under the closedness property of Section 3, the limit is a fixed box of width $O(\sqrt{\epsilon})$ and at most $2n$ witnesses certify it. At $\epsilon = 0$ the only nearby fixed box is $\{x^*\}$. If an iterate reaches it, a finite certificate proves termination. If convergence is only asymptotic, the singleton of an optimal incumbent in the current box is still protected (Corollary 5.14); it bounds the remaining movement, and the bound is exact if the iterates converge to that point (Example 5.11). Otherwise a bound on the remaining movement, such as a residual certificate, justifies a tolerance stop. The upper bounds of order $\sqrt{\epsilon}$ on the width of the limit and of order ϵ on the remaining gap motivate reconsidering OBBT after the incumbent improves; in the two-variable model the limit itself shrinks at exactly these orders. A stored certificate also states exactly when its own proof expires: when the cutoff falls below the face threshold of its pool. The certified box can remain fixed below that threshold, down to an exact box threshold computed by at most $2n$ face optimizations (Section 6).

The computational evidence limits what these guarantees imply. In a prospective comparison of 120 runs on twenty models with two seeds each, an adaptive policy used 20.9% fewer auxiliary LPs than a fixed-order policy but spent more time in its propagator, and neither added policy produced an additional solve or a demonstrated speedup over native SCIP within ten seconds. An earlier study of iterated OBBT as an external presolve on 339 MINLPLib instances found that the tightened boxes reduced node counts, but the savings in the final solve did not offset the cost of OBBT. The measured policy used only current-round screening, not the all-future certificates, so the experiments say nothing about the computational value of those certificates. They show that the number of auxiliary LPs is not a cost measure, and Section 10 shows why a bound on width or on the remaining tightening cannot by itself bound search time. The certificates give finite stopping guarantees for a specified tightening operator; whether acting on them saves time is a separate question that requires evidence of its own.

Attribution. The order properties of the operator, the persistence of fixed boxes, and the existence of a greatest fixed box are elementary consequences of monotonicity; the last is an instance of Tarski’s fixed-point theorem [43]. The limits of iterated domain reduction and their order independence were studied by Caprara and Locatelli [9], and Caprara, Locatelli, and Monaci gave examples in which iterated reduction does not move any bound [10]. The scalar contraction bounds follow the mechanism of marginals-based range reduction [35], and their strict contraction condition has the same numerical threshold as known nonstrict conditions that exclude clustering in branch and bound [25, 46]. Repeated OBBT is also used in adaptive partitioning and power-flow optimization [29, 42]. Convergence-order analysis of McCormick and McCormick–Taylor relaxations is established [5, 6, 30]. Feasibility filtering, dual reuse, and incumbent-triggered reevaluation of stored bounds exist in prior work and in SCIP [15, 40]. LP duality, parametric LP, and contraction principles are classical. Our contribution consists of the OBBT-specific formulations: the tangent model and tangent map with their contraction and stall theorems, the relaxation-specific exact rates and stall conditions, the explicit shape-dependent second-order expansion of composite McCormick relaxations, the finite certificate formulations together with their scope, and a prospective evaluation of an adaptive scheduling policy that uses only current-round screening and heuristic triggers, not the all-future certificates. The local results concern the shape and position dependent tangent map of a specified relaxation family; they do not assert priority for iterated domain reduction, adaptive OBBT, or convergence-order analysis in general.

Organization. Section 2 reviews related work. Section 3 defines the operator and its limits, and Section 4 develops the local theory. Sections 5, 6, and 7 develop the certificates, and Section 8 treats constraints. Sections 9 to 11 describe the implementations, the cost accounting, and the computational evidence, and Section 12 discusses the consequences for starting, stopping, and repeating OBBT. Appendices A and C contain longer proofs, Appendix B gives the numerical bound validation, Appendix D gives scheduling guarantees that keep the baseline solver unchanged, and Appendix E describes the precursor study.

2 Related work

Puranik and Sahinidis survey domain reduction for global NLP and MINLP optimization, including feasibility-based and optimality-based methods [33]. The present paper concerns repeated optimization-based tightening with a specified relaxation and cutoff.

Optimality-based range reduction in solvers. Range reduction with an objective cutoff appears in the work of Ryoo and Sahinidis. Their marginals-based test confines a variable that lies at a bound in an optimal solution of the relaxation, with marginal $\lambda > 0$, to within $(U - L)/\lambda$ of that bound, where L is the relaxation bound [35]. Gleixner, Berthold, Müller, and Weltge proposed three enhancements of LP-based OBBT, implemented in SCIP: filtering of directions by feasible relaxation points, ordering of the auxiliary LPs to exploit warm starts, and Lagrangian variable bounds derived from the dual solutions, which approximate the effect of repeated OBBT throughout the search [15]. They also note that a call that tightens nothing can still yield a useful dual inequality. The SCIP implementation examined here runs OBBT at the root under a budget tied to the root LP effort, and a separate propagator reevaluates the stored dual inequalities when the incumbent improves [40]. Coramin implements filtering by feasible points and by an aggregate objective [14]. Detecting a new incumbent and reevaluating dual information are therefore existing capabilities.

Our current-round witness screening is the filtering of Gleixner et al.; what differs is the persistence check of Section 5, which evaluates the witnesses in the relaxation rebuilt on their own hull.

Iterated domain reduction. Caprara and Locatelli studied repeated optimality-based domain reduction for problems with a nonconvex objective over a convex set. They showed that iterating the reduction on one variable reaches the same limit as a parametric reduction in that variable, and that the limit of cyclic sweeps over all variables does not depend on their order [9]. Caprara, Locatelli, and Monaci identified the hull of the feasible points with objective at most the cutoff as a lower limit for every optimality-based reduction, proved that a mixed reduction attains it for a class of two-variable problems, and gave small examples in which reduction moves no bound at all although this hull is a single point [10]. Our foundations restate their order arguments for Jacobi, sequential, and selective schedules. Our quadratic results give explicit row and face criteria for complete stalling under a specified termwise McCormick relaxation, including for strongly convex objectives. For feasibility-based tightening, Belotti, Cafieri, Lee, and Liberti characterized the limit of propagation as the greatest fixed point of a monotone operator, showed that the iteration can be infinite, and computed the limit by one LP for linear constraints [3, 4]; the greatest fixed point is Tarski’s construction [43]. These works establish existence and characterizations of limits. Section 4 studies local rates through an explicit map on box shapes.

Repeated tightening is also used in application algorithms. Coffrin, Hijazi, and Van Hentenryck iterate feasibility-based bound consistency without an incumbent objective cutoff to strengthen convex relaxations of power-network models [13]. Nagarajan, Lu, Wang, Bent, and Sundar include sequential OBBT in an adaptive partitioning algorithm and repeat it to a prescribed bound-change tolerance [29]. In the preprint version of their power-flow study, Sundar, Nagarajan, Misra, Lu, Coffrin, and Bent rebuild a strengthened quadratic-convex relaxation between OBBT rounds; their globally optimal OBBT variant also imposes an incumbent objective cutoff [42]. Rebuilding, repeating, and using an incumbent cutoff are established procedures. Our local analysis describes the rates and fixed boxes of a specified repeated-tightening map, and the certificates check what all later rounds of that map can remove.

Convergence order and the cluster problem. The rate at which a relaxation approaches the function as the box shrinks is its convergence order. Bompadre and Mitsos developed rules for the pointwise and Hausdorff convergence orders of McCormick relaxations under sums, products, and composition [5], and Najman and Mitsos studied convergence orders of multivariate McCormick relaxations [30]. Bompadre, Mitsos, and Chachuat analyze Taylor and McCormick–Taylor models and give convergence-order rules for arithmetic and composition [6]. The latter combine a Taylor polynomial with a McCormick relaxation of its remainder. Our composite expansion concerns the ordinary clipped McCormick construction: it supplies an explicit second-order coefficient that depends on both box shape and position and then uses that coefficient to define the OBBT tangent map. Recursive composition and convergence-order analysis themselves are established in the cited work. Wechsung, Schaber, and Barton showed how the prefactor of second-order convergence, compared with the smallest eigenvalue of the Hessian, determines the number of boxes that branch and bound visits near a minimizer [46], and Kannan and Barton extended this analysis to constrained problems [25]. The strict contraction condition of Proposition 4.2 has the same numerical threshold as their nonstrict no-clustering conditions. These cluster analyses use a scalar prefactor for the worst case over the box. The tangent model of Section 4 replaces it by the limit of the scaled relaxation gap as a function of the box shape and of the position in the box; this is the information needed to describe a box-to-box map, and the exact rates of Section 4.4 show that the scalar condition can

fail although the iterates contract geometrically.

McCormick relaxations. McCormick introduced the factorable relaxations and the envelopes of bilinear terms [27]. Mitsos, Chachuat, and Barton gave the composition rules for products and univariate functions that we use, in a form that applies to algorithms [28]. Scott, Stuber, and Barton generalized the construction; their standard procedure clips each factor’s relaxations to its interval bounds, and they proved that this procedure is monotone under box restriction when the argument intervals remain in the scalar domains, the scalar factors satisfy their Lipschitz assumptions, and the natural interval enclosures are nested [41]. The expansion of Theorem 4.24 identifies the second-order coefficient of these relaxations; its proof uses their validity and, for the rate statements, their monotonicity.

Selective, learned, and costly tightening. Cengil, Nagarajan, Bent, Eksioglu, and Eksioglu select the variables to tighten in each OBBT round by learned rankings for AC optimal power flow [11]. Gómez-Casares, González-Rodríguez, González-Díaz, and Rodríguez-Fernández compare conic OBBT and dual-information propagation in polynomial optimization and select configurations per instance by learning [16]. Application studies weigh bound quality against computation for ReLU networks [2] and select which integrality restrictions to keep in the tightening subproblems of topology optimization [32], and explicitly stored bounds on lifted variables can improve or worsen complete solver performance [17]. These results support measuring complete solves and charging all tightening work, which our experiments do. Adaptive or learned selection of OBBT directions is therefore not new; the present paper does not claim it.

Mathematical tools. The residual certificates are matrix versions of the a posteriori bound of the contraction principle, of the kind studied for vector-valued metrics [24]. Exact basis regions are the critical regions of multiparametric linear programming [7]. The feasible-repair argument of Section 8 transfers constraint residuals into distances in the manner of the error bounds of Hoffman for linear inequalities [21] and of Robinson for perturbed linear programs [34]. Anderson acceleration [45] can propose candidate limit boxes, but its affine mixing does not produce valid reductions; a proposed box must be checked, for example by the certificate of Section 5. We use these tools without claiming new general fixed-point, contraction, or duality theorems.

Selecting computations. Adaptive control of solver components has been studied with bandit algorithms in SCIP [20], and diving and large-neighborhood-search heuristics have been scheduled jointly by online learning [12]. Hay, Russell, Tolpin, and Shimony formalize the value of further computation and explicit stopping in a probabilistic decision model [19]. Applying such methods to OBBT requires a model of how a strengthening step changes the later search. Section 10 shows what cost accounting alone can and cannot guarantee in this respect.

3 The tightening operator and what limits it

This section fixes the operator studied in the rest of the paper, records the order properties on which all later results rest, and identifies three limits on further tightening. The order arguments are elementary and not new. Monotone iterations of optimality-based domain reduction, their limits, and their independence of the variable order are studied by Caprara and Locatelli [9], and greatest fixed points of feasibility-based tightening by Belotti et al. [3, 4]. We state the facts in the form

used later and make each hypothesis explicit, because several later conclusions fail when one of them is dropped.

3.1 Problem, relaxation family, and order hypotheses

Consider

$$f^* = \min\{f(x) : x \in X \cap B_0\}, \quad (1)$$

where $B_0 = [\ell^0, u^0] \subset \mathbb{R}^n$ is a bounded box and the closed set X collects all other constraints, including integrality. We assume that f^* is attained and write x^* for a global minimizer. A box is a set $B = [\ell, u] = \{x : \ell \leq x \leq u\}$ with $\ell \leq u$; degenerate boxes are allowed, and the empty set is treated separately. We write

$$w(B) = \max_i (u_i - \ell_i), \quad p(B) = (-\ell, u) \in \mathbb{R}^{2n}.$$

For nonempty boxes, $B' \subseteq B$ holds exactly when $p(B') \leq p(B)$ componentwise, so shrinking a box decreases its endpoint vector. When tightening is applied at a node of a branch-and-bound tree, B_0 is the node box and (1) is the node problem.

A *relaxation family* assigns to every box B in a collection $\mathfrak{B}$ a relaxation with projected objective $\phi_B : B \rightarrow \mathbb{R} \cup \{+\infty\}$. The collection $\mathfrak{B}$ contains all subboxes of B_0 ; Section 4 also uses boxes around a minimizer that may extend beyond B_0 , and then $\mathfrak{B}$ consists of all subboxes of a larger domain box (for McCormick relaxations of polynomials, all boxes in $\mathbb{R}^n$). A lifted relaxation has a lifted set $R(B) \subseteq \mathbb{R}^n \times \mathbb{R}^m$ and objective v , and

$$\phi_B(x) = \inf\{v(x, y) : (x, y) \in R(B)\}, \quad \inf \emptyset = +\infty. \quad (2)$$

Points that violate the relaxed constraints therefore have projected value $+\infty$. The two order hypotheses are

$$\phi_B(x) \leq f(x) \quad (x \in X \cap B), \quad (3)$$

$$\phi_{B'}(x) \geq \phi_B(x) \quad (B' \subseteq B, x \in B'). \quad (4)$$

We call (3) validity and (4) monotonicity of the family. In addition, we assume throughout that each ϕ_B is lower semicontinuous, so that its sublevel sets in B are compact. When ϕ_B is given by (2), we assume that $R(B)$ is compact and v is continuous. Then the infimum in (2) is attained whenever it is finite, ϕ_B is lower semicontinuous, and every projected sublevel set is the projection of the corresponding lifted sublevel set. Without attainment, an implementation that optimizes over the lifted cutoff set can compute a smaller set than the projected one, and the two must not be identified.

Monotonicity is a hypothesis on the *whole* relaxation construction: every row, auxiliary bound, retained cut, and the objective. It does not follow from the name of the relaxation. A sufficient condition is *lifted monotonicity*: $B' \subseteq B$ implies $R(B') \subseteq R(B)$, with the same objective v . Lifted monotonicity is strictly stronger than (4); it is what allows the same lifted points to be reused on a larger box, and Section 5 relies on it for that purpose. The following constructions are monotone.

- (a) Termwise relaxations of quadratic objectives that keep convex separable terms exact and replace each bilinear term $x_i x_j$, $i \neq j$, by its McCormick envelope. The four McCormick inequalities describe the convex hull of the graph of $x_i x_j$ over the current rectangle. The graph over a subrectangle is a subset of the graph over the rectangle, so the convex hulls are nested.

- (b) Lifted linear relaxations with fixed linear rows and McCormick inequalities for lifted variables y_k that stand for products $x_i x_j$, including squares $i = j$, for which the four inequalities reduce to the two endpoint tangents and the secant. Section 5 uses this family as its exact reference family and proves its lifted monotonicity.
- (c) Composite McCormick relaxations of factorable functions in which every intermediate relaxation is clipped to its interval bounds; see Section 4.5.

Monotonicity can fail even when the construction is determined by the box alone. Relax x^2 on $[\ell, u]$ by the tangents $y \geq 2px - p^2$ at the five equally spaced points $p = \ell + k(u - \ell)/4$, $k = 0, \dots, 4$, and by the secant $y \leq (\ell + u)x - \ell u$, with objective $v = y$; these are the square rows of the numerical policy of Section 9.4 without its retained points. On the box the tangents lie below x^2 and hence below the secant, so $\phi_{[\ell, u]}(x) = \max_k (2p_k x - p_k^2)$. The five points for $[-1, 1]$ include 0, whereas those for $[-1/2, 1]$ are $-1/2, -1/8, 1/4, 5/8, 1$. Hence $\phi_{[-1, 1]}(0) = 0 > -1/64 = \phi_{[-1/2, 1]}(0)$, contrary to (4). The operator (5) defined below is affected as well: at $U = 1/4$ it maps $[-1, 1]$ to $[-1/2, 1/2]$ but the smaller box $[-1/2, 1]$ to $[-1/2, 41/80]$, because the point $(x, y) = (41/80, 1/4)$ satisfies every row on $[-1/2, 1]$ but violates the tangent at $1/2$ of $[-1, 1]$. Tangents at the two endpoints alone, as in (b), do not have this defect. Constructions that depend on more than the box need not be monotone either. Cutting planes separated at earlier relaxation solutions and kept in a pool depend on the search history, and rows dropped to save time, or a relaxation that is solved only approximately, can produce a larger feasible set on a smaller box. For such constructions the statements below require a separate argument.

The relaxation bound on a box is

$$L(B) = \inf_{x \in B} \phi_B(x).$$

Monotonicity gives $L(B') \geq L(B)$ whenever $B' \subseteq B$: for $x \in B'$ one has $\phi_{B'}(x) \geq \phi_B(x) \geq L(B)$.

3.2 The operator and update schemes

For a cutoff U , which in practice is the value of an incumbent, define

$$K_U(B) = \{x \in B : \phi_B(x) \leq U\}, \quad T_U(B) = \text{box } K_U(B), \quad (5)$$

where $\text{box } S$ is the smallest closed box containing S , and $\text{box } \emptyset = \emptyset$. We also set $K_U(\emptyset) = T_U(\emptyset) = \emptyset$, so that repeated rounds remain defined after an empty cutoff set; the inclusions below then hold with $\emptyset$ as the smallest element. If $K_U(B) \neq \emptyset$, then $T_U(B) = [\ell', u']$ with

$$\ell'_i = \min\{x_i : x \in K_U(B)\}, \quad u'_i = \max\{x_i : x \in K_U(B)\}, \quad i = 1, \dots, n.$$

These $2n$ support problems are convex programs when ϕ_B is convex, and linear programs for lifted polyhedral relaxations. The cutoff slack of U is $\epsilon = U - f^*$. It is nonnegative when U is the value of a point of $X \cap B_0$; at a node, a global incumbent outside the node box can have a value below the node optimum, and then the node can be pruned. Results that need a nonempty limit or a minimizer in every iterate assume $U \geq f^*$ explicitly.

Solvers organize the support problems in different ways, and the difference matters for exact rate statements.

- A *Jacobi round* solves all $2n$ support problems on the one frozen relaxation built on B and returns $T_U(B)$.

- A *directional update* replaces one endpoint of the current box by the corresponding support over the cutoff set of the relaxation built on the current box. If that set is empty, the update returns $\emptyset$, and every later update of $\emptyset$ returns $\emptyset$.
- A *sequential round* applies $2n$ directional updates, one per signed coordinate direction, rebuilding the relaxation after each update. We write $S_U(B)$ for its result; it depends on the order of directions.
- A *selective round* applies directional updates to a subset of the directions.

Lemma 3.1 (Containment and order). *For all boxes B', B and cutoffs $U' \leq U$:*

- (a) $\{x \in X \cap B : f(x) \leq U\} \subseteq K_U(B) \subseteq T_U(B) \subseteq B$;
- (b) if $B' \subseteq B$, then $K_U(B') \subseteq K_U(B)$ and $T_U(B') \subseteq T_U(B)$;
- (c) $T_{U'}(B) \subseteq T_U(B)$.

Proof. (a) For a feasible $x \in B$ with $f(x) \leq U$, validity gives $\phi_B(x) \leq f(x) \leq U$. The remaining inclusions hold because $K_U(B) \subseteq B$ and B is a closed box. (b) For $x \in K_U(B')$, monotonicity gives $\phi_B(x) \leq \phi_{B'}(x) \leq U$, so $x \in K_U(B)$; the box hull preserves inclusion. (c) $K_{U'}(B) \subseteq K_U(B)$. $\square$

Lemma 3.2 (Update schedules). *Let $C_0 = B$ and let $C_1, C_2, \dots$ be the boxes produced by a sequence of directional updates at cutoff U .*

- (a) $T_U^m(B) \subseteq C_m$ for every m .
- (b) If every one of the $2n$ directions is updated at least once among the updates $j+1, \dots, m$, then $C_m \subseteq T_U(C_j)$. In particular, every sequential round satisfies $T_U^{2n}(B) \subseteq S_U(B) \subseteq T_U(B)$.
- (c) If every direction is updated infinitely often, then $\bigcap_m C_m = \bigcap_k T_U^k(B)$.

Proof. (a) A directional update of a box C changes one endpoint to the corresponding endpoint of $T_U(C)$ and keeps the others, so its result contains $T_U(C)$. Induction with Lemma 3.1(b) gives $C_m \supseteq T_U(C_{m-1}) \supseteq T_U(T_U^{m-1}(B)) = T_U^m(B)$. (b) Each update after step j optimizes over a cutoff set contained in $K_U(C_j)$, by Lemma 3.1(b). An endpoint updated in this way is therefore at least as tight as the corresponding endpoint of $T_U(C_j)$, and later updates never move it outward. Every endpoint is updated at least once, which proves $C_m \subseteq T_U(C_j)$; part (a) with $m = 2n$ gives the other inclusion for a sequential round. (c) Partition the schedule into consecutive finite blocks, each containing every direction at least once, and let t_j be the end of block j , with $t_0 = 0$. By (b) and monotonicity of T_U , $C_{t_j} \subseteq T_U(C_{t_{j-1}}) \subseteq T_U^j(B)$, and by (a), $T_U^{t_j}(B) \subseteq C_{t_j}$. Each block is nonempty, so $t_j \geq j$. The sequences $(T_U^{t_j}(B))_j$ and $(T_U^j(B))_j$ are therefore subsequences of the nested sequence $(T_U^k(B))_k$ with indices tending to infinity, and both have the intersection $\bigcap_k T_U^k(B)$. The two inclusions give the same intersection for $(C_{t_j})_j$, and this equals $\bigcap_m C_m$ because the boxes C_m are nested. $\square$

Part (c) is the order independence of the limit studied by Caprara and Locatelli [9], stated here for every fair schedule of directional updates and for Jacobi rounds; it uses only monotonicity. The lemma also determines which bounds transfer between update schemes. Every upper bound on the box produced by a Jacobi round also bounds the box produced by a complete sequential round, so upper bounds on widths and rates transfer. A selective round that leaves some directions unprocessed need not be contained in $T_U(B)$ and inherits no upper bound, but after m updates it still contains

$T_U^m(B)$, so lower enclosures transfer with the number of updates. Lower bounds on the box after a complete round do not transfer from Jacobi to sequential rounds, because a sequential round can be strictly faster. For example, let $n = 1$, $f(x) = x^2$, $U = 0$, and $\phi_{[\ell, u]}(x) = x^2 - (u - \ell)^2/16$, a valid, convex, and monotone family. A Jacobi round maps $[-h, h]$ to $[-h/2, h/2]$. A sequential round that updates the lower and then the upper endpoint maps $[-a, b]$ to $[-a', b']$ with $a' = (a + b)/4$ and $b' = (a' + b)/4 = (a + 5b)/16$, provided $b \leq 3a$ and $a \leq 11b$; the ratio b/a stays in $[1/2, 1]$ along the iteration from $a = b$, so these conditions hold throughout. The sequential widths therefore decrease like λ_s^k with $\lambda_s = (9 + \sqrt{17})/32 \approx 0.410$, the larger eigenvalue of $\begin{bmatrix} 1/4 & 1/4 \\ 1/16 & 5/16 \end{bmatrix}$, whereas Jacobi widths decrease exactly like 2^{-k} .

3.3 Iterates, fixed boxes, and the limit

Let $B_{k+1} = T_U(B_k)$, starting from B_0 , and let $B_\infty = \bigcap_k B_k$. By Lemma 3.1 the iterates are nested. If $U \geq f^*$, every iterate contains every global minimizer, so B_∞ is a nonempty compact box. The following lemma describes the part of the box that no valid tightening at cutoff U can remove.

Lemma 3.3 (Hull of the original sublevel set). *Let $H_U = \text{box}\{x \in X \cap B_0 : f(x) \leq U\}$. Every iterate contains H_U , and $T_U(H_U) = H_U$ if $H_U \neq \emptyset$. In particular, if x^* and x^{**} are two global minimizers, then $H = \text{box}\{x^*, x^{**}\} \subseteq B_\infty$ for every $U \geq f^*$ and*

$$L(B_\infty) \leq L(H).$$

Proof. By Lemma 3.1(a) and induction, every iterate contains each feasible point with objective at most U , hence their hull, because the iterates are closed boxes. The same points lie in $K_U(H_U)$, so $H_U \subseteq \text{box } K_U(H_U) = T_U(H_U) \subseteq H_U$. Finally $H \subseteq H_{f^*} \subseteq H_U \subseteq B_\infty$, and monotonicity of L gives $L(B_\infty) \leq L(H)$. $\square$

Caprara, Locatelli, and Monaci call H_U the lower limit of optimality-based domain reduction [10]. They identify a two-variable class for which iterated reduction reaches it and give examples in which reduction does not move any bound although H_U is a single point. Lemma 3.3 also limits iterated tightening at the root of a symmetric model: if the problem has several global minimizers related by a symmetry, the limit box contains their hull for every $U \geq f^*$, so the iterates cannot converge to a single point, and the relaxation bound after tightening is at most the relaxation bound on that hull. Tightening can still remove parts of the box outside the hull.

The box H_U depends only on the original problem and the cutoff. The relaxation can keep more than H_U .

Definition 3.4 (Fixed box). A nonempty box P is a *fixed box* of T_U if $T_U(P) = P$.

Since $T_U(P) \subseteq P$ always, P is fixed exactly when $P \subseteq T_U(P)$, that is, when every face of P is attained by a point of $K_U(P)$.

Lemma 3.5 (Persistence of fixed boxes). *Let P be a fixed box of T_U and $P \subseteq B$. Then $P \subseteq T_{U'}(B)$ for every $U' \geq U$, and P is contained in the result of every directional update of B at a cutoff $U' \geq U$. Consequently every Jacobi, sequential, or selective iteration started at a box containing P , with cutoffs at least U , keeps P in all its iterates.*

Proof. By Lemma 3.1, $K_U(P) \subseteq K_U(B) \subseteq K_{U'}(B)$. The point of $K_U(P)$ that attains the lower face a_i of P in coordinate i lies in $K_{U'}(B)$, so the minimum of x_i over $K_{U'}(B)$ is at most a_i ; the upper faces are analogous. Hence no directional update, and no Jacobi round, moves an endpoint of B past the corresponding face of P . Induction over the updates proves the last claim. $\square$

Lemma 3.5 is the order-theoretic core of the certificates in Section 5: finitely many points of $K_U(P)$ that attain all faces of P prove that P is fixed, and hence bound the total movement of every endpoint under all later rounds of the same relaxation family at cutoffs at least U . Section 5 lowers this cutoff to the largest relaxation value of the points. Fixed boxes can also be combined.

Lemma 3.6 (Greatest fixed box). *Let $\mathcal{P}$ be a nonempty family of fixed boxes of T_U contained in B_0 . Then $\text{box} \bigcup_{P \in \mathcal{P}} P$ is a fixed box. Consequently, the hull P_U of all fixed boxes in B_0 , with $P_U = \emptyset$ if there is none, contains every fixed box in B_0 and, when nonempty, is itself the greatest fixed box in B_0 . Every Jacobi iterate contains P_U , and $H_U \subseteq P_U$.*

Proof. The hull $Q = \text{box} \bigcup_{P \in \mathcal{P}} P$ is a nonempty box in B_0 . Each $P \in \mathcal{P}$ lies in Q , so Lemma 3.1(b) gives $P = T_U(P) \subseteq T_U(Q)$. The closed box $T_U(Q)$ therefore contains Q , and Lemma 3.1(a) gives $T_U(Q) = Q$. Applied to the family of all fixed boxes in B_0 , when it is nonempty, this shows that P_U is a fixed box; by construction it contains every fixed box in B_0 . Lemma 3.5 places P_U in every Jacobi iterate; for $P_U = \emptyset$ there is nothing to prove. If $H_U \neq \emptyset$, it is a fixed box in B_0 by Lemma 3.3, so $H_U \subseteq P_U$. $\square$

Proposition 5.9 in Section 5 is the finite-certificate form of the first statement. In lattice terms, the subboxes of B_0 together with $\emptyset$ form a complete lattice under inclusion, T_U is monotone on it, and P_U is its greatest fixed point; the lemma is the elementary case of Tarski's fixed-point theorem [43] that we need. It uses monotonicity alone. Identifying P_U with the limit of the iteration needs a closedness condition.

Proposition 3.7 (The limit and the greatest fixed box). *Suppose every Jacobi iterate is nonempty, for example because $U \geq f^*$. The limit B_∞ of the Jacobi iterates is a nonempty compact box. It contains P_U , and hence H_U and every fixed box in B_0 , and it is also the limit of every sequence of directional updates that updates each direction infinitely often. Suppose in addition that the family is closed along decreasing sequences in the following sense:*

$$C_k \downarrow C, \quad x_k \in K_U(C_k), \quad x_k \rightarrow x \implies x \in K_U(C), \quad (6)$$

where $C_k \downarrow C$ means that the boxes are nested with intersection C . Then B_∞ is a fixed box, and hence $B_\infty = P_U$.

Proof. The iterates are nested nonempty compact boxes, so their intersection is a nonempty compact box. The containments are Lemma 3.6, and the statement about schedules is Lemma 3.2(c).

Assume (6). For each k and each direction, say the upper endpoint of coordinate i , the endpoint of B_{k+1} is $\max\{x_i : x \in K_U(B_k)\}$, attained at some $y^k \in K_U(B_k)$ by compactness. A subsequence of (y^k) converges to some $\bar{y}$, and (6) gives $\bar{y} \in K_U(B_\infty)$. Its coordinate $\bar{y}_i$ is the limit of the upper endpoints of B_{k+1} , which is the upper endpoint of B_∞ in coordinate i . Repeating the argument for every direction shows that every face of B_∞ is attained in $K_U(B_\infty)$, so B_∞ is a fixed box in B_0 . Hence $B_\infty \subseteq P_U$, and the first part gives equality. $\square$

The same argument shows that (6) makes T_U continuous along decreasing sequences of boxes, $T_U(\bigcap_k C_k) = \bigcap_k T_U(C_k)$, which is the property that identifies the limit of the iterates with the greatest fixed box. Condition (6) holds whenever the projected objective, as a function $(x, \ell, u) \mapsto \phi_{[\ell, u]}(x)$ on $\{\ell \leq x \leq u\}$, is lower semicontinuous: if $C_k \downarrow C$, $x_k \in K_U(C_k)$, and $x_k \rightarrow x$, then $x \in C$, the endpoints of C_k converge to those of C , and $\phi_C(x) \leq \liminf_k \phi_{C_k}(x_k) \leq U$. The termwise relaxations of (a) are jointly continuous in this sense. For lifted families, suppose $R(B) = \{z \in Z : g_j(z; \ell, u) \leq 0, j = 1, \dots, m\}$ with the box rows included, a compact set Z ,

functions g_j continuous in (z, ℓ, u) , and a continuous objective v . If $x_k \in K_U(C_k)$, choose lifts $z_k \in R(C_k)$ with $v(z_k) \leq U$. A subsequence converges to some z , the endpoints of C_k converge monotonically to those of C , and continuity gives $z \in R(C)$ and $v(z) \leq U$; its projection is the limit x . McCormick rows have coefficients that are polynomial in the box endpoints, so they satisfy this condition, as do tangent rows at points that depend continuously on the endpoints. Closedness and monotonicity are independent hypotheses: the five-point tangent family above satisfies (6) but is not monotone, and the next example is monotone but violates (6).

Without (6), the limit of iterated tightening need not be a fixed box, and can be strictly larger than the greatest fixed box.

Example 3.8 (A limit that is not a fixed box). Let $n = 1$, $X = \{0\}$, $f \equiv 0$, $B_0 = [0, 1]$, and $U = 0$. For a box $B = [r, s] \subseteq B_0$ let $\phi_B(x) = 0$ if $x \leq g(s)$ and $\phi_B(x) = +\infty$ otherwise, where $g(s) = s/4$ for $s \leq 1/2$ and $g(s) = (2s + 1)/4$ for $s > 1/2$. The function g is nondecreasing with $g(s) \leq s$ on $[0, 1]$. Hence the family is valid ($\phi_B(0) = 0$ whenever $0 \in B$), monotone, and has closed sublevel sets, and $T_0([0, s]) = [0, g(s)]$. From $s_0 = 1$ the iterates satisfy $s_k - 1/2 = 2^{-k-1}$, so $B_\infty = [0, 1/2]$, whereas $T_0(B_\infty) = [0, 1/8]$. A box $[r, s] \subseteq B_0$ is fixed only if $g(s) \geq s$, that is, $s = 0$. Hence $P_U = H_U = \{0\}$. Continuing from B_∞ gives $[0, 1/8], [0, 1/32], \dots$, which reach P_U only as the limit of a second infinite sequence.

3.4 Three limits on further tightening

Lemmas 3.3, 3.5, and 3.6 and Proposition 3.7 identify three things that limit how much further rounds can remove from a current iterate B_k , for a fixed relaxation family and cutoff:

- (i) the hull H_U of the original sublevel set, which no valid tightening at cutoff U removes and which only a better incumbent or branching shrinks;
- (ii) the fixed boxes of T_U , which no number of further rounds of the same relaxation family removes, and which only a smaller cutoff, a stronger relaxation, or branching can shrink; their hull P_U , the greatest fixed box when one exists and empty otherwise, contains H_U and every fixed box in B_0 , and equals the limit B_∞ under (6) (if some iterate is empty, both are empty);
- (iii) the rate at which the iterates approach B_∞ , which determines how many rounds are needed for a given part of the possible reduction.

These objects satisfy $H_U \cup P \subseteq P_U \subseteq B_\infty \subseteq B_k$ for every fixed box P in B_0 , although H_U and P need not contain each other. For example, with $f(x) = x^2$ on $[-1, 1]$, $U = 0$, and the valid family $\phi_B \equiv 0$, one has $H_U = \{0\}$ and the fixed box $\{1\}$, and there $P_U = B_\infty = [-1, 1]$. If $P = [a, b]$ is a fixed box inside the current box $B_k = [\ell^k, u^k]$, the total future movement of each endpoint satisfies

$$\ell_i^\infty - \ell_i^k \leq a_i - \ell_i^k, \quad u_i^k - u_i^\infty \leq u_i^k - b_i.$$

The first item is a property of the problem, the second of the relaxation family, and the third of the iteration. Section 4 gives sufficient conditions near a minimizer under which fixed boxes exist at every sufficiently small scale, so that tightening stalls, and sufficient conditions under which the iterates contract geometrically, with a limit of width $O(\sqrt{\epsilon})$; these conditions do not cover every relaxation family. Section 5 turns the second item into finite certificates, and Section 7 bounds the remaining movement under a uniform sensitivity bound. Sections 9 to 11 examine how far this information supports decisions about when to start, stop, and repeat tightening in a solver.

4 Local rates and stalling near a minimizer

This section quantifies items (ii) and (iii) of Section 3.4 near a global minimizer x^* . It answers three questions. How fast do the iterates contract? How large is their limit as a function of the cutoff slack $\epsilon = U - f^*$? When do fixed boxes exist at every small scale, so that no number of rounds tightens anything? The answers depend on the relaxation family through a second-order *tangent model*, the limit of the relaxation rescaled to boxes that shrink to x^* , and on an associated map on box shapes. They take the form of sufficient conditions for geometric contraction and for stalling; Example 4.12 shows that a valid monotone family can satisfy neither.

Throughout, $U = f^* + \epsilon$ with $\epsilon \geq 0$, and the order hypotheses of Section 3 hold on all boxes used, including the comparison boxes $x^* + tD(d)$ introduced below. By Lemmas 3.2 and 3.5, the upper bounds on widths and rates below hold for Jacobi rounds and for complete sequential rounds, the stall statements hold for every update scheme, and lower bounds and exact rates are statements about Jacobi rounds. All results involve x^* , f^* , and tangent data that a solver does not know. Their role is to identify which limit dominates and how it scales; Section 5 supplies computable certificates.

4.1 Two scalar bounds

The simplest estimates compare the relaxation with a growth function of the distance to x^* . They fix two regimes.

Proposition 4.1 (Sharp growth). *Suppose $\kappa > 0$ and $\tau \geq 0$ satisfy*

$$\phi_B(x) \geq f^* + \kappa\|x - x^*\|_\infty - \tau w(B)^2 \quad (7)$$

for every box B with $x^ \in B \subseteq B_0$ and every $x \in B$. Then*

$$w(T_U(B)) \leq \frac{2}{\kappa}(\epsilon + \tau w(B)^2)$$

for every such box. Suppose some iterate satisfies $w(B_K) \leq \kappa/(4\tau)$ (any iterate if $\tau = 0$). If $\epsilon = 0$, then $w(B_{k+1}) \leq (2\tau/\kappa)w(B_k)^2$ for $k \geq K$ and $B_\infty = \{x^\}$. If $\epsilon > 0$, then $w(B_{k+1}) \leq w(B_k)/2 + 2\epsilon/\kappa$ for $k \geq K$ and $w(B_\infty) \leq 4\epsilon/\kappa$.*

Proof. A point $x \in K_U(B)$ satisfies $\kappa\|x - x^*\|_\infty \leq \phi_B(x) - f^* + \tau w(B)^2 \leq \epsilon + \tau w(B)^2$, and every coordinate range of $K_U(B)$ is at most twice this radius. If $w(B_k) \leq \kappa/(4\tau)$, then $(2\tau/\kappa)w(B_k)^2 \leq w(B_k)/2$; since the iterates are nested, all later iterates stay in this region. Hence $w(B_{k+1}) \leq w(B_k)/2 + 2\epsilon/\kappa$ for $k \geq K$, and the limit of the nonincreasing widths is at most $4\epsilon/\kappa$. For $\epsilon = 0$ the widths tend to zero, the quadratic recurrence holds, and $x^* \in B_k$ for every k . $\square$

The one-round bound is the mechanism of marginals-based range reduction: a relaxation whose value grows at least linearly with slope κ away from a bound cannot keep points farther than $(U - L)/\kappa$ from that bound [35]. Condition (7) is a statement about all relaxed points, not only feasible ones. It holds, for example, in a problem with only bound constraints whose objective satisfies $f(x) - f^* \geq \kappa\|x - x^*\|_\infty$ on B_0 and whose relaxation satisfies $\phi_B \geq f - \tau w(B)^2$ on B . It also holds on small boxes at a vertex minimizer at which every partial derivative is nonzero and makes the objective increase into the box (strict complementarity in every coordinate), for the composite relaxations of Section 4.5: on boxes containing such a vertex the first-order term is at least $\kappa\|x - x^*\|_\infty$ with κ the smallest absolute partial derivative, and Theorem 4.24 bounds the rest by a multiple of $w(B)^2$.

Proposition 4.2 (Quadratic growth). *Suppose $\mu > 0$ and $\tau \geq 0$ satisfy*

$$\phi_B(x) \geq f^* + \mu \|x - x^*\|_\infty^2 - \tau w(B)^2$$

for every box B with $x^ \in B \subseteq B_0$ and every $x \in B$. Then $w(T_U(B))^2 \leq 4\epsilon/\mu + q^2 w(B)^2$ with $q^2 = 4\tau/\mu$. If $q < 1$, the iterates satisfy*

$$w(B_k)^2 \leq q^{2k} w(B_0)^2 + \frac{4\epsilon}{\mu} \frac{1 - q^{2k}}{1 - q^2}, \quad w(B_\infty) \leq 2\sqrt{\frac{\epsilon}{\mu - 4\tau}},$$

and for $\epsilon = 0$ the widths contract at least by the factor q per round.

Proof. A point of $K_U(B)$ lies within ∞ -norm distance $\sqrt{(\epsilon + \tau w(B)^2)/\mu}$ of x^* . Bound each coordinate range by twice this radius, square, and iterate the affine recurrence for $w(B_k)^2$. $\square$

For $\epsilon > 0$ the proposition bounds the widths by a scalar envelope; it does not assert that the distance to the actual limit box contracts by the factor q . The strict condition $\tau < \mu/4$ has the same numerical threshold as conditions excluding the cluster problem in branch and bound: with $\mu = \lambda_1/2$, where λ_1 is the smallest eigenvalue of the Hessian at x^* , the threshold is $\lambda_1/8$, compared with $K \leq \lambda_1/8$ in Wechsung, Schaber, and Barton [46], and with $\mu = \gamma/2$ it is $\gamma/8$, compared with $\tau^* \leq \gamma/8$ in Kannan and Barton [25]. Their prefactors bound the gap between minimum values; here τ bounds the pointwise gap over the box, which is at least as large. Both bounds are far from necessary. For $f(x, y) = x^2 + y^2 + xy$ with the McCormick envelope of xy , the best constants are $\mu = 3/4$ (on the face $x = 1$ of the unit cube the minimum of f is attained at $y = -1/2$) and $\tau = 1/4$, so $q^2 = 4/3 > 1$, yet Jacobi rounds contract by the factor 0.7247 (Proposition 4.15). The scalar bound charges the largest relaxation gap at every point and the weakest growth in every direction. The tangent model keeps the dependence on the shape of the box and on the position inside it.

4.2 The tangent model and the tangent map

A *shape* is a vector $d = (d^-, d^+) \in \mathbb{R}_{\geq 0}^{2n}$, and $D(d) = \prod_{i=1}^n [-d_i^-, d_i^+]$. Every box containing x^* can be written as $x^* + tD(d)$ with $t > 0$. For a shape $u \gg 0$, meaning that all $2n$ components are positive, the *gauge* of a box $B = [\ell, u'] \ni x^*$ is

$$g_u(B) = \min\{t \geq 0 : B \subseteq x^* + tD(u)\} = \max_i \max\left\{\frac{x_i^* - \ell_i}{u_i^-}, \frac{u_i' - x_i^*}{u_i^+}\right\}.$$

The endpoint vector of $x^* + tD(d)$ is $p(\{x^*\}) + td$; shapes are endpoint vectors measured from x^* .

Assumption 4.3 (Tangent model). The point $x^* \in X \cap B_0$ is a global minimizer, and there is a function $Q(d, \xi) \in \mathbb{R}$, defined for shapes d and $\xi \in D(d)$, with the following property. For every compact set $\mathcal{K}$ of shapes there is $t_{\mathcal{K}} > 0$ such that the boxes $x^* + tD(d)$, $d \in \mathcal{K}$, $0 < t \leq t_{\mathcal{K}}$, belong to $\mathfrak{B}$ and

$$\varrho_{\mathcal{K}}(t) = \sup_{d \in \mathcal{K}, \xi \in D(d)} \frac{|\phi_{x^* + tD(d)}(x^* + t\xi) - f^* - t^2 Q(d, \xi)|}{t^2} \rightarrow 0 \quad (t \downarrow 0). \quad (8)$$

The function Q is the *tangent model* of the relaxation family at x^* . Assumption 4.3 requires finite relaxation values on the comparison boxes and no first-order term, so it describes interior minimizers and minimizers on faces at which the objective has zero derivative across the face. For quadratic objectives with McCormick relaxations the remainder in (8) is zero (Section 4.4);

for composite McCormick relaxations of factorable functions, Q is computed by a recursion over the factors (Theorem 4.24). Minimizers on the boundary with nonzero derivatives are treated in Section 4.6. The restricted tangent contraction for exactly represented linear equations closes Section 4.3.

For $c \geq 0$ and a shape d , let

$$S_c(d) = \{\xi \in D(d) : Q(d, \xi) \leq c\},$$

and define the shape $\Phi_c(d) \in \mathbb{R}_{\geq 0}^{2n}$ by $\text{box } S_c(d) = D(\Phi_c(d))$, that is, $\Phi_c(d)_i^- = -\min_{\xi \in S_c(d)} \xi_i$ and $\Phi_c(d)_i^+ = \max_{\xi \in S_c(d)} \xi_i$. The *tangent map* is $\Phi = \Phi_0$. In scaled coordinates, Φ describes one Jacobi round on a box $x^* + tD(d)$ at zero cutoff slack, up to the remainder in (8).

Lemma 4.4 (Properties of the tangent model). *Under Assumption 4.3:*

- (a) $Q(sd, s\xi) = s^2 Q(d, \xi)$ for $s > 0$;
- (b) if $d \leq d'$, then $Q(d, \xi) \geq Q(d', \xi)$ for $\xi \in D(d)$;
- (c) $Q(d, 0) \leq 0$;
- (d) $Q(d, \cdot)$ is lower semicontinuous on $D(d)$, and convex if every ϕ_B is convex;
- (e) $S_c(d)$ is compact and contains 0, so $\Phi_c(d)$ is well defined; moreover $\Phi_c(d) \leq d$, $\Phi_c(d) \leq \Phi_c(d')$ if $d \leq d'$, $\Phi_c(d) \leq \Phi_{c'}(d)$ if $c \leq c'$, and $\Phi_c(sd) = s\Phi_{c/s^2}(d)$; in particular Φ is order preserving and positively homogeneous of degree one;
- (f) $\Phi_c(d) \rightarrow \Phi(d)$ componentwise as $c \downarrow 0$.

Proof. (a) The box $x^* + tD(sd)$ equals $x^* + (ts)D(d)$, and the point $x^* + t(s\xi)$ equals $x^* + (ts)\xi$. Apply (8) with $\mathcal{K} = \{d, sd\}$ to both descriptions, divide by t^2 , and let $t \downarrow 0$. (b) For $d \leq d'$ the box $x^* + tD(d)$ is contained in $x^* + tD(d')$; apply monotonicity at $x^* + t\xi$, subtract f^* , divide by t^2 , and let $t \downarrow 0$. (c) Validity at the feasible point x^* gives $\phi_B(x^*) \leq f^*$. (d) For fixed d , $Q(d, \cdot)$ is the uniform limit on $D(d)$ of the real-valued functions $\xi \mapsto t^{-2}(\phi_{x^*+tD(d)}(x^* + t\xi) - f^*)$, which are lower semicontinuous, and convex if ϕ_B is. Uniform limits of lower semicontinuous functions are lower semicontinuous, and pointwise limits of convex functions are convex. (e) Compactness follows from (d), and $0 \in S_c(d)$ from (c). The inclusion $S_c(d) \subseteq D(d)$ gives $\Phi_c(d) \leq d$; for $d \leq d'$, (b) gives $S_c(d) \subseteq S_c(d')$; and $S_c(d)$ increases with c . By (a), $\xi \in S_c(sd)$ if and only if $\xi/s \in S_{c/s^2}(d)$, so $S_c(sd) = sS_{c/s^2}(d)$. (f) The sets $S_c(d)$ are compact, decrease as $c \downarrow 0$, and have intersection $S_0(d)$. For a fixed component, choose a maximizer ξ^c of ξ_i over $S_c(d)$; every cluster point of ξ^c as $c \downarrow 0$ lies in $S_0(d)$, so the maxima converge to the maximum over $S_0(d)$. Minima are analogous. $\square$

The following constant measures contraction. It is an upper Collatz–Wielandt number of the order-preserving homogeneous map Φ , made robust to a small cutoff slack.

Definition 4.5 (Contraction factor of the tangent map).

$$r^* = \inf\{\lambda \geq 0 : \Phi_\delta(u) \leq \lambda u \text{ for some } u \gg 0 \text{ and some } \delta > 0\}.$$

Lemma 4.6. *Under Assumption 4.3, $0 \leq r^* \leq 1$ and $r^* = \inf\{\lambda \geq 0 : \Phi(u) \leq \lambda u \text{ for some } u \gg 0\}$.*

Proof. $\Phi_\delta(u) \leq u$ gives $r^* \leq 1$. Since $\Phi \leq \Phi_\delta$, the right side is at most r^* . Conversely, if $\Phi(u) \leq \lambda u$ with $u \gg 0$ and $\lambda' > \lambda$, then $\lambda u < \lambda' u$ componentwise, and Lemma 4.4(f) gives $\Phi_\delta(u) \leq \lambda' u$ for small $\delta > 0$; hence $r^* \leq \lambda'$. $\square$

If the remainder in (8) is zero and $\epsilon = 0$, a Jacobi round maps $x^* + tD(d)$ exactly to $x^* + tD(\Phi(d))$, so the exact asymptotic rate is the growth rate of $\Phi^k(d)$. This rate never exceeds r^* : if $\Phi(u) \leq \lambda u$ and $d \leq su$, then $\Phi^k(d) \leq s\lambda^k u$. Proposition 4.9 proves equality when Φ has a positive eigenvector. Equality in general would require further properties of Φ , such as continuity, which we do not establish; we use r^* only as an upper bound.

4.3 Contraction, exact rates, and stalling

Theorem 4.7 (Local contraction). *Let Assumption 4.3 hold, and let $u \gg 0$, $\delta > 0$, and $\lambda \in (0, 1)$ satisfy $\Phi_\delta(u) \leq \lambda u$. There is $\bar{t} > 0$ with the following properties.*

- (a) *If B is a box with $x^* \in B$ and $g_u(B) = t \leq \bar{t}$, and $t^2 \geq 2\epsilon/\delta$, then $g_u(T_U(B)) \leq \lambda t$.*
- (b) *If $g_u(B_0) \leq \bar{t}$, the Jacobi iterates, and the iterates of complete sequential rounds, satisfy*

$$g_u(B_k) \leq \max\{\lambda^k g_u(B_0), \sqrt{2\epsilon/\delta}\}, \quad B_\infty \subseteq x^* + \sqrt{2\epsilon/\delta} D(u).$$

In particular, for $\epsilon = 0$ the iterates converge to $\{x^\}$ and $g_u(B_k) \leq \lambda^k g_u(B_0)$.*

Proof. Choose $\bar{t} \leq t_{\{u\}}$ such that $\varrho_{\{u\}}(t) \leq \delta/2$ for $t \leq \bar{t}$. (a) If $t = 0$, then $B = \{x^*\}$ and there is nothing to prove. Otherwise let $C = x^* + tD(u)$, so that $B \subseteq C$. A point $x = x^* + t\xi \in K_U(C)$ satisfies

$$t^2 Q(u, \xi) - \frac{\delta}{2} t^2 \leq \phi_C(x) - f^* \leq \epsilon \leq \frac{\delta}{2} t^2,$$

so $Q(u, \xi) \leq \delta$ and $\xi \in S_\delta(u)$. Hence $T_U(C) \subseteq x^* + tD(\Phi_\delta(u)) \subseteq x^* + \lambda t D(u)$, and $T_U(B) \subseteq T_U(C)$ by Lemma 3.1. (b) The gauges of the nested iterates are nonincreasing. If $g_u(B_k) \geq \sqrt{2\epsilon/\delta}$, part (a) gives $g_u(B_{k+1}) \leq \lambda g_u(B_k)$; otherwise $g_u(B_{k+1}) \leq g_u(B_k) < \sqrt{2\epsilon/\delta}$. Induction gives the displayed bound, and $B_\infty \subseteq B_k$ gives the inclusion. Complete sequential rounds are covered by Lemma 3.2. $\square$

Corollary 4.8 (Rate and floor). *Suppose Assumption 4.3 holds and $r^* < 1$. For every $\lambda \in (r^*, 1)$ there are $u \gg 0$, $\delta > 0$, and $\bar{t} > 0$ such that the conclusions of Theorem 4.7 hold for every start box $B_0 \ni x^*$ with $w(B_0) \leq \bar{t} \min_j u_j$. For $\epsilon = 0$ the iterates converge to x^* linearly with factor λ in the gauge g_u . For $\epsilon > 0$, $w(B_\infty) \leq \sqrt{2\epsilon/\delta} \max_i (u_i^- + u_i^+)$. If moreover $X \cap B_0$ contains a neighborhood of x^* on which $f \leq f^* + M\|x - x^*\|_\infty^2$ for some $M > 0$, then $w(B_\infty) \geq 2\sqrt{\epsilon/M}$ for small ϵ , so under these additional premises the width of the limit is of order exactly $\sqrt{\epsilon}$.*

Proof. By the definition of r^* there are $u \gg 0$, $\delta > 0$, and $\lambda' < \lambda$ with $\Phi_\delta(u) \leq \lambda' u \leq \lambda u$. A box containing x^* with width at most $\bar{t} \min_j u_j$ has gauge at most $\bar{t}$. The upper bound on $w(B_\infty)$ is Theorem 4.7(b). For the lower bound, every point of the cube $x^* + \sqrt{\epsilon/M} [-1, 1]^n$ is feasible for small ϵ and has objective at most U , so the cube lies in H_U , which lies in B_∞ by Lemma 3.3. $\square$

The upper bound on the rate is exact in the presence of an eigenvector.

Proposition 4.9 (Exact rate from a positive eigenvector). *Suppose the tangent model is exact, $\phi_{x^*+tD(d)}(x^* + t\xi) = f^* + t^2 Q(d, \xi)$ for all $t > 0$ and all shapes d for which the box belongs to $\mathfrak{B}$, and let $\epsilon = 0$. If $\Phi(u) = \rho u$ for some $u \gg 0$, then $T_U^k(x^* + tD(u)) = x^* + \rho^k t D(u)$ and $r^* = \rho$. If the Jacobi iterates start from a box with $x^* + aD(u) \subseteq B_0 \subseteq x^* + bD(u)$, $a > 0$, and all these comparison boxes belong to $\mathfrak{B}$, then*

$$x^* + a\rho^k D(u) \subseteq B_k \subseteq x^* + b\rho^k D(u).$$

Hence, if $\rho > 0$, $w(B_k) = \Theta(\rho^k)$ and $w(B_k)^{1/k} \rightarrow \rho$; if $\rho = 0$, then $B_1 = \{x^\}$.*

Proof. With zero remainder and $\epsilon = 0$, $K_U(x^* + tD(d)) = x^* + tS_0(d)$, so $T_U(x^* + tD(d)) = x^* + tD(\Phi(d))$, and induction proves the first identity. Monotonicity of T_U gives the sandwich. For r^* , Lemma 4.6 gives $r^* \leq \rho$. If $\Phi(v) \leq \lambda v$ for some $v \gg 0$ and $\lambda < \rho$, choose $a > 0$ with $au \leq v$; then $a\rho^k u = \Phi^k(au) \leq \Phi^k(v) \leq \lambda^k v$ for all k , which is impossible. Lemma 4.6 then gives $r^* \geq \rho$. $\square$

The lower bound uses Jacobi rounds; sequential rounds can be strictly faster, as the example after Lemma 3.2 shows exactly.

We now give a sufficient condition for fixed boxes to exist at every sufficiently small scale.

Theorem 4.10 (Stalling). *Let Assumption 4.3 hold. Let v be a nonzero shape and $\delta > 0$, and suppose that for every positive component v_i^+ there is $\xi \in D(v)$ with $\xi_i = v_i^+$ and $Q(v, \xi) \leq -\delta$, and likewise for every positive component v_i^- with $\xi_i = -v_i^-$. Then there is $\bar{t} > 0$ such that $x^* + tD(v)$ is a fixed box of T_U for every $t \in (0, \bar{t}]$ and every $U \geq f^*$. Every Jacobi, sequential, or selective iteration started from a box containing $x^* + \bar{t}D(v)$, with cutoffs at least f^* , keeps this box; in particular $w(B_k) \geq \bar{t} \max_i(v_i^- + v_i^+)$ for all k , whatever $\epsilon \geq 0$. Moreover $r^* = 1$.*

Proof. Choose $\bar{t}$ with $\varrho_{\{v\}}(t) < \delta$ for $t \leq \bar{t}$. For a witness ξ as in the hypothesis, $\phi_{x^*+tD(v)}(x^* + t\xi) < f^* + t^2(-\delta + \delta) = f^* \leq U$, so $x^* + t\xi \in K_U(x^* + tD(v))$. A component of v that is zero needs no witness: the face then passes through x^* , which lies in K_U by validity. Thus every face of $x^* + tD(v)$ is attained, and the box is fixed. Lemma 3.5 gives the persistence. For r^* , the same witnesses lie in $S_0(v)$, so $\Phi(v) = v$. If $\Phi(u) \leq \lambda u$ with $u \gg 0$ and $\lambda < 1$, choose s with $v \leq su$; then $v = \Phi^k(v) \leq s\lambda^k u \rightarrow 0$, a contradiction. Lemma 4.6 gives $r^* = 1$. $\square$

Under the hypotheses of Theorem 4.10, the limit box does not shrink to x^* as $\epsilon \downarrow 0$. Improving the incumbent cannot remove the fixed boxes $x^* + tD(v)$ or force convergence to x^* , although it can still remove parts of a larger box outside them; the obstruction is the relaxation, not the near-optimal set. With zero remainder, the same proof needs only $Q(v, \xi) \leq 0$ at the witnesses (Proposition 4.19). With a nonzero remainder the strict margin δ is needed; Example 4.12 has tangent witnesses with value zero and no positive-width fixed box.

Both theorems are governed by the values of Q on the faces of a shape. For a shape d and a component $d_i^+ > 0$, define the face value $m_i^+(d) = \min\{Q(d, \xi) : \xi \in D(d), \xi_i = d_i^+\}$, and $m_i^-(d)$ analogously; the minima exist by Lemma 4.4(d).

Corollary 4.11 (Face test). *Let Assumption 4.3 hold and let d be a nonzero shape.*

- (a) *If $d \gg 0$ and every face value of d is positive, then for every δ with $0 < \delta < \min_{i,\pm} m_i^\pm(d)$ there is $\lambda < 1$ with $\Phi_\delta(d) \leq \lambda d$; hence $r^* < 1$ and Theorem 4.7 applies with $u = d$.*
- (b) *If every face value of d for a positive component is negative, then Theorem 4.10 applies with $v = d$.*

If every ϕ_B is convex, each face value is a convex program, and $2n$ convex programs evaluate both tests at a given shape. The tests are sufficient conditions; zero or mixed signs leave the question open.

Proof. (a) If $\Phi_\delta(d)_i^+ = d_i^+$, a maximizer ξ of ξ_i over $S_\delta(d)$ has $\xi_i = d_i^+$ and $Q(d, \xi) \leq \delta < m_i^+(d)$, which contradicts the definition of $m_i^+(d)$. Hence every component of $\Phi_\delta(d)$ is strictly smaller than the corresponding component of d , and $\lambda = \max_{i,\pm} \Phi_\delta(d)_i^\pm / d_i^\pm < 1$. (b) The minimizers defining the face values are the witnesses of Theorem 4.10. $\square$

When the signs of the face values are mixed, a tangent round at zero cutoff slack, that is, one application of Φ , moves the endpoints with positive face value and keeps the others; the shape changes, and the test must be repeated on the new shape. Example 4.22 shows that the same objective can stall on one shape and tighten on another. For relaxations with a nonzero remainder, strict signs give the same conclusion on sufficiently small boxes at zero cutoff slack, whereas a positive slack can keep a face whose tangent value is positive. The next example shows that the two sufficient conditions do not exhaust the possibilities.

Example 4.12 (A critical tangent map). Let $n = 1$, $X = \mathbb{R}$, $f(x) = x^2$, $B_0 = [-1/2, 1/2]$, and for every subbox B

$$\phi_B(x) = x^2 - E(w(B)), \quad E(w) = \frac{w^2}{4} - \frac{w^3}{8}.$$

On $[0, 1]$, $E \geq 0$ and $E'(w) = w/2 - 3w^2/8 \geq 0$, so the family is valid, convex, continuous, and monotone. At $U = f^* = 0$ the cutoff set of a box of width $w > 0$ lies in an interval of width $2\sqrt{E(w)} = w\sqrt{1 - w/2} < w$, so no box of positive width is fixed. On $tD(d)$, with $s = d^- + d^+$, $\phi(t\xi) = t^2(\xi^2 - s^2/4) + t^3s^3/8$, so Assumption 4.3 holds with $Q(d, \xi) = \xi^2 - s^2/4$ and a remainder of order t^3 . For $c \geq 0$, $\Phi_c(d) = (\min\{d^-, q\}, \min\{d^+, q\})$ with $q = (s^2/4 + c)^{1/2} \geq s/2$, so Φ_c keeps the smaller component of every shape and $r^* = 1$. The face values of $d = (1, 1)$ are zero, so neither test applies. The iteration is neither stalled nor geometric. From $[-h_0, h_0]$ with $0 < h_0 \leq 1/2$, the Jacobi radii satisfy $h_{k+1} = h_k\sqrt{1 - h_k}$; they decrease to zero and

$$\frac{1}{h_{k+1}} - \frac{1}{h_k} = \frac{1}{\sqrt{1 - h_k}(1 + \sqrt{1 - h_k})} \rightarrow \frac{1}{2},$$

so $h_k \sim 2/k$. At a cutoff slack $\epsilon \in (0, h_0^3)$ the radii follow $h_{k+1} = \min\{h_k, g(h_k)\}$ with $g(h) = (\epsilon + h^2 - h^3)^{1/2}$. The function g is increasing on $(0, 1/2]$, $g(\epsilon^{1/3}) = \epsilon^{1/3}$, and $\epsilon^{1/3} < g(h) < h$ for $h > \epsilon^{1/3}$, so the radii decrease to $\epsilon^{1/3}$ and the limit has width $2\epsilon^{1/3}$, larger than any $O(\sqrt{\epsilon})$ bound for small ϵ . Finally, the actual face values $\phi_{[-t, t]}(\pm t) = t^3$ are positive although the tangent face values vanish, which is why Theorem 4.10 requires a strict margin.

Remark 4.13 (Linear equations kept exactly). A projected relaxation that keeps an affine equality exactly is infinite off the feasible affine set, so Assumption 4.3 must be restricted to that set. The following proposition gives the contraction result; Proposition 8.16 in Section 8.6 computes one exact constrained map.

Let the feasible set be $X = \{x : Cx = c_0\}$, intersected with the box B_0 , and let the relaxation on B be

$$\phi_B(x) = \phi_B^0(x) \text{ if } Cx = c_0, \quad \phi_B(x) = +\infty \text{ otherwise,}$$

where ϕ_B^0 is a valid and monotone relaxation of f that is finite on B . This construction is valid and monotone because the equality does not depend on B . We use the shapes $D(u)$ and the gauge g_u defined above.

Proposition 4.14 (Restricted tangent contraction). *Let x^* lie in the interior of B_0 and minimize f on $X \cap B_0$, let f be differentiable at x^* , and let $U = f^* + \epsilon$ with $\epsilon \geq 0$. Fix a shape $u \gg 0$ and suppose that the boxes $x^* + tD(u)$ belong to $\mathfrak{B}$ for small $t > 0$ and that a function $Q(u, \cdot)$ on $N_u = D(u) \cap \ker C$ satisfies*

$$\sup_{\xi \in N_u} \frac{|\phi_{x^* + tD(u)}^0(x^* + t\xi) - f^* - t\nabla f(x^*)^T \xi - t^2 Q(u, \xi)|}{t^2} \rightarrow 0 \quad (t \downarrow 0).$$

Let $S_\delta^C(u) = \{\xi \in N_u : Q(u, \xi) \leq \delta\}$ and define the shape $\Phi_\delta^C(u)$ by $\text{box } S_\delta^C(u) = D(\Phi_\delta^C(u))$. If $\delta > 0$ and $\lambda \in (0, 1)$ satisfy $\Phi_\delta^C(u) \leq \lambda u$, there is $\bar{t} > 0$ with the following properties.

(a) If $B \ni x^*$ has $g_u(B) = t \leq \bar{t}$ and $t^2 \geq 2\epsilon/\delta$, then $g_u(T_U(B)) \leq \lambda t$.

(b) If $g_u(B'_0) \leq \bar{t}$, the Jacobi iterates from B'_0 , and the iterates of complete sequential rounds, satisfy $g_u(B_k) \leq \max\{\lambda^k g_u(B'_0), \sqrt{2\epsilon/\delta}\}$; at zero cutoff slack they converge to x^* with one-round factor at most λ in the gauge g_u .

Proof. For $\xi \in \ker C$ and small $s > 0$, the points $x^* \pm s\xi$ are feasible, so minimality gives $\nabla f(x^*)^T \xi = 0$; the first-order term vanishes on N_u . Taking $\xi = 0$ in the expansion and using validity, $\phi_B^0(x^*) \leq f^*$, gives $Q(u, 0) \leq 0$, so $S_\delta^C(u)$ contains 0. Choose $\bar{t}$ such that $x^* + \bar{t}D(u) \subseteq B_0$ and the remainder is at most $\delta \bar{t}^2/2$ on N_u for $t \leq \bar{t}$. (a) If $t = 0$, then $B = \{x^*\}$, and validity gives $T_U(B) = \{x^*\}$. Otherwise $t > 0$; put $B' = x^* + tD(u) \supseteq B$. Lemma 3.1 gives $T_U(B) \subseteq T_U(B')$. A point of $K_U(B')$ has the form $x^* + t\xi$ with $\xi \in D(u)$ and $C\xi = 0$, and $f^* + t^2Q(u, \xi) - \delta t^2/2 \leq \phi_{B'}(x^* + t\xi) \leq f^* + \epsilon$. Hence $Q(u, \xi) \leq \epsilon/t^2 + \delta/2 \leq \delta$, so every retained ξ lies in $S_\delta^C(u)$, and $T_U(B') \subseteq x^* + tD(\Phi_\delta^C(u)) \subseteq x^* + \lambda tD(u)$. Thus $g_u(T_U(B)) \leq \lambda t$. (b) Validity keeps x^* in every iterate, so the gauges are defined and, by nesting, nonincreasing. Apply (a) while $g_u(B_k) \geq \sqrt{2\epsilon/\delta}$; below that value nesting keeps $g_u(B_{k+1}) \leq g_u(B_k)$. Complete sequential rounds end inside the Jacobi image by Lemma 3.2(b). $\square$

The proposition is the constrained counterpart of Theorem 4.7, with the tangent support problems restricted to the nullspace of C . The first-order term of the expansion need not vanish on the whole space; minimality on the affine set removes it on the nullspace. In Proposition 8.16, $x^* = 0$, the expansion has zero remainder, and $Q(\mathbf{1}, (\xi_1, -\xi_1)) = 2\xi_1^2 + 4|\xi_1| - 2$ on the unit cube shape $\mathbf{1}$. Its zero sublevel set on the nullspace is $|\xi_1| \leq \sqrt{2} - 1$, so $\Phi_0^C(\mathbf{1}) = (\sqrt{2} - 1)\mathbf{1}$ is an eigenvector shape, and the factor $\sqrt{2} - 1$ of Proposition 8.16(a) is attained exactly. The proposition is conditional on a restricted expansion that is uniform on the nullspace and on a strict contraction of the restricted map. Nonlinear equality constraints cannot simply be replaced by their first-order tangent spaces: the curvature of the constraint and the remainder on finite boxes must be controlled. Section 8.7 obtains such control from a feasible repair.

4.4 Quadratic objectives with McCormick relaxations

Let

$$f(x) = f^* + \frac{1}{2}(x - x^*)^T H(x - x^*)$$

with H symmetric and $H_{ii} \geq 0$, and let the relaxation keep the diagonal terms exactly and replace each off-diagonal term $H_{ij}(x_i - x_i^*)(x_j - x_j^*)$, $i < j$, by H_{ij} times the McCormick underestimator of the product if $H_{ij} > 0$ and the overestimator if $H_{ij} < 0$. McCormick planes commute with translation: the translation $x = x^* + y$ adds the same affine function to every plane, so maxima and minima of planes commute with it. The relaxation is therefore the one a solver builds for f written in the original variables. The planes also scale quadratically, so the tangent model is exact:

$$\phi_{x^*+tD(d)}(x^* + t\xi) = f^* + t^2Q(d, \xi), \quad Q(d, \xi) = \sum_i \frac{H_{ii}}{2} \xi_i^2 + \sum_{i < j} H_{ij} m_{ij}^d(\xi), \quad (9)$$

where m_{ij}^d is the McCormick estimator of $\xi_i \xi_j$ over $D(d)$ of the sign just described. On the unit cube $D(\mathbf{1}) = [-1, 1]^n$, with $s_{ij} = \text{sign } H_{ij}$,

$$Q(\mathbf{1}, \xi) = \sum_i \frac{H_{ii}}{2} \xi_i^2 + \sum_{i < j} |H_{ij}| (|\xi_i + s_{ij} \xi_j| - 1). \quad (10)$$

A centered box with half-widths h_i reduces to the unit cube by the substitution $\xi_i = h_i \eta_i$, which replaces H by $H' = \text{diag}(h)H \text{diag}(h)$, because the McCormick estimator of $\xi_i \xi_j$ on the box is

$h_i h_j$ times the estimator of $\eta_i \eta_j$ on the cube. If $H_{ii} < 0$, a convex relaxation uses the chord $\frac{H_{ii}}{2}((d_i^+ - d_i^-)\xi_i + d_i^- d_i^+)$ of the concave term; the model is still exact but different, and we do not consider this case. If $X \supseteq B_0$, an interior global minimizer forces H to be positive semidefinite; an indefinite H requires constraints that exclude its directions of negative curvature from the local feasible set, for example linear equations kept exactly as in Remark 4.13.

Proposition 4.15 (Two variables). *Let $f(x, y) = x^2 + y^2 + axy$ with $0 < |a| < 2$, and let $U = f^* = 0$. On every centered cube, a Jacobi round multiplies the half-width by*

$$\rho(a) = \frac{\sqrt{2a^2 + 4|a|} - |a|}{2}.$$

The function ρ increases with $|a|$ from 0 to 1; for example $\rho(0.5) = 0.5406$, $\rho(1) = 0.7247$, $\rho(1.5) = 0.8702$, and $\rho(1.9) = 0.9748$. Every box containing 0 in its interior has Jacobi iterates with $w(B_k) = \Theta(\rho(a)^k)$, and $r^* = \rho(a)$. For $f = \alpha x^2 + \beta y^2 + cxy$ with $\alpha, \beta > 0$ and $|c| < 2\sqrt{\alpha\beta}$, the same holds with $a = c/\sqrt{\alpha\beta}$, and the boxes with $\sqrt{\alpha} h_x = \sqrt{\beta} h_y$ play the role of the cubes.

Proof. Let $a > 0$. On $[-1, 1]^2$ the underestimator of xy is $|x + y| - 1$, so $Q(1, \xi) = \xi_1^2 + \xi_2^2 + a(|\xi_1 + \xi_2| - 1)$. Fix $\xi_1 = t \geq 0$ and write $s = \xi_1 + \xi_2$. Minimizing $t^2 + (s - t)^2 + a|s| - a$ over s gives the minimizer $s = 0$ for $t \leq a/2$ and $s = t - a/2$ for $t \geq a/2$, with values

$$m_a(t) = 2t^2 - a \quad (0 \leq t \leq a/2), \quad m_a(t) = t^2 + at - \frac{a^2}{4} - a \quad (t \geq a/2).$$

The minimizing ξ_2 is $-t$ or $-a/2$, which lies in $[-1, 1]$. The function m_a is continuous and increasing for $t > 0$, so the largest t with $m_a(t) \leq 0$ is the positive root of the second expression, $\rho(a)$; the second expression applies because $\rho(a) > a/2$, which is equivalent to $a < 2$, and $\rho(a) \leq 1$ because $a^2 \leq 4$. The symmetries $(\xi_1, \xi_2) \mapsto (\xi_2, \xi_1)$ and $\xi \mapsto -\xi$ preserve the cube and $Q(1, \cdot)$, so $\Phi(1) = \rho(a)1$. The derivative of ρ is positive because $(2a + 2)^2 > 2a^2 + 4a$. For $a < 0$, the reflection $\xi_2 \mapsto -\xi_2$ preserves the cube and turns the overestimator term $|a|(|\xi_1 - \xi_2| - 1)$ into $|a|(|\xi_1 + \xi_2| - 1)$.

The relaxation is defined and monotone on all boxes of $\mathbb{R}^2$, so the comparison cubes may extend beyond B_0 . A box containing 0 in its interior lies between two centered cubes, and Proposition 4.9 gives the rate and r^* . For the last claim, the substitution $z_1 = \sqrt{\alpha}x$, $z_2 = \sqrt{\beta}y$ maps the objective to $z_1^2 + z_2^2 + (c/\sqrt{\alpha\beta})z_1 z_2$, and McCormick estimators commute with positive scaling of the variables. $\square$

The proposition proves $w(B_k) = \Theta(\rho^k)$ from every interior box; it does not prove that the successive width ratios converge from arbitrary boxes. The tangent map on other shapes is also explicit.

Proposition 4.16 (Centered rectangles). *Let $f = x^2 + y^2 + axy$ with $0 < a < 2$ and $U = 0$. For $h, k > 0$ and $R = k/h$, a Jacobi round maps $[-h, h] \times [-k, k]$ to $[-hr(R), hr(R)] \times [-kr(1/R), kr(1/R)]$, where*

$$r(R) = \sqrt{\frac{aR}{1 + R^2}} \quad \text{if } 4R^3 \leq a(1 + R^2), \quad r(R) = \frac{\sqrt{a^2(1 + R^2) + 4aR} - aR}{2} \quad \text{otherwise.}$$

In particular $r(1) = \rho(a)$.

Proof. On the rectangle the underestimator of xy is $|kx + hy| - hk$. With $x = h\tau$, $y = kz$, and $\tau \geq 0$, the relaxation is $h^2\tau^2 + k^2z^2 + ahk(|\tau + z| - 1)$. Minimizing over z gives $z = -\min\{\tau, a/(2R)\} \in [-1, 0]$ and, after division by h^2 , the value $(1 + R^2)\tau^2 - aR$ for $\tau \leq a/(2R)$ and $\tau^2 + aR\tau - a^2/4 - aR$ for $\tau \geq a/(2R)$. The positive zero of the first expression lies in its branch exactly when $4R^3 \leq a(1 + R^2)$;

otherwise the zero of the second expression applies. The zero is below one: at $\tau = 1$ the point lies on a face of the rectangle, where the McCormick estimator equals the product, so the relaxation equals $f > 0$. Central symmetry gives the negative extent, and interchanging the coordinates gives the second factor. $\square$

For $a = 1$ and the rectangle $[-1, 1] \times [-1/4, 1/4]$, the first round multiplies the larger width by $r(1/4) = \sqrt{4/17} \approx 0.485$ and the smaller one by $r(4) = (\sqrt{33} - 4)/2 \approx 0.872$. The first observed ratio of maximal widths is thus far below the asymptotic rate 0.7247. Two further exact facts explain the behavior from asymmetric boxes and at a positive cutoff slack.

Proposition 4.17 (Eigenvectors are not unique). *Let $0 < a < 2$, $\rho = \rho(a)$, and $r = 1 - \rho + a/2$. For $b, c > 0$, the box $[-c, b]^2$ satisfies $T_0([-c, b]^2) = [-\rho c, \rho b]^2$ if and only if $c \geq rb$ and $b \geq rc$.*

Proof. On $[-c, b]^2$ the underestimator of xy is the maximum of the planes $-c(x + y) - c^2$ and $b(x + y) - b^2$, which agree on the line $x + y = b - c$. Fix $x = t \geq 0$ and minimize over y the function $q_2(y) = t^2 + y^2 + a(b(t + y) - b^2)$, obtained by using the second plane alone. Its unique minimizer is $y = -ab/2$, with value $t^2 + abt - a^2b^2/4 - ab^2$, whose positive root is $t = \rho b$. The full objective q satisfies $q \geq q_2$, with equality exactly where $x + y \geq b - c$. Hence the largest feasible t equals ρb if the point $(\rho b, -ab/2)$ lies in that region and in the box. Otherwise it is strictly smaller: at $t = \rho b$, $q_2(y) > 0$ except at $y = -ab/2$, and at that point either the point lies outside the box or the first plane is strictly larger, so that $q > q_2 = 0$; thus $q > 0$ on the whole cross-section $\{x = \rho b\}$ of the box, and the feasible values of t form a compact interval containing 0. The region condition reads $(\rho - a/2)b \geq b - c$, that is, $c \geq rb$; it implies $ab/2 < c$ because $r - a/2 = 1 - \rho > 0$, so the point lies in the box. The reflection $\xi \mapsto -\xi$ maps $[-c, b]^2$ to $[-b, c]^2$ and shows that the lower extent equals ρc exactly when $b \geq rc$. The second coordinate follows by interchanging the coordinates. $\square$

For $b = 1 - c$, the condition reads $r/(1 + r) \leq c \leq 1/(1 + r)$; for $a = 1$, $r = (4 - \sqrt{6})/2$ and the interval is approximately $[0.4367, 0.5633]$. The tangent map thus has a continuum of positive eigenvectors with the same eigenvalue. Iterates from an asymmetric box can approach a translated square rather than the centered one, while still contracting at the rate $\rho(a)$.

Proposition 4.18 (Exact limit at a positive cutoff slack). *Let $f = x^2 + y^2 + axy$ with $0 < a < 2$, $\epsilon > 0$, and $h_\epsilon = \sqrt{\epsilon/(1 - a^2/4)}$. Write $C_h = [-h, h]^2$.*

- (a) $T_\epsilon(C_h) = C_{g(h)}$ with $g(h) = \min\{h, \frac{1}{2}(\sqrt{(2a^2 + 4a)h^2 + 4\epsilon} - ah)\}$.
- (b) C_h is a fixed box if and only if $h \leq h_\epsilon$. The four points $h_\epsilon(\pm 1, \mp a/2)$ and $h_\epsilon(\mp a/2, \pm 1)$ lie in $K_\epsilon(C_{h_\epsilon})$ and attain all its faces.
- (c) Every bounded start box $B_0 \supseteq C_{h_\epsilon}$ has $B_\infty = C_{h_\epsilon}$. Hence $w(B_\infty) = 2\sqrt{\epsilon/(1 - a^2/4)}$ and $f^* - L(B_\infty) = a\epsilon/(1 - a^2/4)$.
- (d) From a centered cube with $h_0 > h_\epsilon$, the half-widths decrease to h_ϵ , and $(h_{k+1} - h_\epsilon)/(h_k - h_\epsilon) \rightarrow a/2$, while $h_{k+1}/h_k \rightarrow 1$.

Proof. (a) Scale by h . With cutoff value $c = \epsilon/h^2 \geq 0$, the largest feasible $t = x/h$ is the root of $m_a(t) = c$ from the proof of Proposition 4.15. It lies in the second branch, because the first branch is at most $a^2/2 - a < 0 \leq c$, which gives the stated formula after multiplying by h and truncating at h ; symmetry gives a cube. (b) The cube is fixed exactly when $m_a(1) = 1 - a^2/4 \leq \epsilon/h^2$. At $\xi = (1, -a/2)$ one has $Q(1, \xi) = 1 - a^2/4$, so the scaled points satisfy $\phi = h_\epsilon^2(1 - a^2/4) = \epsilon$. (c) Let $G(h)$ be the untruncated expression in (a). A direct computation gives $G(h_\epsilon) = h_\epsilon$ and $G'(h_\epsilon) = a/2$;

G' increases with h and tends to $\rho(a) < 1$. Hence $h_\epsilon < G(h) < h$ and $G(h) - h_\epsilon \leq \rho(a)(h - h_\epsilon)$ for $h > h_\epsilon$, so the half-widths from any centered cube C_H with $H > h_\epsilon$ decrease to h_ϵ . A bounded box B_0 lies in some C_H , so $B_\infty \subseteq C_{h_\epsilon}$; since C_{h_ϵ} is a fixed box in B_0 , Lemma 3.5 gives the reverse inclusion. On C_h the relaxation is $x^2 + y^2 + ah|x + y| - ah^2$, whose minimum is $-ah^2$, at the origin. (d) follows from $G'(h_\epsilon) = a/2$ and $h_k \rightarrow h_\epsilon > 0$. $\square$

Part (d) shows that near a positive floor the ratio of successive widths tends to one even though the distance to the limit contracts by $a/2$ per round, faster than the rate $\rho(a)$ at zero slack. A ratio near one is therefore not evidence that the zero-slack rate is near one.

For more variables, the face test of Corollary 4.11 reduces to simple row conditions on the unit cube. For each k let

$$S_k = \sum_{i < j, i \neq k \neq j} |H_{ij}|, \quad c_{kj} = \min_{0 \leq s \leq 1} \left(\frac{H_{jj}}{2} s^2 - |H_{kj}|s \right),$$

the total off-diagonal weight of the pairs that do not involve k , and the best gain from adjusting coordinate j against the pair (k, j) ; explicitly $c_{kj} = -H_{kj}^2/(2H_{jj})$ if $0 < H_{jj}$ and $|H_{kj}| \leq H_{jj}$, and $c_{kj} = H_{jj}/2 - |H_{kj}|$ otherwise.

Proposition 4.19 (Row conditions). *In the setting of (9):*

- (a) *If $H_{kk}/2 \leq S_k$ for every k , then $T_U(B) = B$ for every centered cube $B = x^* + t[-1, 1]^n$ and every $U \geq f^*$.*
- (b) *If $H_{kk}/2 + \sum_{j \neq k} c_{kj} > S_k$ for every k , then $\Phi(1) \leq \lambda 1$ for some $\lambda < 1$; hence $r^* < 1$ and the conclusions of Theorem 4.7 hold with $u = 1$.*

For a centered box with half-widths h_i , the same statements hold with H replaced by $D_h H D_h$, where $D_h = \text{diag}(h)$: statement (a) for $h_i \geq 0$, and statement (b) for $h_i > 0$, after removing the coordinates with zero half-width, whose bounds are already fixed.

Proof. (a) At $\xi = \pm e_k$, every pair containing k contributes $|H_{kj}|(|\pm 1| - 1) = 0$ to (10), and every other pair contributes $-|H_{ij}|$, so $Q(1, \pm e_k) = H_{kk}/2 - S_k \leq 0$. Since the model is exact, $\phi(x^* \pm t e_k) = f^* + t^2 Q(1, \pm e_k) \leq f^* \leq U$. These $2n$ points attain every face of the cube. (b) On the face $\xi_k = \pm 1$, a pair (k, j) contributes at least $-|H_{kj}||\xi_j|$, because $|\pm 1 + s \xi_j| \geq 1 - |\xi_j|$, and every other pair contributes at least $-|H_{ij}|$. Hence

$$Q(1, \xi) \geq \frac{H_{kk}}{2} + \sum_{j \neq k} \left(\frac{H_{jj}}{2} \xi_j^2 - |H_{kj}||\xi_j| \right) - S_k \geq \frac{H_{kk}}{2} + \sum_{j \neq k} c_{kj} - S_k > 0,$$

so every face value is positive and Corollary 4.11(a) applies. Scaling was explained after (10); a zero half-width multiplies the corresponding row and column by zero. $\square$

Condition (a) is the row condition $H_{kk}/2 + \sum_{j \neq k} |H_{kj}| \leq \sum_{i < j} |H_{ij}|$. It says that the McCormick gaps of the pairs not involving k , each of which is maximal at the center of the cube, outweigh the curvature in direction k . The two conditions differ by the term $\sum_j c_{kj}$, which accounts for moving the other coordinates. Neither is necessary; for the following family the first is.

Example 4.20 (A family with an exact stall threshold). Let $f(x) = \sum_i x_i^2 + a \sum_{i < j} x_i x_j$ with $0 < a < 2$ and $n \geq 3$. The eigenvalues of H are $2 - a$ and $2 + a(n - 1)$, so f is strongly convex. On the unit cube,

$$\Phi(1) = \min\{1, t_n(a)\} 1, \quad t_n(a) = \frac{\sqrt{a^2(n-1)^2 + 2an(n-1)} - a(n-1)}{2}.$$

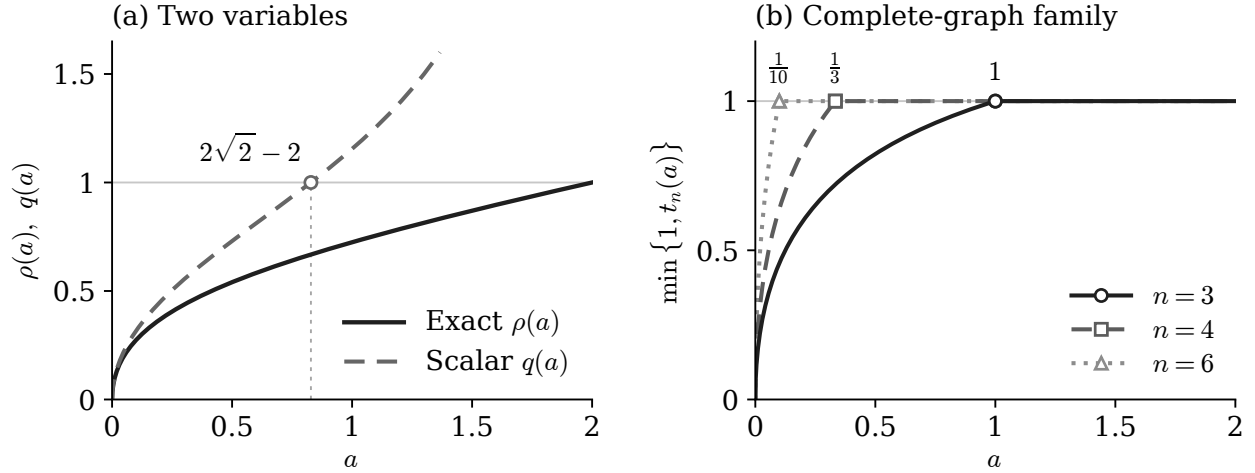


Figure 1: Analytic Jacobi factors at $U = f^* = 0$ for termwise McCormick relaxations with exact square terms. (a) For $f = x^2 + y^2 + axy$, the exact cube factor $\rho(a)$ is below one for $0 < a < 2$ (Proposition 4.15). The scalar quadratic-growth bound with $\mu = 1 - a^2/4$ and $\tau = a/4$ gives $q(a) = \sqrt{a/(1 - a^2/4)}$; its sufficient condition $q < 1$ holds only for $a < 2\sqrt{2} - 2$ (Proposition 4.2). The scalar curve is shown through $q = 1.6$. (b) For $f = \sum_i x_i^2 + a \sum_{i < j} x_i x_j$, $n \geq 3$, the exact cube factor $\min\{1, t_n(a)\}$ reaches one at $a = 2/[(n-1)(n-2)]$ (Example 4.20). Markers identify the thresholds 1, $1/3$, and $1/10$ for $n = 3, 4, 6$. All objectives are strongly convex for $0 < a < 2$; the plateaus denote fixed cubes at every scale. Endpoints show continuous limits.

To see this, fix $\xi_1 = t \geq 0$. The function $Q(1, \cdot)$ is convex and invariant under permutations of $\xi_2, \dots, \xi_n$, so its minimum over these coordinates is attained at a point with $\xi_2 = \dots = \xi_n = s$, where

$$Q = t^2 + (n-1)s^2 + a(n-1)(|t+s| - 1) + \frac{a}{2}(n-1)(n-2)(2|s| - 1).$$

For $s < 0$ its derivative is at most $(n-1)(2s - a(n-3)) < 0$, and for $s > 0$ it is positive, so the minimum is at $s = 0$ and equals $t^2 + a(n-1)t - an(n-1)/2$, whose positive root is $t_n(a)$. Symmetry gives all $2n$ components. Moreover $t_n(a) \geq 1$ exactly when $a(n-1)(n-2) \geq 2$, which is condition (a) of Proposition 4.19. Hence either the cube stalls at every scale, or it is an eigenvector with eigenvalue $t_n(a) < 1$ and, by Proposition 4.9, Jacobi rounds at zero cutoff slack from every box containing a neighborhood of 0 contract at the exact rate $t_n(a)$. For example, $t_3(1/2) = (\sqrt{7} - 1)/2 \approx 0.8229$. For $a = 1/10$ the stall occurs exactly for $n \geq 6$. At $a = 1/10$ and $n = 6$, H is positive definite and strictly diagonally dominant ($\sum_{j \neq i} |H_{ij}| = 1/2 < 2 = H_{ii}$), yet no round tightens any bound. The relaxed objective at the center is $-an(n-1)/2$ on the unit cube: the bilinear gaps accumulate. The mechanism belongs to the termwise relaxation; a solver that recognizes the convex quadratic and relaxes it as a whole does not stall.

Example 4.21 (The sign pattern matters). Let $f = x_1^2 + x_2^2 + x_3^2 + \frac{3}{4}(x_1x_2 + x_1x_3 - x_2x_3)$ on $[-1, 1]^3$. Its Hessian $2I + \frac{3}{4}A$ with $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & -1 \\ 1 & -1 & 0 \end{bmatrix}$ has eigenvalues $1/2, 11/4, 11/4$, because $(-1, 1, 1)$, $(0, 1, -1)$, and $(2, 1, 1)$ are eigenvectors of A with eigenvalues $-2, 1, 1$. The row condition fails in every row ($5/2 > 9/4$). By (10),

$$Q(1, \xi) = \|\xi\|_2^2 + \frac{3}{4}(|\xi_1 + \xi_2| + |\xi_1 + \xi_3| + |\xi_2 - \xi_3| - 3),$$

and at $z = (1, -3/8, -3/8)$ the value is $41/32 - 18/32 - 24/32 = -1/32$. The signed permutations $(\xi_1, \xi_2, \xi_3) \mapsto (\xi_2, \xi_1, -\xi_3)$ and $(\xi_1, \xi_2, \xi_3) \mapsto (\xi_3, -\xi_2, \xi_1)$ preserve the cube and permute the three absolute values, so they preserve $Q(1, \cdot)$; they map z to points on the positive second and third faces, and $Q(1, -\xi) = Q(1, \xi)$ gives the negative faces. Theorem 4.10 applies with a strict margin. For this objective, condition (b) of Proposition 4.19 gives the lower bound $-1/32$ on every face value, which z attains.

Example 4.22 (The shape changes the outcome). Let $f = \sum_{i=1}^3 x_i^2 + \sum_{i < j} x_i x_j$, the case $a = 1$, $n = 3$ of Example 4.20, for which the cube stalls with equality in the row condition. On the thinner box $[-1/4, 1/4] \times [-1, 1]^2$, use the scaled matrix $H' = \text{diag}(1/4, 1, 1) H \text{diag}(1/4, 1, 1)$. For $k = 1$, the witnesses $\pm e_1$ of part (a) of Proposition 4.19 give face values at most $H'_{11}/2 - S_1 = 1/16 - 1 < 0$. For $k = 2, 3$, the estimate in the proof of part (b), which holds coordinate by coordinate, gives face values at least $1 - 3/16 - 1/4 - 1/4 = 5/16 > 0$. Hence the first coordinate keeps its bounds and the other two contract. A round changes the shape, and the face test must then be applied to the new shape.

Example 4.23 (Finite tangent relaxations of squares). Let $f = \sum_i x_i^2$ and relax each square x_i^2 by a lifted variable y_i subject to the McCormick inequalities for the product $x_i \cdot x_i$, which are the endpoint tangents $y_i \geq 2\ell_i x_i - \ell_i^2$ and $y_i \geq 2u_i x_i - u_i^2$ and the secant $y_i \leq (\ell_i + u_i)x_i - \ell_i u_i$; the objective is $\sum_i y_i$. On a centered box with half-widths $h_i > 0$, $\phi_B(x) = \sum_i (2h_i |x_i| - h_i^2)$, and a Jacobi round at $U = 0$ gives the exact map

$$h'_i = \min\left\{h_i, \frac{1}{2h_i} \sum_j h_j^2\right\} :$$

the coordinates $j \neq i$ can be zero with $y_j = -h_j^2$, and the endpoint is attained with these values. For $n = 1$ the half-width halves each round. For $n \geq 2$ every cube is fixed, although the objective is strictly convex with a unique minimizer; the tangent model is $Q(1, \xi) = \sum_i (2|\xi_i| - 1)$, whose face values $2 - n$ are nonpositive. Adding the valid rows $y_i \geq 0$ changes the map to $h'_i = h_i/2$, and the exact convex squares give $\{0\}$ after one round. The rate is a property of the relaxation, not of the problem. Section 5 uses this finite family as its exact reference family.

4.5 Factorable functions and composite McCormick relaxations

For general smooth objectives the tangent model is computed by a recursion over a factorable representation. A *factorable representation* is a sequence $v_1, \dots, v_m$ of functions of x with $v_i = x_i$ for $i \leq n$ and, for $k > n$, one of: a constant; $v_a + v_b$; $c v_a$ with $c \in \mathbb{R}$; $v_a v_b$ (with $a = b$ allowed); or $h_k(v_a)$ for a function h_k of one variable, where $a, b < k$. The objective is $f = v_m$. On a box B , natural interval arithmetic gives intervals $[v_k^L(B), v_k^U(B)]$: endpoint operations for sums and scalings, the minimum and maximum of the four endpoint products for products, and the exact range of h_k over the interval of its argument for univariate factors. The composite McCormick relaxation [27, 28] computes convex and concave relaxations $cv_k \leq v_k \leq cc_k$ on B factor by factor: sums and scalings exactly (a negative scaling exchanges the convex and concave sides); products by the rule

$$\begin{aligned} cv_k &= \max\{\alpha_1 + \alpha_2 - v_a^L v_b^L, \beta_1 + \beta_2 - v_a^U v_b^U\}, \\ \alpha_1 &= \min\{v_b^L cv_a, v_b^L cc_a\}, \quad \alpha_2 = \min\{v_a^L cv_b, v_a^L cc_b\}, \\ \beta_1 &= \min\{v_b^U cv_a, v_b^U cc_a\}, \quad \beta_2 = \min\{v_a^U cv_b, v_a^U cc_b\}, \end{aligned}$$

and, with $\gamma_1 = \max\{v_b^L cv_a, v_b^L cc_a\}$, $\gamma_2 = \max\{v_a^U cv_b, v_a^U cc_b\}$, $\gamma_3 = \max\{v_b^U cv_a, v_b^U cc_a\}$, and $\gamma_4 = \max\{v_a^L cv_b, v_a^L cc_b\}$,

$$cc_k = \min\{\gamma_1 + \gamma_2 - v_a^U v_b^L, \gamma_3 + \gamma_4 - v_a^L v_b^U\};$$

univariate factors are relaxed by $cv_k = h_Z^{cv}(\text{mid}\{cv_a, cc_a, z^{\min}\})$, where h_Z^{cv} is the convex envelope of h_k on $Z = [v_a^L, v_a^U]$, $z^{\min}$ is a minimizer of h_Z^{cv} on Z , and mid is the median of three numbers; the concave side uses the concave envelope and a maximizer. The relaxed objective is $\phi_B = cv_m$. The *clipped* variant replaces cv_k by $\max\{cv_k, v_k^L\}$ and cc_k by $\min\{cc_k, v_k^U\}$ before they are used by later factors. Scott, Stuber, and Barton prove that the clipped construction is monotone in the sense of (4) when the natural interval of every univariate argument lies in the domain of the corresponding scalar function and these functions are Lipschitz there, and they note that the unclipped rule can violate monotonicity [41]. On small boxes around a point at which the scalar functions are twice continuously differentiable, these conditions hold. Validity and convexity of both variants are standard.

The tangent model is built from first-order data. Write $v_k^* = v_k(x^*)$ and $\pi_k(\xi) = \nabla v_k(x^*)^T \xi$, and, for $t \in \mathbb{R}$, $t^+ = \max\{t, 0\}$ and $t^- = \max\{-t, 0\}$. For a shape d define intervals $[\underline{\pi}_k(d), \bar{\pi}_k(d)]$ and nonnegative functions $E_k^{cv}(d, \xi)$, $E_k^{cc}(d, \xi)$, $\xi \in D(d)$, by the following recursion.

- Variable x_i : interval $[-d_i^-, d_i^+]$, $E^{cv} = E^{cc} = 0$. Constant: interval $[0, 0]$, $E^{cv} = E^{cc} = 0$.
- Sum: intervals and both functions add.
- Scaling $c v_a$: for $c \geq 0$, interval $c[\underline{\pi}_a, \bar{\pi}_a]$, $E^{cv} = cE_a^{cv}$, $E^{cc} = cE_a^{cc}$; for $c < 0$, interval $[c\bar{\pi}_a, c\underline{\pi}_a]$, $E^{cv} = |c|E_a^{cc}$, $E^{cc} = |c|E_a^{cv}$.
- Product $v_a v_b$: interval $v_b^*[\underline{\pi}_a, \bar{\pi}_a] + v_a^*[\underline{\pi}_b, \bar{\pi}_b]$ in interval arithmetic, and

$$\begin{aligned} E^{cv} &= (v_b^*)^+ E_a^{cv} + (v_b^*)^- E_a^{cc} + (v_a^*)^+ E_b^{cv} + (v_a^*)^- E_b^{cc} \\ &\quad + \min\{(\pi_a - \underline{\pi}_a)(\pi_b - \underline{\pi}_b), (\bar{\pi}_a - \pi_a)(\bar{\pi}_b - \pi_b)\}, \\ E^{cc} &= (v_b^*)^+ E_a^{cc} + (v_b^*)^- E_a^{cv} + (v_a^*)^+ E_b^{cc} + (v_a^*)^- E_b^{cv} \\ &\quad + \min\{(\bar{\pi}_a - \pi_a)(\pi_b - \underline{\pi}_b), (\pi_a - \underline{\pi}_a)(\bar{\pi}_b - \pi_b)\}. \end{aligned}$$

- Univariate $h_k(v_a)$, with $h_1 = h'_k(v_a^*)$, $h_2 = h''_k(v_a^*)$, and $\sigma_a = (\pi_a - \underline{\pi}_a)(\bar{\pi}_a - \pi_a)$: interval $h_1[\underline{\pi}_a, \bar{\pi}_a]$, and

$$E^{cv} = \frac{1}{2} h_2^- \sigma_a + h_1^+ E_a^{cv} + h_1^- E_a^{cc}, \quad E^{cc} = \frac{1}{2} h_2^+ \sigma_a + h_1^+ E_a^{cc} + h_1^- E_a^{cv}.$$

Here π_a stands for $\pi_a(\xi)$ and the interval endpoints for their values at d . The interval $[\underline{\pi}_k(d), \bar{\pi}_k(d)]$ is natural interval arithmetic applied to the linearized factor sequence over $D(d)$, so it contains $\pi_k(\xi)$ for $\xi \in D(d)$; consequently every factor of the form $\pi - \underline{\pi}$ or $\bar{\pi} - \pi$ is nonnegative and $E_k^{cv}, E_k^{cc} \geq 0$. By induction, the interval endpoints are continuous and positively homogeneous of degree one in d , and the functions E_k are continuous and positively homogeneous of degree two in (d, ξ) .

Theorem 4.24 (Second-order expansion of composite McCormick relaxations). *Let $x^* \in \mathbb{R}^n$, and suppose that for every univariate factor $v_k = h_k(v_a)$, the function h_k is twice continuously differentiable on an open interval containing $v_a(x^*)$. Let $\mathcal{K}$ be a compact set of shapes. There is $t_{\mathcal{K}} > 0$ such that for $0 < t \leq t_{\mathcal{K}}$ all intervals and relaxations on $B_t = x^* + tD(d)$, $d \in \mathcal{K}$, are defined, and, uniformly over $d \in \mathcal{K}$ and $\xi \in D(d)$, with $x_t = x^* + t\xi$, for every factor k ,*

$$v_k^L(B_t) = v_k^* + t\underline{\pi}_k(d) + O(t^2), \quad v_k^U(B_t) = v_k^* + t\bar{\pi}_k(d) + O(t^2), \quad (11)$$

$$cv_k(x_t) = v_k(x_t) - t^2 E_k^{cv}(d, \xi) + o(t^2), \quad cc_k(x_t) = v_k(x_t) + t^2 E_k^{cc}(d, \xi) + o(t^2), \quad (12)$$

$$v_k^L(B_t) - cv_k(x_t) \leq o(t^2), \quad cc_k(x_t) - v_k^U(B_t) \leq o(t^2), \quad (13)$$

where the relaxations are those built on B_t . If each h_k'' is Lipschitz near $v_a(x^*)$, every $o(t^2)$ can be replaced by $O(t^3)$. The clipped relaxations have the same expansions. Consequently,

$$\phi_{B_t}(x_t) = f(x^*) + t\nabla f(x^*)^T \xi + t^2 Q(d, \xi) + o(t^2), \quad Q(d, \xi) = \frac{1}{2} \xi^T \nabla^2 f(x^*) \xi - E_m^{\text{cv}}(d, \xi),$$

uniformly on $\mathcal{K}$, and Q is continuous. At a minimizer with $\nabla f(x^*) = 0$ at which the relaxations are monotone, for example in the clipped construction, Assumption 4.3 holds.

The proof is in Appendix A.1. Its main step concerns univariate factors: the median rule can clip the argument of the envelope at an endpoint of its interval, and (13) shows that this clipping does not change the second-order term. The expansion covers zero factor values, critical points $h'_k(v_a^*) = 0$, inflection points $h''_k(v_a^*) = 0$, first-order intervals of zero width, and shapes with zero components. The box B_0 plays no role, so the theorem also applies at minimizers on the boundary of B_0 , with shapes restricted to the admissible ones; this is used in Section 4.6. The smoothness hypothesis concerns each scalar factor at its own argument, and smoothness of f does not replace it. For example, $f(x) = |x| - |x| + x^2 = x^2$, with the square kept exact, has on $[-t, t]$ the composite relaxation $x^2 + |x| - t$, with or without clipping: the concave envelope of $|x|$ is the constant t , and the subtraction uses it. The gap at $x^* = 0$ is t , of first order. The direct relaxation of $|x|$ itself is exact, so nonsmoothness alone does not force such an error; the theorem simply does not apply at a kink. At a nonzero argument, $|\cdot|$ is linear nearby and the theorem applies.

Two consequences are worth stating. First, for quadratic objectives with the squares represented as univariate factors and the products of distinct variables as products, E_m^{cv} is the McCormick gap and Theorem 4.24 reproduces (9). Representing x_i^2 instead as the product $x_i \cdot x_i$ gives, at $x_i^* = 0$, the gap $\min\{(\xi_i + d_i^-)^2, (d_i^+ - \xi_i)^2\}$ and the model $2|\xi_i| - 1$ on the unit interval, which is the finite tangent relaxation of Example 4.23. The tangent model therefore depends on the factorable representation as well as on the rules. Second, since E_m^{cv} is bounded on compact sets, the theorem gives $\phi_B \geq f - Cw(B)^2$ on small boxes containing x^* . This recovers, near x^* , the second-order pointwise convergence of McCormick relaxations [5]; the theorem adds the limit of the scaled gap as a function of the box shape and of the position in the box, which is what the tangent map needs.

Solvers usually relax lifted reformulations with an auxiliary variable per factor. For quadratic problems whose auxiliary variables occur only in the objective, minimizing over the auxiliaries decouples the terms. With exact convex squares and McCormick envelopes for the products of distinct variables, this yields the termwise relaxation, so (9) applies exactly; with finite tangent relaxations of the squares it yields the model of Example 4.23. When auxiliary variables also occur in constraints, or when cuts are added, Theorem 4.24 does not apply, and Assumption 4.3 is a hypothesis to be verified for the construction at hand.

4.6 Minimizers on the boundary of the box

At a minimizer on the boundary of B_0 with nonzero derivatives across the active faces, the first-order term dominates in the active coordinates. Let x^* be a global minimizer with $x_i^* = \ell_i^0$ for i in an active set A and $\ell_i^0 < x_i^* < u_i^0$ for i in the free set F ; active upper bounds reduce to this case by reflecting the coordinate. We assume that A and F are both nonempty; the case $F = \emptyset$ is discussed after Theorem 4.27. Let $g \in \mathbb{R}^n$ satisfy $g_i > 0$ for $i \in A$ and $g_i = 0$ for $i \in F$. For a differentiable objective with $g = \nabla f(x^*)$ and no constraints other than the bounds, these are the first-order optimality conditions with strict complementarity; with other constraints, g is the first-order coefficient of the projected relaxation in Assumption 4.25 below. Admissible shapes form the closed cone $\mathcal{S} = \{d \geq 0 : d_i^- = 0 \text{ for } i \in A\}$; every box B with $x^* \in B \subseteq B_0$ equals $x^* + tD(d)$ with $d \in \mathcal{S}$. Write $w_i(B) = u_i - \ell_i$ for the width of coordinate i .

Assumption 4.25 (Boundary tangent model). There is a continuous function $Q(d, \xi)$ on $\{(d, \xi) : d \in \mathcal{S}, \xi \in D(d)\}$ such that for every compact $\mathcal{K} \subseteq \mathcal{S}$ the comparison boxes belong to $\mathfrak{B}$ for small t and

$$\varrho_{\mathcal{K}}(t) = \sup_{d \in \mathcal{K}, \xi \in D(d)} \frac{|\phi_{x^*+tD(d)}(x^* + t\xi) - f^* - tg^T\xi - t^2Q(d, \xi)|}{t^2} \longrightarrow 0.$$

The essential point is that $\mathcal{K}$ may contain shapes whose active components are zero or small. Theorem 4.24 supplies Assumption 4.25 for clipped composite McCormick relaxations with $g = \nabla f(x^*)$. For a free shape d_F let $\iota(d_F)$ be the shape with free part d_F and zero active components, define ι on points in the same way, and set

$$Q^F(d_F, \xi_F) = Q(\iota(d_F), \iota(\xi_F)), \quad G^F(d_F) = - \min_{\xi_F \in D_F(d_F)} Q^F(d_F, \xi_F) \geq 0.$$

The reduced tangent map Φ_c^F and its factor $r^*(\Phi^F)$ are defined from Q^F as in Section 4.2. For the boundary theorem fix a free shape $u_F \gg 0$ and a bound $M \geq 0$ on the initial active components, and let

$$G_0 = G^F(u_F), \quad G_M = - \min\{Q(d, \xi) : d \in \mathcal{S}, d_F = u_F, d_A^+ \leq M, \xi \in D(d)\}, \quad g_{\min} = \min_{i \in A} g_i.$$

Since $\iota(u_F)$ is admissible, $0 \leq G_0 \leq G_M$.

Lemma 4.26 (One round near a boundary minimizer). *Let Assumption 4.25 hold, let $\delta > 0$, and let $\mathcal{K}_F$ be a compact set of free shapes. There are $\bar{t}, \bar{\sigma} > 0$ such that for every box $B = x^* + tD(d)$ with $d \in \mathcal{S}$, $d_F \in \mathcal{K}_F$, $\max_{i \in A} d_i^+ \leq \bar{\sigma}$, and $t \leq \bar{t}$, and every $\epsilon \geq 0$:*

- (i) *if $\epsilon \leq \delta t^2/2$, the free part of $T_U(B)$ lies in $x_F^* + tD_F(\Phi_\delta^F(d_F))$;*
- (ii) *$x^* + t\iota(\xi_F) \in K_U(B)$ for every $\xi_F \in D_F(d_F)$ with $Q^F(d_F, \xi_F) \leq -\delta$;*
- (iii) *$w_i(T_U(B)) \leq (\epsilon + (G^F(d_F) + \delta/2)t^2)/g_i$ for $i \in A$;*
- (iv) *$w_i(T_U(B)) \geq \min\{td_i^+, (\epsilon + (G^F(d_F) - \delta/2)t^2)/g_i\}$ for $i \in A$.*

Parts (i) and (ii) say that in scaled coordinates the free part of one round lies between the hulls of the sublevel sets of Q^F at levels $-\delta$ and δ . Parts (iii) and (iv) show that the bound on the active widths is sharp to first order, since δ is arbitrary. The proof is in Appendix A.2.

Theorem 4.27 (Boundary minimizers). *Let Assumption 4.25 hold, and let $u_F \gg 0$, $\delta > 0$, and $\lambda \in (0, 1)$ satisfy $\Phi_\delta^F(u_F) \leq \lambda u_F$. Let $M \geq 0$. There is $t > 0$ with the following property. Let the start box satisfy $x^* \in B_0 \subseteq x^* + t_0 D(d^0)$ with an admissible shape $d^0 \in \mathcal{S}$, $d_F^0 = u_F$, $\max_{i \in A} (d^0)_i^+ \leq M$, and $0 < t_0 \leq \bar{t}$, and consider Jacobi iterates, or complete sequential rounds, from B_0 at cutoff $U = f^* + \epsilon$. Put $t_k = \lambda^{k-1}t_0$ for $k \geq 1$, and let $N = \min\{k \geq 1 : t_k^2 < 2\epsilon/\delta\}$, with $N = \infty$ if $\epsilon = 0$.*

- (a) *For $1 \leq k \leq N$, the free part of B_k lies in $x_F^* + t_k D_F(u_F)$.*
- (b) *$w_i(B_1) \leq (\epsilon + (G_M + \delta/4)t_0^2)/g_i$ for $i \in A$. If $N \geq 2$, then for $1 \leq k \leq N$ and $i \in A$,*

$$w_i(B_{k+1}) \leq (\epsilon + (G_0 + \eta_k)t_k^2)/g_i,$$

with $0 \leq \eta_k \leq \delta/2$, and $\eta_k \rightarrow 0$ as $t_k \rightarrow 0$.

- (c) *If $\epsilon = 0$, then $B_\infty = \{x^*\}$, the free widths are $O(\lambda^k)$ by (a), and $w_i(B_{k+1}) \leq (G_0/g_i + o(1))t_k^2$. If $\epsilon > 0$, the free part of B_∞ lies in $x_F^* + \sqrt{2\epsilon/\delta} D_F(u_F)$ and $w_i(B_\infty) \leq (2 + 2G_M/\delta)\epsilon/g_i$ for $i \in A$.*

Every $\lambda > r^*(\Phi^F)$ admits such u_F and δ .

The free coordinates thus obey the upper bounds of the interior theory, with the reduced map Φ^F and a floor of order at most $\sqrt{\epsilon}$, while the active widths are at most of order ϵ plus the square of the free scale t_k . The statement bounds the free parts and the active widths by the common geometric scale t_k ; it does not assert that the actual free gauge contracts in every round. It need not: for $f = x + y^2$ on $x \geq 0$ with the valid monotone family $\phi_B = f - h^2$, where h is the upper extent of B in x , a Jacobi round at zero cutoff maps the active extent h to $\min\{h, h^2\}$ and the free radius r to $\min\{r, h\}$, so from $[0, t_0] \times [-t_0^3, t_0^3]$ with $0 < t_0 < 1$ the free radius does not move in the first two rounds. The proof is in Appendix A.2. If $g_i = 0$ for some active i , that coordinate has no first-order term. It can be treated as a free coordinate restricted to shapes with $d_i^- = 0$: the same proof applies to the enlarged free set, provided the contraction hypothesis $\Phi_\delta^F(u_F) \leq \lambda u_F$ is verified on this one-sided cone of shapes. If all coordinates are active ($F = \emptyset$) with $g_i > 0$, Proposition 4.1 applies on small boxes: at zero cutoff slack the convergence is quadratic, and at a positive slack the widths satisfy the forced recurrence of that proposition and end at most $4\epsilon/\kappa$.

The refined estimate in (b) requires $N \geq 2$. If the initial scale already lies below the cutoff floor, the active components of B_1 need not be small relative to t_1 , and the refined constant can fail. For $f(x, y) = x - x^2 + y^2$ on $[0, 1/4] \times [-1/4, 1/4]$, keep y^2 exact and use the secant $-hx$ of $-x^2$ on $[0, h]$. On $B = [0, h] \times [-r, r]$ the relaxation is $(1 - h)x + y^2$; the expansion is exact with $g = (1, 0)$ and $Q(d, \xi) = -d_x^+ \xi_x + \xi_y^2$, so $G_0 = 0$, and $u_F = 1$, $\delta = 1/4$, $\lambda = 3/4$, $M = 1$ satisfy the hypotheses. Start from $[0, w_0] \times [-w_0, w_0]$ with $\epsilon = w_0/2$ and w_0 small. Then $N = 1$, the free bounds do not move in the first two rounds, and the active widths are $h_1 = \epsilon/(1 - w_0)$ and $h_2 = \epsilon/(1 - h_1)$. Hence $(h_2 - \epsilon)/w_0^2 = 1/(4 - 6w_0) > 1/4$, whereas the refined estimate would give at most $\delta/2 = 1/8$.

Example 4.28 (An exact boundary example). Let $f = x_1 + x_2^2 + x_3^2 + ax_2x_3$ with $0 < a < 2$ on $[0, 1] \times [-1, 1]^2$, with exact squares and the McCormick underestimator of x_2x_3 . The minimizer is 0, with $A = \{1\}$, $g_1 = 1$, and $G_0 = a$. On $B = [0, h] \times [-s, s]^2$ the relaxation is $x_1 + \psi_s(x_2, x_3)$ with $\psi_s = x_2^2 + x_3^2 + as|x_2 + x_3| - as^2$, whose minimum is $-as^2$. A Jacobi round at cutoff ϵ therefore maps (h, s) to

$$(\min\{h, \epsilon + as^2\}, g(s)),$$

with g as in Proposition 4.18: the largest free extent is attained with $x_1 = 0$, and the largest active extent with $(x_2, x_3) = 0$. For $\epsilon = 0$ the free half-width is $\rho(a)^k s_0$ and the active width is $a\rho(a)^{2k-2} s_0^2$ once it falls below h_0 . For $\epsilon > 0$ and $s_0 \geq h_\epsilon$, the free half-width decreases to h_ϵ and the active width to $\min\{h_0, \epsilon(1 + a/(1 - a^2/4))\}$: a free floor of order $\sqrt{\epsilon}$ and an active floor of order ϵ . For $a = 1$ the limit widths are $2h_\epsilon = (4/\sqrt{3})\sqrt{\epsilon} \approx 2.309\sqrt{\epsilon}$ and, for large h_0 , $7\epsilon/3$.

4.7 The relaxation gap after tightening, and stopping

The width of the limit box matters because it controls the relaxation gap that remains after tightening.

Corollary 4.29 (Gap after tightening). *Under the hypotheses of Theorem 4.7(b), let*

$$G = - \min_{\xi \in D(u)} Q(u, \xi) \geq 0.$$

If $\epsilon > 0$ and $\sqrt{2\epsilon/\delta} \leq \bar{t}$, then

$$f^* - L(B_\infty) \leq \left(\frac{2G}{\delta} + 1\right)\epsilon,$$

and $f^* - L(B_\infty) \leq (2G/\delta)\epsilon + o(\epsilon)$ as $\epsilon \downarrow 0$; for $\epsilon = 0$, $L(B_\infty) = f^*$. Under the hypotheses of Proposition 4.1, once an iterate has width at most $\kappa/(4\tau)$ (any iterate if $\tau = 0$), $f^* - L(B_\infty) \leq 16\tau\epsilon^2/\kappa^2$.

Proof. Let $t_\epsilon = \sqrt{2\epsilon/\delta}$ and $C = x^* + t_\epsilon D(u)$. By Theorem 4.7, $B_\infty \subseteq C$, so $L(B_\infty) \geq L(C)$. For $x = x^* + t_\epsilon \xi \in C$, $\phi_C(x) \geq f^* + t_\epsilon^2 Q(u, \xi) - \varrho_{\{u\}}(t_\epsilon) t_\epsilon^2 \geq f^* - G t_\epsilon^2 - \varrho_{\{u\}}(t_\epsilon) t_\epsilon^2$. Since $\varrho_{\{u\}}(t_\epsilon) \leq \delta/2$, this gives the first bound, and since $\varrho_{\{u\}}(t_\epsilon) \rightarrow 0$, the second. For $\epsilon = 0$, $B_\infty = \{x^*\}$ lies in every comparison box, so the same estimate with $t \downarrow 0$ gives $L(B_\infty) \geq f^*$, and validity at x^* gives equality. In the sharp case, (7) gives $L(B) \geq f^* - \tau w(B)^2$ for boxes containing x^* , and $w(B_\infty) \leq 4\epsilon/\kappa$. $\square$

Under the contraction hypotheses, the gap left after tightening to the limit is at most proportional to the incumbent's suboptimality ϵ ; for the model problem it equals $a\epsilon/(1 - a^2/4)$ (Proposition 4.18). Together with the $O(\sqrt{\epsilon})$ bound on the width, this has several consequences for deciding when to tighten.

Rounds to the cutoff floor. Under the hypotheses of Theorem 4.7, the gauge falls at least by the factor λ per round until it is below $\sqrt{2\epsilon/\delta}$. For $\epsilon > 0$, zero rounds suffice when the starting gauge satisfies $t_0 \leq \sqrt{2\epsilon/\delta}$. Otherwise at most $\lceil \log(t_0 \sqrt{\delta/(2\epsilon)}) / \log(1/\lambda) \rceil$ rounds bring the iterates inside $x^* + \sqrt{2\epsilon/\delta} D(u)$, which contains the limit; further rounds change the box only inside this region and change the gap by at most the $O(\epsilon)$ bound above. This bound on the number of rounds grows only logarithmically as ϵ decreases.

When to repeat after an incumbent improves. Lowering the cutoff slack from ϵ to ϵ' lowers the bounds on the limit width by the factor $\sqrt{\epsilon'/\epsilon}$ and the bounds on the remaining gap by the factor ϵ'/ϵ ; in the model problem the limit itself shrinks by exactly these factors. This is the local reason to repeat tightening after an incumbent improves, and the reason why a cutoff far above f^* limits the value of any number of rounds. In the stall regime of Theorem 4.10, by contrast, no improvement of the incumbent removes the fixed boxes around x^* .

What observed ratios show. The successive width ratios of an iteration are a natural signal for stopping, and they estimate the contraction rate in favorable cases. They are not a certificate. From boxes that are not eigenvectors, early ratios can lie far below the asymptotic rate (the example after Proposition 4.16); near a positive floor the ratios tend to one while the distance to the limit contracts quickly (Proposition 4.18(d)); a ratio near one can thus indicate a rate near one, the approach to a floor, or a stall. Example 4.12 has ratios tending to one and still converges to a point, sublinearly. Proposition 7.10 gives valid monotone relaxation families whose iterations share arbitrarily long histories and then either stall or contract to a point. Observed ratios therefore support stopping heuristics, not guarantees.

Which regimes admit finite certificates. In the stall regime, the witnesses of Theorem 4.10 are finitely many points whose coordinate hull is a fixed box at every small scale; they are a certificate of the kind developed in Section 5. In the contracting regime with $\epsilon > 0$, the limit box is a fixed box of width $O(\sqrt{\epsilon})$ under (6), and $2n$ points attaining its faces certify it, as in Proposition 4.18(b). In the contracting regime with $\epsilon = 0$, the only fixed box containing x^* in a small neighborhood is $\{x^*\}$; no box of positive width can certify that the remaining movement is zero. An iterate that reaches $\{x^*\}$ can be certified as fixed. If convergence is only asymptotic, as in the model problem, a bound on the remaining movement, such as the residual certificate of Section 7, justifies a tolerance stop. When the optimal incumbent lies in the current box, its protected singleton (Corollary 5.14) already

bounds the remaining movement, and the bound is exact if the iterates converge to it. These cases do not exhaust the possibilities: Example 4.12 has no fixed box of positive width and no geometric contraction.

Symmetry and spread minimizers. By Lemma 3.3, several global minimizers keep their hull in every iterate, whatever the local rates. Tightening at the root of a symmetric model cannot converge to a single point until branching separates the minimizers. At a node that contains a single minimizer, the conclusions of this section apply when their hypotheses hold: a tangent model, monotonicity on the comparison boxes, a contraction shape, and a start box in the local region. A single minimizer does not supply them; Example 4.20 has a unique minimizer and stalls.

5 Finite witness certificates

Before paying for another OBBT round, a solver would like to know how much that round, or all remaining rounds, can still change. This section bounds the change from above with finite sets of points that are feasible for the relaxation. The bounds concern a specified tightening procedure and a specified relaxation family. They are not new variable bounds, and they do not by themselves predict solving time. Three statements must be kept apart.

1. A point feasible for the current relaxation bounds the improvement of one frozen round (Section 5.2).
2. Points feasible for the relaxation rebuilt on an inner box P bound the improvement of every later round, directional update, and compatible cutoff, provided the box being tightened contains P (Sections 5.3–5.7).
3. A residual and a verified sensitivity bound limit the remaining movement of an exact iteration (Section 7).

Statement 1 is the feasibility filtering of Gleixner et al. [15]. Statement 2 rests on the persistence of fixed boxes (Lemma 3.5), the order argument behind monotone domain-reduction iterations [9, 43]. The OBBT-specific content of this section is a finite certificate for a fixed box, checked exactly on the rebuilt relaxation; a description of the operations it covers and of those it does not; its exactness at the limit of the iteration; and the difference between finding such a certificate and checking it.

5.1 Witnesses and lifted reuse

We use the setting of Section 3: the relaxation family is valid (3) and monotone (4), each ϕ_B is lower semicontinuous, and a lifted family has compact sets $R(B)$ and a continuous objective v . The cutoff sets $K_U(B)$, the operator T_U , the relaxation bound $L(B)$, fixed boxes, and the Jacobi, directional, sequential, and selective update schemes are as defined there.

A *lifted witness* on a box P at cutoff U is a complete lifted vector $z \in R(P)$, including every auxiliary coordinate, that satisfies every row of the relaxation rebuilt on P and has $v(z) \leq U$. By (2), its projection $\pi(z)$ satisfies $\phi_P(\pi(z)) \leq v(z) \leq U$, so $\pi(z) \in K_U(P)$. This direction needs no attainment premise; a checked lifted witness always certifies a projected one.

Two order properties play different roles below. All coordinate and objective-value ceilings in this section need only the monotonicity (4) of the projected objectives. Lifted monotonicity, $R(B') \subseteq R(B)$ for $B' \subseteq B$ with a common objective, is needed for a different use: to treat a stored lifted point $z \in R(P)$ itself, with its stored value $v(z)$, as a feasible point of the linear

program on a box $B \supseteq P$. Such reuse lets a solver screen a later frozen round, mix witnesses for a new cutoff (Section 6), or warm-start a later solve without recomputing auxiliary coordinates. Under monotonicity alone, $\phi_B(\pi(z)) \leq \phi_P(\pi(z)) \leq v(z)$ still holds, but a feasible lift on B must be recomputed.

The reference family. Several examples use the lifted linear family with McCormick product rows listed in Section 3. Its rows are of three kinds. Fixed rows $Gz \leq g$ do not depend on the box; they include the original linear constraints, quadratic constraints written linearly in lifted product variables, and valid inequalities such as $y_k \geq 0$ for a lifted square. Bound rows are $\ell \leq x \leq u$. For each lifted variable y_k standing for a product $x_i x_j$, with $i = j$ allowed, the four McCormick inequalities [27] on $B = [\ell, u]$ can be written as

$$\begin{aligned} y_k - x_i x_j &\geq -\min\{(x_i - \ell_i)(x_j - \ell_j), (u_i - x_i)(u_j - x_j)\}, \\ y_k - x_i x_j &\leq \min\{(u_i - x_i)(x_j - \ell_j), (x_i - \ell_i)(u_j - x_j)\}. \end{aligned} \quad (14)$$

After the product $x_i x_j$ cancels, each of the four inequalities is affine in (x, y_k) . For example, the first reads $y_k \geq \ell_j x_i + \ell_i x_j - \ell_i \ell_j$. When $i = j$ the two upper inequalities coincide, and the rows are the tangents of x_i^2 at both endpoints together with the secant. This is a finite linear relaxation of the square, not its convex epigraph. The objective is a fixed affine function $v(z) = c^T z + c_0$.

Lemma 5.1 (The reference family). *The reference family is lifted monotone with a common objective, and hence monotone. Each $R(B)$ is a polytope. If the fixed rows are valid for the original problem and v evaluated at the graph lift $y_k = x_i x_j$ equals $f(x)$, the family is valid.*

Proof. Let $B' = [\ell', u'] \subseteq B = [\ell, u]$ and $x \in B'$. Then $0 \leq x_i - \ell'_i \leq x_i - \ell_i$ and $0 \leq u'_i - x_i \leq u_i - x_i$ for every i . Each product in (14) multiplies two such nonnegative distances, so its value on B' is at most its value on B . Consequently each lower bound on $y_k - x_i x_j$ on B' is at least the corresponding lower bound on B , and each upper bound is at most the corresponding upper bound. Fixed rows and the objective do not depend on the box, and bound rows on B' imply those on B . Thus $R(B') \subseteq R(B)$. The set $R(B)$ is a polyhedron; it is bounded because $x \in B$ and each y_k lies between affine functions of x . At the graph lift the middle terms in (14) vanish, and all bounds have the correct sign because every distance is nonnegative on B . Together with valid fixed rows, this puts the graph lift of each $x \in X \cap B$ in $R(B)$, so $\phi_B(x) \leq f(x)$. $\square$

Example 5.2 (Monotonicity without lifted monotonicity). Relax $s = x^2$ on a box $[\ell, u]$ by the convex-combination formulation common in piecewise-linear relaxations: auxiliary weights $\lambda_0, \lambda_1 \geq 0$ with $\lambda_0 + \lambda_1 = 1$ and $x = \lambda_0 \ell + \lambda_1 u$, the secant row $s \leq \lambda_0 \ell^2 + \lambda_1 u^2$, the two endpoint tangents, and the objective $v = s$. In every case $\lambda_0 \ell^2 + \lambda_1 u^2 = (\ell + u)x - \ell u$, and for $\ell < u$ the weights are determined by x . The projection onto (x, s) is therefore the square relaxation of the reference family, and since the objective depends only on s , the projected objectives are those of that family, which are monotone by Lemma 5.1. Lifted monotonicity fails: on $P = [0, 1/2]$ the point $x = 1/2$ has weights $(0, 1)$, while on $B = [0, 1]$ it requires $(1/2, 1/2)$. A stored lifted witness for P is not a feasible point of the relaxation on B , although its projection remains in every projected set that the certificates below use.

5.2 The current frozen round

Proposition 5.3 (Current-round ceilings). *Let $B = [\ell, u]$, let W be a nonempty finite subset of $K_U(B)$, and write $T_U(B) = [\ell', u']$. For every coordinate i ,*

$$0 \leq \ell'_i - \ell_i \leq \min_{x \in W} x_i - \ell_i, \quad 0 \leq u_i - u'_i \leq u_i - \max_{x \in W} x_i. \quad (15)$$

Proof. The coordinate minimum over $K_U(B)$ is at most the minimum over its subset W , and the maximum is at least the maximum over W . Since $K_U(B) \subseteq B$, the new endpoints lie in $[\ell_i, u_i]$. $\square$

A witness attaining an endpoint proves that this direction cannot move in the present round, and one point can screen several directions. The ceilings bound possible benefit; a witness coordinate is not a valid new bound. Points produced by earlier support solves of the same round, or an optimal solution of the relaxation itself, are typical witnesses. Checking k lifted points against N nonzero row coefficients costs $O(kN)$ arithmetic operations. Finding good points can cost as much as the round itself. A ceiling on coordinate movement also says nothing about side products of a support solve, such as dual information used for Lagrangian variable bounds [15].

The ceiling (15) concerns the frozen relaxation on B . After rebuilding on a smaller box, the same point may be infeasible, and later rounds may move much further.

Example 5.4 (A frozen witness does not bound later rounds). Minimize $f(x) = x^2$ over $B_0 = [-r_0, r_0]$, $r_0 > 0$, with the square relaxation of the reference family, objective $v = s$, and the cutoff $U = 0$ given by the incumbent $x = 0$; this is the case $n = 1$ of Example 4.23. On $[-r, r]$ the rows are $s \geq 2r|x| - r^2$ (the two tangents) and $s \leq r^2$ (the secant). With $s \leq 0$ they give $|x| \leq r/2$, and every such x has the lift $s = 0$. Hence $T_0([-r, r]) = [-r/2, r/2]$, and the exact Jacobi iterates are $B_k = [-2^{-k}r_0, 2^{-k}r_0]$. In the first round the witness $(r_0/2, 0)$ bounds the decrease of the upper endpoint by $r_0/2$, which is exact for that round. After rebuilding on $[-r_0/2, r_0/2]$, the new upper tangent requires $s \geq r_0x - r_0^2/4 = r_0^2/4 > 0$ at $x = r_0/2$, so the witness is excluded. The total decrease over all rounds is r_0 .

5.3 Protected boxes

A witness remains useful in all later rounds if it is feasible for the relaxation built on a box that every later box contains. The following definitions make this precise and describe the operations that preserve such a box.

Definition 5.5 (Relaxation-sound step). A box C' is obtained from a box C by a relaxation-sound step at cutoff U' if $T_{U'}(C) \subseteq C' \subseteq C$.

A Jacobi round is a relaxation-sound step with $C' = T_{U'}(C)$. So is a directional update, and so is the simultaneous update of any selected set of endpoints on one frozen relaxation. Sequential and selective rounds are sequences of such steps. Conservative versions, which move each selected endpoint at most to its exact support, are also relaxation-sound. This includes OBBT whose endpoint bounds are proved by safe dual bounds evaluated with directed rounding (Proposition B.2). Rounding an approximate optimum outward alone does not prove such a bound. The identity step is relaxation-sound.

Definition 5.6 (Witness pool and protected box). Let $P = [a, b]$ be a nonempty box. A *witness pool* for P is a finite set $W \subseteq K_U(P)$ with box $W = P$; that is, for every coordinate i some $x \in W$ has $x_i = a_i$ and some $x' \in W$ has $x'_i = b_i$. The *threshold* of W is $U_W = \max_{x \in W} \phi_P(x) \leq U$. A box with a witness pool is *protected* at every cutoff $U' \geq U_W$.

The definition evaluates ϕ_P , the relaxation *rebuilt on* P , not the relaxation on a larger current box. In lifted form, a pool consists of lifted witnesses on P ; their stored values $v(z)$ bound the threshold from above. At most $2n$ points are needed, one for each endpoint, and fewer suffice when one point attains several endpoints. A protected box is a fixed box; the word *protected* refers to the finite certificate and its threshold.

Theorem 5.7 (Protection). *Let W be a witness pool for $P = [a, b]$ with threshold U_W . Let $C_0 \supseteq P$ be a box in B_0 , and for $k \geq 0$ let C_{k+1} be obtained from C_k by a relaxation-sound step at a cutoff $U_k \geq U_W$. Then:*

- (a) $T_{U'}(P) = P$ for every $U' \geq U_W$;
- (b) $P \subseteq C_k$ and $W \subseteq K_{U_k}(C_k)$ for every k ; in particular no step proves C_k infeasible at its cutoff;
- (c) writing $C_k = [\ell^k, u^k]$, for every i and k ,

$$0 \leq \ell_i^k - \ell_i^0 \leq a_i - \ell_i^0, \quad 0 \leq u_i^0 - u_i^k \leq u_i^0 - b_i. \quad (16)$$

Proof. (a) For $U' \geq U_W$, $W \subseteq K_{U_W}(P) \subseteq K_{U'}(P)$, so $P = \text{box } W \subseteq T_{U'}(P) \subseteq P$. (b) By (a), P is a fixed box of T_{U_W} . If $P \subseteq C_k$, then Lemma 3.5 gives $P \subseteq T_{U_k}(C_k) \subseteq C_{k+1}$, and induction gives $P \subseteq C_k$ for every k . Lemma 3.1 gives $W \subseteq K_{U_W}(P) \subseteq K_{U_k}(C_k)$. (c) The inequalities restate $P \subseteq C_k \subseteq C_0$. $\square$

The theorem needs no continuity, convergence, or fairness in the choice of directions; the steps may mix Jacobi rounds and directional updates and may use different cutoffs above U_W . The witnesses need not be feasible for the original problem, and none of them needs to be an optimal solution of a support problem. The essential requirement is feasibility in the relaxation rebuilt on P . In Example 5.4 the first-round witnesses are feasible on B_0 but not on their own hull, and indeed that hull is not retained.

A protected box limits what the specified OBBT procedure can remove. It is not a domain reduction and never replaces the search domain. For instance, minimize $f(x) = x$ over $[0, 1]$ with the exact relaxation $\phi_B(x) = x$ and the cutoff $U = 1$ of the incumbent $\hat{x} = 1$. The singleton $\{1\}$ is protected by the witness $\hat{x}$, and every OBBT iterate contains it, but the minimizer 0 lies outside it. Optimal solutions may lie anywhere in the current box. A protected box need not contain any original feasible point either: its pool may consist of relaxation-only points.

The finite certificate loses nothing when a prescribed box is tested: a box is fixed exactly when a short witness pool exists.

Proposition 5.8 (Fixed boxes have short certificates). *A box with a witness pool is a fixed box. Conversely, every fixed box P of T_U has a witness pool $W \subseteq K_U(P)$ with $|W| \leq 2n$, and for a lifted family the pool can be chosen to consist of at most $2n$ lifted witnesses. If, moreover, $R(P)$ is a polytope given by rational data, v has rational coefficients, and the endpoints of P and U are rational, then these lifted witnesses can be chosen rational, so an exact rational check accepts them.*

Proof. The first claim is Theorem 5.7(a). Let P be fixed. The set $K_U(P)$ is a sublevel set of the lower semicontinuous function ϕ_P on the compact box P , hence compact, and $T_U(P) = P$ means that each coordinate attains its minimum a_i and maximum b_i on it. Choosing one minimizer and one maximizer per coordinate gives at most $2n$ points whose hull is P . For a lifted family, the infimum in (2) is attained, so each chosen point has a lift in $R(P)$ with value at most U . In the rational case, the set of lifts attaining a given endpoint, such as $\{z \in R(P) : v(z) \leq U, x_i = a_i\}$, is a nonempty polytope defined by rational data, and any of its vertices is the solution of a nonsingular rational linear system. $\square$

The proposition makes the certificate complete as a *test* for a prescribed box. It is not a search procedure: it says nothing about how to find a fixed box, and Section 5.8 shows that natural proposals can miss one.

Certificates from different sources can be combined.

Proposition 5.9 (Combining certificates). *If W_1 and W_2 are witness pools for P_1 and P_2 with thresholds U_1 and U_2 , then $W_1 \cup W_2$ is a witness pool for $\text{box}(P_1 \cup P_2)$ with threshold at most $\max\{U_1, U_2\}$.*

Proof. Let $P = \text{box}(P_1 \cup P_2)$ and $U' = \max\{U_1, U_2\}$. For $x \in W_j$, $\phi_P(x) \leq \phi_{P_j}(x) \leq U_j \leq U'$ by monotonicity. Every endpoint of P is an endpoint of P_1 or P_2 and is attained by the corresponding pool. $\square$

For example, the singleton of an incumbent that lies in the current box, which is protected (Corollary 5.14), can be merged with a stalled box that does not contain the incumbent. The ceilings of the merged box are no larger than those of either part, and some are smaller.

5.4 Exact ceilings at the limit

Theorem 5.7 gives upper bounds on remaining movement. The best of these bounds is exact whenever the limit of the iteration is a fixed box.

Corollary 5.10 (Exact ceilings at the limit). *Let $U \geq f^*$, let B_k be the Jacobi iterates from B_0 , and assume the closedness condition (6). Then the limit B_∞ has a witness pool of at most $2n$ points. For every k , the ceilings (16) of $P = B_\infty$ from $C_0 = B_k$ equal the exact remaining movements of the Jacobi iteration. The same holds for complete sequential rounds $S_{k+1} = S_U(S_k)$ with $S_0 = B_0$: $B_\infty \subseteq S_k$, the iterates S_k converge to B_∞ , and the ceilings of B_∞ from S_k equal their remaining movements.*

Proof. By Proposition 3.7, B_∞ is a fixed box, and Proposition 5.8 gives the pool. The ceilings of B_∞ from B_k are the distances between corresponding endpoints of B_k and B_∞ , which are the remaining movements because B_k converges to B_∞ . A complete sequential round is a sequence of directional updates, so Theorem 5.7 gives $B_\infty \subseteq S_k$, and Lemma 3.2 shows that the S_k have the same intersection B_∞ . $\square$

The reference family satisfies (6) by the argument following Proposition 3.7, with the compact set $R(B_0)$ containing every $R(B)$ by Lemma 5.1. Under this condition, no information is lost by measuring remaining benefit through protected boxes: the exact remaining movement is a protected-box ceiling. The difficulty lies entirely in finding the right box. Without (6) the corollary can fail. In Example 3.8 the greatest fixed box is $\{0\}$, while the iterates converge to $[0, 1/2]$; from $B_k = [0, s_k]$ the ceiling of $\{0\}$ is s_k , whereas the remaining movement is $s_k - 1/2$.

Example 5.11 (Exact ceilings without finite termination). Return to Example 5.4. The limit is $B_\infty = \{0\}$. It is protected by the lifted witness $(x, s) = (0, 0)$; on the degenerate box the square roots force $s = 0$. The ceilings (16) from B_0 are r_0 in both directions, and they equal the total movement, although no finite number of rounds reaches $\{0\}$. The relaxation bound behaves in the same way: $L(B_k) = -4^{-k}r_0^2$, attained at $(0, -4^{-k}r_0^2)$, while Corollary 5.12 below gives the ceiling 0, which is the limit. The witnesses produced by any single round, $(\pm r/2, 0)$ on $[-r, r]$, never pass the rebuilt check.

5.5 Objective ceilings

The relaxation bound $L(C) = \inf_{x \in C} \phi_C(x)$ of Section 3 is also controlled by a protected box.

Corollary 5.12 (Objective ceiling). *Let W , P and (C_k) be as in Theorem 5.7. For every k and every $x \in P$,*

$$L(C_0) \leq L(C_k) \leq L(P) \leq \phi_P(x). \quad (17)$$

In particular, a lifted point $z \in R(P)$ of any objective value gives $L(C_k) \leq v(z)$. If $\lambda \leq L(C_0)$, the relaxation bound can increase by at most $v(z) - \lambda$ along the sequence.

Proof. Since $C_k \subseteq C_0$, monotonicity of L gives $L(C_k) \geq L(C_0)$. Since $P \subseteq C_k$ and $\phi_{C_k} \leq \phi_P$ on P , $L(C_k) \leq \inf_{x \in P} \phi_P(x) = L(P)$. $\square$

The objective point need not be feasible for the original problem, and it need not be optimal for the relaxation on P ; any lifted point feasible on P gives a valid ceiling, and the minimum of v over $R(P)$ gives the best one. Monotonicity suffices for this value statement. Under lifted monotonicity the simpler argument applies: z itself stays feasible in every $R(C_k)$ with the same value. The lower bound λ must bound the value of *this* relaxation on C_0 . A lower bound on the original optimum cannot be substituted, because the relaxation value can lie far below it; Example 5.13 has $L = -3$ and original optimum 0, and substituting 0 would give a negative ceiling.

The practical reading concerns pruning. If $L(P) < U$, then $U - L(C_k) \geq U - L(P) > 0$ for every k : no amount of OBBT with this family closes the gap at this node. Branching, cuts, or a stronger relaxation are needed. The corollary bounds only the relaxation value along a sequence of relaxation-sound steps. A child node created by branching need not contain P , and a stronger relaxation has a different value.

Example 5.13 (A strongly convex objective with a permanent gap). Minimize

$$f(x) = x_1^2 + x_2^2 + x_3^2 + x_1x_2 + x_1x_3 + x_2x_3 = \frac{1}{2}(\|x\|_2^2 + (x_1 + x_2 + x_3)^2)$$

over $[-1, 1]^3$, the case $a = 1$, $n = 3$ of Example 4.20. The unique minimizer is 0, with $f^* = 0$. Use the reference family with square variables s_i , product variables y_{ij} for $i < j$, fixed rows $s_i \geq 0$, objective $v = \sum_i s_i + \sum_{i < j} y_{ij}$, and the cutoff $U = 0$ of the optimal incumbent. On the cube $P = [-1, 1]^3$ the square rows are $s_i \geq 2|x_i| - 1$ and $s_i \leq 1$, and the product rows are $|x_i + x_j| - 1 \leq y_{ij} \leq 1 - |x_i - x_j|$.

For the endpoint $x_1 = 1$, take $x = (1, 0, 0)$, $s = (1, 0, 0)$, $y_{12} = y_{13} = 0$ and $y_{23} = -1$. Every row holds: $s_1 = 1$ meets its tangent and secant, the other squares satisfy $0 \geq -1$, the incident products satisfy $0 \leq y_{1j} \leq 0$, and the opposite product satisfies $-1 \leq y_{23} \leq 1$. Its value is $1 - 1 = 0 \leq U$. The symmetric points for the other five endpoints have the same value. Hence the cube is protected with threshold $0 = f^*$. Every relaxation-sound step at every cutoff produced by a feasible incumbent leaves the whole cube unchanged.

The point $x = 0$, $s = 0$, $y_{ij} = -1$ has value -3 , so $L(C_k) \leq -3$ for every such sequence. The value is exactly -3 : $s_i \geq 0$ and $y_{ij} \geq |x_i + x_j| - 1 \geq -1$. The gap $U - L = 3$ therefore persists. All these points satisfy $s_i \geq x_i^2$; replacing the finite square rows by the convex epigraphs would not remove the stall, in agreement with the tangent-model analysis of Example 4.20. The cause is the accumulated McCormick gap of the three products. This is the equality case of condition (a) of Proposition 4.19: $H_{kk}/2 = S_k = 1$ for every k .

5.6 Original feasible points

Validity gives protected boxes without any relaxation solve. Lemma 3.3 shows that every iterate contains the hull H_U of the whole original sublevel set. A finite part of that set gives a certificate that a checker can verify.

Corollary 5.14 (Hulls of original feasible points). *Let S be a finite nonempty subset of $X \cap B_0$ with $f(x) \leq U$ for all $x \in S$, and let $P = \text{box } S$. Then S is a witness pool for P with threshold at most $\max_{x \in S} f(x)$. Moreover, P is retained by every box operation that keeps all points of $\{x \in X \cap C : f(x) \leq U'\}$ in the current box C , for cutoffs $U' \geq \max_{x \in S} f(x)$.*

Proof. Each $x \in S$ lies in $X \cap P$, so validity gives $\phi_P(x) \leq f(x) \leq U$, and $\text{box } S = P$. An operation that keeps every original feasible point with objective at most U' keeps S , hence its hull. $\square$

The second statement covers more operations than Theorem 5.7: every reduction that is valid for the original problem, including cuts, propagation of original constraints, stronger relaxations, and integer rounding when S respects integrality. In that case the integer-coordinate endpoints of P are automatically integral. The singleton of the incumbent is the simplest instance: OBBT never removes the incumbent from a box that contains it. At a node whose box excludes the incumbent, the incumbent still supplies the cutoff but no protected singleton. With several known solutions, for example images of the incumbent under a symmetry of the model, the hull can be large. A lifted checker needs exact lifts of the points in S , which can be hard to obtain for nonlinear equalities; a numerically accepted incumbent is not exactly feasible in general.

5.7 Rounding, cuts, and branching

A solver interleaves OBBT with integer rounding, cut generation, propagation, and branching. A certificate for a fixed relaxation family covers these operations only under the conditions below.

Proposition 5.15 (Integer rounding). *Let I be the set of integer-constrained coordinates, and let rounding replace ℓ_i by $\lceil \ell_i \rceil$ and u_i by $\lfloor u_i \rfloor$ for $i \in I$. If $P = [a, b]$ is protected and $a_i, b_i \in \mathbb{Z}$ for all $i \in I$, then the conclusions of Theorem 5.7 remain true when roundings are interleaved with the relaxation-sound steps.*

Proof. Rounding a box $C \supseteq P$ keeps P : from $\ell_i \leq a_i \in \mathbb{Z}$ follows $\lceil \ell_i \rceil \leq a_i$, and from $u_i \geq b_i \in \mathbb{Z}$ follows $\lfloor u_i \rfloor \geq b_i$. The induction in Theorem 5.7 therefore continues through rounding steps. $\square$

Fractional faces can be removed. Minimize x^2 over $[-1, 1]$ with the square relaxation and the cutoff $U = 1/4$. The box $[-1/2, 1/2]$ is protected by the lifted witnesses $(\pm 1/2, 1/4)$: on it the tangents read $s \geq |x| - 1/4$ and the secant reads $s \leq 1/4$. It is the limit of the exact iteration from $[-1, 1]$, whose first round gives $[-5/8, 5/8]$. If x is integer, the first rounding gives $\{0\}$, removing the protected faces; here this is correct, because 0 is the only integer point with $x^2 \leq 1/4$. If the support problems themselves enforce integrality, the operator changes: K_U is replaced by its subset of points satisfying integrality, which is again monotone, and a pool must consist of such points. Its endpoints are then integral on I .

New rows change the family. If a lifted monotone family receives the same row on every box, lifted monotonicity is preserved; if every lifted witness of a pool satisfies the row, the pool remains a certificate for the modified family. Otherwise a new check is required. A cut can remove an objective witness while keeping the face witnesses. On $[-1/2, 1/2]$ at cutoff $1/4$ above, the lifted point $(0, -1/4)$ is feasible and gives $L \leq -1/4$, although $x^2 \geq 0$. The valid row $s \geq 0$ excludes this point, and the strengthened relaxation has bound 0, while the face witnesses $(\pm 1/2, 1/4)$ survive. Reductions valid for the original problem but stronger than the relaxation can remove relaxation-only faces altogether. A solver that recognizes convexity of f in Example 5.13 reduces the cube to $\{0\}$ at cutoff 0. Pools of original feasible points (Corollary 5.14) are guaranteed to survive every valid reduction that keeps their points.

A certificate applies at a node only if the node box contains P . The intersection of P with a child box need not be protected. Example 4.22 shows the same objective tightening on a thinner box for the tangent model; the next example shows it for a child created by branching in the lifted family.

Example 5.16 (A protected box does not pass to a child). In Example 5.13, branch on $x_1 \geq 0$, giving the child $C = [0, 1] \times [-1, 1]^2 = P \cap C$. On C the product rows for y_{12} are $y_{12} \geq \max\{-x_1, x_1 + x_2 - 1\}$ and the two upper rows, and similarly for y_{13} . At the endpoint $x_2 = 1$, every relaxed point satisfies $s_2 \geq 1$, $y_{12} \geq x_1$, $y_{13} \geq -x_1$, $y_{23} \geq x_2 + x_3 - 1 = x_3$, $s_1 \geq 0$ and $s_3 \geq \max\{0, -2x_3 - 1\}$. Hence

$$v \geq 1 + x_3 + \max\{0, -2x_3 - 1\} \geq \frac{1}{2},$$

where the last bound follows by considering $x_3 \geq -1/2$ and $x_3 \leq -1/2$. The value $1/2$ is attained at $x = (0, 1, -1/2)$ with $y_{23} = -1/2$ and all other lifted variables at their lower bounds. No point of $K_0(C)$ attains $x_2 = 1$, so OBBT on the child tightens u_2 . The parent witness for this endpoint has $y_{13} = -1$, which violates the child row $y_{13} \geq -x_1 = 0$.

5.8 Finding a certificate versus checking it

Checking a proposed certificate is inexpensive. Given a candidate box P and a lifted pool of $k \leq 2n + 1$ points, one builds the relaxation on P and evaluates every row at every point: with N nonzero row coefficients, this is $O(kN)$ arithmetic operations, plus $O(kn)$ comparisons for endpoint attainment and k objective evaluations. No optimization problem is solved. The check must be exact, for example in rational arithmetic. A feasibility test with a tolerance can accept a point that violates a rebuilt row slightly, and then the conclusion that no later round moves an endpoint is unfounded.

Finding a certificate is a different matter. By Proposition 5.8, a candidate must be a fixed box of T_U , and useful ceilings require a fixed box close to the limit of the iteration. Computing such a box can require the support solves that OBBT itself performs, and particular proposals can fail even at a fixed box. Two concrete proposals are available. The first uses the points that an exact round produces anyway.

Proposition 5.17 (Checking a completed round). *Let $B' = T_U(B)$ be nonempty and put $R_U(C) = \{z \in R(C) : v(z) \leq U\}$ for a box C . Let $\widehat{W} \subseteq R_U(B)$ be a finite set of complete lifted points whose projections $W = \pi_x(\widehat{W})$ attain all $2n$ endpoints of B' , where $\pi_x(x, y) = x$, such as lifted optimal solutions of the support problems.*

- (a) *If $\widehat{W} \subseteq R_U(B')$, then W is a witness pool for B' , and Theorem 5.7 applies with $P = B'$.*
- (b) *If $T_U(B) = B$, the hypothesis of (a) holds.*
- (c) *The hypothesis of (a) can fail although $T_U(B') = B'$.*

Consequently, if exact Jacobi iterates reach a fixed box after finitely many rounds, the test in (a) succeeds no later than in the round that starts at the fixed box. If they never reach one, the test never succeeds.

Proof. (a) The lifted check gives $W \subseteq K_U(B')$, and by construction box $W = B'$. (b) Then $B' = B$, so $\widehat{W} \subseteq R_U(B) = R_U(B')$. (c) See Example 5.18. For the consequence, if B_j is the first fixed box, part (b) applies to the round that starts at B_j . If the test succeeded in some round, the resulting box would be fixed by part (a). $\square$

Example 5.18 (A fixed box missed by its own round). Minimize $f(x) = x^2$ over $B = [0, 1]$ subject to $x \leq 1/2$, kept as a fixed row, with the square relaxation and the cutoff $U = 1/4$ of the incumbent $x = 1/2$. On B the rows are $s \geq 0$, $s \geq 2x - 1$, $s \leq x$, $x \leq 1/2$ and $s \leq 1/4$. The lower support is 0, attained at $(0, 0)$. The upper support is $1/2$, attained on the segment $\{1/2\} \times [0, 1/4]$, with vertices $(1/2, 0)$ and $(1/2, 1/4)$. Thus $B' = [0, 1/2]$. On B' the tangent at $1/2$ and the secant force $s = 1/4$ at $x = 1/2$. The vertex $(1/2, 0)$ fails the rebuilt check, while $(1/2, 1/4)$ passes, so the outcome depends on which optimal vertex the solver returns. The box B' is fixed either way; it is the hull of the original feasible points 0 and $1/2$.

In exact arithmetic, a Jacobi round that returns its input box already proves a fixed box. The rebuilt check adds three things. It can detect the fixed box one round earlier and save that round's $2n$ support solves. It produces a portable certificate, which also covers directional updates, other cutoffs above its threshold, and every containing box. Finally, it needs only primal points. A valid *reduction* of a bound needs a dual proof that no relaxed point lies beyond the new bound. Proving the *absence* of further reduction needs only primal witnesses, which is exactly what a pool supplies. Section 9 describes an exact driver that verifies each round by rational primal–dual proofs and then applies the test of Proposition 5.17; Theorem 9.2 states its guarantees. When the iterates converge only asymptotically, as in Example 5.4, this proposal never succeeds, although Corollary 5.10 guarantees that the limit is protected.

The second proposal uses original feasible points in the current box: the incumbent, other solutions found by heuristics, and their images under known symmetries (Corollary 5.14). It needs no relaxation solve, only exact lifts, and it survives all valid reductions. Its ceilings are exact when the hull equals the limit. In Example 5.11 the incumbent singleton $\{0\}$ is the limit, and in Example 5.18 the hull of 0 and $1/2$ is the fixed box. In general the hull of known solutions can be far inside the limit. Proposition 5.9 merges certificates from both proposals.

A protected box therefore does not make the round that finds it cheaper. Its value lies in later work that it makes unnecessary: rounds after a certified stall, repeated tightening after cutoff changes above its threshold, repeated tightening at nodes that contain it, and certified bounds on the remaining relaxation gap. Whether these savings exceed the cost of discovery depends on how often such events occur during search.

6 Cutoff changes and cached witnesses

During search the incumbent improves and the cutoff U decreases. Since $T_{U'}(B) \subseteq T_U(B)$ for $U' \leq U$, a lower cutoff can make further tightening possible, and certificates computed for the old cutoff may expire. This section answers three questions about stored primal information. At which cutoff does a certificate expire? Which current-round witnesses can be produced for a new cutoff without solving a new support problem? How do stored points behave when the box changes?

Primal witnesses give upper bounds on the possible benefit of tightening. Stored dual solutions give the opposite information, a guaranteed reduction, for a relaxation whose constraint matrix does not change. Section 8 treats dual information and combines both kinds in a bracket for one frozen relaxation. Throughout this section, a *lifted pool* $\widehat{W}$ is a finite set of complete lifted points with stored objective values. When a lifted pool is used on one box B , all of its points lie in $R(B)$; mixing and frontiers also require $R(B)$ to be convex and the objective v to be affine, as in the reference family.

6.1 Cutoff thresholds of a protected box

Let $P = [a, b]$ be a protected box with a lifted pool $\widehat{W} \subseteq R(P)$ whose projections attain every endpoint of P , and write $q_z = v(z)$ for $z \in \widehat{W}$. The $2n$ faces of P are $\{x \in P : x_i = a_i\}$ and $\{x \in P : x_i = b_i\}$; when $a_i = b_i$ the two coincide, which does no harm below. The certificate of Theorem 5.7 remains valid at every cutoff $U' \geq U_{\widehat{W}} = \max_{z \in \widehat{W}} q_z$, the largest stored value. This threshold can be larger than necessary, because an expensive point may be redundant. The exact threshold for reusing part of the pool is

$$\tau_{\widehat{W}}(P) = \max_F \min\{q_z : z \in \widehat{W}, \pi(z) \in F\}, \quad (18)$$

where F ranges over the faces of P . Each inner minimum is over a nonempty set because $\widehat{W}$ attains every face.

Proposition 6.1 (Face threshold of a pool). *For a cutoff U' , the subpool $\{z \in \widehat{W} : q_z \leq U'\}$ attains every face of P if and only if $U' \geq \tau_{\widehat{W}}(P)$. Convex combinations do not lower this threshold: the projection of $\text{conv}(\widehat{W}) \cap \{v \leq U'\}$ has box hull P if and only if $U' \geq \tau_{\widehat{W}}(P)$.*

Proof. The subpool attains a face F exactly when some pool point on F has value at most U' , that is, when the inner minimum in (18) is at most U' . Requiring this for all faces gives the first claim. For the second, the subpool is contained in $\text{conv}(\widehat{W}) \cap \{v \leq U'\}$, which proves one direction. Conversely, let $\bar{z} = \sum_z \lambda_z z$ be a convex combination with $v(\bar{z}) \leq U'$ whose projection lies on the face $x_i = b_i$. Every $z \in \widehat{W}$ has $x_i(z) \leq b_i$, and the weighted average equals b_i , so every z with $\lambda_z > 0$ lies on that face. Hence $v(\bar{z}) = \sum_z \lambda_z q_z$ is at least the inner minimum for that face. The same argument applies to every face. $\square$

For example, let $R(P) = P = [0, 1]^2$ with no auxiliary variables, $v(x) = x_1 + x_2$, and $\widehat{W}$ the four corners. Then $U_{\widehat{W}} = 2$ but $\tau_{\widehat{W}}(P) = 1$, because the three corners $(0, 0)$, $(1, 0)$ and $(0, 1)$ attain all four faces. Mixing can produce witnesses for a *smaller* box at a lower cutoff, but such a box needs its own check on its rebuilt relaxation (Section 6.2).

The pool threshold depends on the stored points. The following threshold depends only on the box and the relaxation.

Proposition 6.2 (Exact expiry cutoff of a box). *Let P be a nonempty box in a lifted family. For each face F of P let $\mu_F(P) = \min\{v(z) : z \in R(P), \pi(z) \in F\}$, with value $+\infty$ when no lift projects to F , and let $\tau(P) = \max_F \mu_F(P)$. Then*

$$T_U(P) = P \iff U \geq \tau(P).$$

Moreover, $\tau(P) \leq \tau_{\widehat{W}}(P)$ for every lifted pool $\widehat{W} \subseteq R(P)$ that attains every face of P .

Proof. Because $R(P)$ is compact and v continuous, $K_U(P) = \pi(R_U(P))$, and the minimum defining $\mu_F(P)$ is attained when finite. A face F is therefore attained by a point of $K_U(P)$ exactly when $\mu_F(P) \leq U$. If $K_U(P)$ is nonempty, Proposition 5.8 turns this face-by-face statement into the claimed equivalence. If $K_U(P)$ is empty, then $T_U(P) = \emptyset \neq P$, and every face has $\mu_F(P) > U$. Pool points are feasible for the face problems, which gives $\tau(P) \leq \tau_{\widehat{W}}(P)$. $\square$

The quantities $\mu_F(P)$ are the finite-box counterparts of the face values of Corollary 4.11. The two thresholds have different costs and uses. The face threshold $\tau_{\widehat{W}}(P)$ uses only the stored coordinates and objective values, without an optimization solve. When the incumbent value drops below it, the stored certificate expires. This is an exact event at which reconsidering OBBT is

justified; it does not show that tightening will pay off. The box threshold $\tau(P)$ requires up to $2n$ face-restricted minimizations of the objective, comparable to one OBBT round. It gives a positive statement: if the current box is P and the cutoff U falls below $\tau(P)$, the next exact round either finds $K_U(P)$ empty or, since some face is no longer attained and $K_U(P)$ is compact, moves at least one endpoint strictly. The size of that movement is not bounded below by this argument; Section 8 gives guaranteed reductions from dual information. For cutoffs between $\tau(P)$ and $\tau_{\widehat{W}}(P)$ the pool is inconclusive. None of these thresholds is necessary for the existence of a different protected box inside P . Proposition 8.6 is the single-direction counterpart of $\tau(P)$ for one frozen support problem.

6.2 Witnesses at a lower cutoff in one relaxation

A stored current-round witness whose value exceeds a new cutoff can be moved below it by convex combination with any relaxed point of lower value.

Proposition 6.3 (Mixing to a new cutoff). *Let $R(B)$ be convex with an affine objective v , and let $a, z \in R(B)$ with $v(a) \leq U' < v(z)$. Then*

$$\theta = \frac{U' - v(a)}{v(z) - v(a)} \in [0, 1), \quad z_{U'} = a + \theta(z - a) \quad (19)$$

satisfies $z_{U'} \in R(B)$ and $v(z_{U'}) = U'$.

Proof. The point $z_{U'}$ is a convex combination of two points of the convex set $R(B)$. Since v is affine, $v(z_{U'}) = v(a) + \theta(v(z) - v(a)) = U'$. $\square$

The anchor a needs only relaxation feasibility. A minimizer of v over $R(B)$, with value $L(B)$, or a lift of the new incumbent are natural choices. The original coordinates move to $x(a) + \theta(x(z) - x(a))$, and Proposition 5.3 then bounds the frozen round at the new cutoff without a new support solve. The construction is confined to one frozen relaxation, and the hull of mixed points is generally not protected.

Example 6.4 (Mixed witnesses at a new cutoff). Relax $s = x^2$ on $[-1, 1]$ by the square rows $s \geq 2|x| - 1$ and $s \leq 1$, with objective $v = s$. Mixing the graph points $(\pm 1, 1)$ with the anchor $(0, 0)$ to the cutoff $U' = 1/4$ gives $\theta = 1/4$ and the points $(\pm 1/4, 1/4)$, which are feasible on $[-1, 1]$. On their hull $[-1/4, 1/4]$ the secant reads $s \leq 1/16$, so both stored lifts fail the rebuilt check. Their projections nevertheless have feasible rebuilt lifts $(\pm 1/4, 1/16)$, so this hull is protected at cutoff $1/4$. On $[-1, 1]$ the mixed points bound the decrease of each endpoint in the frozen round at cutoff $1/4$ by $3/4$; the exact round gives $[-5/8, 5/8]$, a decrease of $3/8$.

A hull of mixed points need not be protected. Mixing the same graph points with the relaxation minimizer $(0, -1)$ to cutoff 0 gives $\theta = 1/2$ and points $(\pm 1/2, 0)$. These attain the supports of the current cutoff relaxation. On their hull $[-1/2, 1/2]$, however, $\phi(\pm 1/2) = 1/4 > 0$, so no feasible lift at cutoff 0 reaches either face. The hull is not fixed, and the next round gives $[-1/4, 1/4]$.

6.3 The cutoff frontier of a cached pool

A nonempty lifted pool $\widehat{W} = \{z_1, \dots, z_k\} \subseteq R(B)$ with values $q_j = v(z_j)$ supplies the current-round witnesses

$$\mathcal{C}_U = \text{conv}(\widehat{W}) \cap \{v \leq U\} \subseteq R_U(B)$$

at every cutoff U . The best ceilings that the pool can give are the coordinate extrema over $\mathcal{C}_U$. They require no new support solve, because they are attained at explicit points. The proof below is standard extreme-point reasoning; its use here is to generate the candidates from a retained pool.

Proposition 6.5 (Exact frontier of a pool). *If $\mathcal{C}_U$ is nonempty, which holds exactly when $\min_j q_j \leq U$, every minimum and maximum of an original or lifted coordinate over $\mathcal{C}_U$ is attained at one of the following points:*

- (a) *a pool point z_j with $q_j \leq U$;*
- (b) *for a pair with $q_j < U < q_l$, the mixture of z_j and z_l with value exactly U .*

There are at most $k + k(k - 1)/2$ such points.

Proof. Let $\Lambda_U = \{\lambda \in \mathbb{R}^k : \lambda \geq 0, \sum_j \lambda_j = 1, \sum_j \lambda_j q_j \leq U\}$. Because v is affine, the map $\lambda \mapsto \sum_j \lambda_j z_j$ sends Λ_U onto $\mathcal{C}_U$, and Λ_U is nonempty exactly when some $q_j \leq U$. A coordinate is a linear function of λ , so its extrema over the polytope Λ_U are attained at vertices. Let λ be a vertex with support $J = \{j : \lambda_j > 0\}$. If $|J| \geq 2$ and the cutoff inequality is slack, a nonzero h supported on J with $\sum_j h_j = 0$ gives $\lambda \pm \epsilon h \in \Lambda_U$ for small $\epsilon > 0$, which is impossible at a vertex. If the cutoff is tight and $|J| \geq 3$, a nonzero h supported on J satisfies the two equations $\sum_j h_j = 0$ and $\sum_j h_j q_j = 0$, with the same contradiction. If the cutoff is tight, $J = \{j, l\}$ and $q_j = q_l$, the first equation implies the second, and again a perturbation exists. Hence either λ is a unit vector e_j with $q_j \leq U$, or $J = \{j, l\}$, $q_j \neq q_l$, and the cutoff is tight. In the latter case two positive weights average q_j and q_l to U , so U lies strictly between them and the image of λ is the stated mixture. $\square$

The proof covers repeated points, equal values, and degenerate hulls. The proposition optimizes over the convex hull of the pool, not over the whole relaxation; an empty frontier says only that the pool cannot supply a witness at that cutoff.

The frontier also describes exactly how the pooled ceilings respond to the cutoff. Fix a coordinate i and let $g(U) = \max\{x_i(z) : z \in \mathcal{C}_U\}$ for $U \geq q_{\min} = \min_j q_j$. The pooled ceiling on the decrease of u_i is $u_i - g(U)$.

Proposition 6.6 (Cutoff response of a pool). *For a nonempty pool, the function g is nondecreasing, concave, and continuous on $[q_{\min}, \infty)$. It is affine between consecutive distinct values among $q_1, \dots, q_k$ and constant for $U \geq \max_j q_j$. The coordinate minimum over $\mathcal{C}_U$ has the same properties with nonincreasing and convex in place of nondecreasing and concave.*

Proof. The set Λ_U grows with U , so g is nondecreasing. If λ and λ' are optimal at U and U' , then for $t \in [0, 1]$ the combination $t\lambda + (1 - t)\lambda'$ is feasible at $tU + (1 - t)U'$, which gives concavity. A concave function is continuous on the interior of its interval. At $q_{\min}$, let $\Delta > 0$ be the gap to the next larger pool value; if there is none, g is constant. For $q_{\min} < U < q_{\min} + \Delta$, every $\lambda \in \Lambda_U$ puts total weight $\epsilon \leq (U - q_{\min})/\Delta$ on points with $q_j > q_{\min}$. Writing $c = \max_j |x_i(z_j)|$, this gives $g(U) \leq g(q_{\min}) + 2c\epsilon$, and continuity follows. On an open interval between consecutive distinct pool values, the sets of eligible points and of straddling pairs in Proposition 6.5 do not change. There g is the maximum of finitely many affine functions of U : constants for eligible points, and for a pair (j, l) the coordinate $x_i(z_j) + \frac{U - q_j}{q_l - q_j}(x_i(z_l) - x_i(z_j))$. Hence g is convex there; being also concave, it is affine, and by continuity it is affine on the closed interval. For $U \geq \max_j q_j$, Λ_U is the whole simplex. The statements for the minimum follow by applying these arguments to $-x_i$. $\square$

Proposition 6.6 turns a pool into an exact screening rule for future incumbents. For a tolerance $\delta \geq 0$, the upper direction of x_i is screened at cutoff U when $g(U) \geq u_i - \delta$. Because g is nondecreasing and continuous, the set of such cutoffs is empty or an interval $[\sigma_i, \infty)$. The threshold σ_i is found by evaluating g at the distinct pool values and solving one affine equation. When the incumbent value falls below σ_i , the pool can no longer screen this direction, and a support solve or new witnesses are needed. The computation involves no optimization problem; at each pool value it evaluates at most $k + k(k - 1)/2$ candidates.

Example 6.7 (A frontier and its cutoff response). Let $R(B) = B = [0, 2] \times [0, 3]$ with no auxiliary variables, $v(x) = x_2$, and the pool $z_1 = (0, 0)$, $z_2 = (2, 3)$, $z_3 = (1, 1)$ with values 0, 3, 1. At $U = 2$ the frontier consists of z_1 , z_3 , the mixture $(4/3, 2)$ of z_1 and z_2 , and the mixture $(3/2, 2)$ of z_3 and z_2 . The best pooled witness for the upper endpoint of x_1 uses the anchor z_3 , not the point of lowest value. For this endpoint,

$$g(U) = \begin{cases} U, & 0 \leq U \leq 1, \\ (U + 1)/2, & 1 \leq U \leq 3, \\ 2, & U \geq 3. \end{cases}$$

At $U = 2$ the pooled ceiling on the decrease of $u_1 = 2$ is $1/2$, whereas the exact decrease is 0 because $(2, 0) \in R_2(B)$. The pool bounds the round correctly but not tightly. With tolerance $\delta = 1/4$, the pool screens this direction exactly for the cutoffs $U \geq 5/2$.

Changed boxes. Before a pool is used on a new box, every point must pass the rows of the relaxation built on that box. For the filtered pool, Propositions 6.5 and 6.6 are again exact. They are not exact for the old convex hull intersected with the new rows: mixtures of excluded points can be feasible, and additional inequalities create coefficient vertices with more than two positive weights. Under lifted monotonicity, a point feasible on a box P remains feasible on every box containing P , so the witnesses of a protected box can join the pool at every containing node without a check. Points computed on a parent box need not be feasible on a child box; in Example 5.4, the witness $(r/2, 0)$ from $[-r, r]$ fails on $[-r/2, r/2]$.

6.4 Continuing after a cutoff decrease

When the cutoff decreases, a solver can either continue tightening from its current box or restart from the starting box. Continuing yields boxes contained in the restart iterates. If the current box was obtained by a finite number of relaxation-sound steps at compatible cutoffs, both sequences have the same intersection, by monotonicity alone.

Proposition 6.8 (Continuing versus restarting). *Let $U' \leq U$, let $B'_k = T_{U'}^k(B_0)$, and let $C_0 \subseteq B_0$ be obtained from B_0 by $m \geq 0$ relaxation-sound steps at cutoffs at least U' , for example by OBBT at the old cutoff U . Then $C_k = T_{U'}^k(C_0)$ satisfies*

$$B'_{k+m} \subseteq C_k \subseteq B'_k \quad (k \geq 0).$$

In particular, $\bigcap_k C_k = \bigcap_k B'_k$.

Proof. A relaxation-sound step $C \rightarrow C'$ at cutoff $U'' \geq U'$ gives $C' \supseteq T_{U''}(C) \supseteq T_{U'}(C)$. Induction and monotonicity give $C_0 \supseteq T_{U'}^m(B_0)$, and hence $C_k \supseteq T_{U'}^{k+m}(B_0)$. The upper inclusion follows from $C_0 \subseteq B_0$ and monotonicity. The two bounding subsequences have the same intersection, proving the last statement. $\square$

If the iterates are nonempty and (6) holds at the new cutoff, the common intersection is also the greatest fixed box, by Proposition 3.7. These extra assumptions are needed for that identification, not for the equality of intersections.

After an incumbent improvement, therefore, no tightening is lost by resuming from the current box. A box that is protected at the new cutoff, for example one whose face threshold $\tau_{\widehat{W}}(P)$ does not exceed it, still bounds all remaining movement.

6.5 Conditions for reuse

Table 1 collects the conditions under which stored primal information remains valid. The entries restate Propositions 5.3, 6.1 and 6.2, Theorem 5.7, Corollary 5.14, and the residual certificates of Section 7.

Information	Bounds	Valid for	Recheck or expiry
Pool in $K_U(B)$	movement in the frozen round on B	the same box and rows; lower cutoffs through mixing and the frontier	any change of box or rows
Protected box P with lifted pool $\widehat{W}$	all movement of relaxation-sound steps; relaxation bound through $L(P)$	boxes containing P ; cutoffs at least $\tau_{\widehat{W}}(P)$ with the stored pool, or at least $\tau(P)$ with a recomputed pool; the same family	box not containing P ; cutoff below the threshold; new rows violated by a witness; rounding of fractional faces
Hull of original feasible points S	movement under every reduction valid for the original problem	boxes containing S ; cutoffs at least $\max_{x \in S} f(x)$	cutoff below $\max_{x \in S} f(x)$; box not containing S
Residual and comparison matrix	total movement of exact Jacobi iteration	one cutoff and one family, inside the invariant region on which the matrix is verified	leaving that region; a change of cutoff or family without a new residual bound and comparison matrix

Table 1: Conditions for reusing stored primal information. A failed or expired condition makes the certificate inapplicable; it does not show that further tightening is useful.

7 Residual certificates for remaining movement

A protected box bounds all remaining movement, but a good bound needs an inner box close to the limit, and such a box is hard to find when the iterates converge only asymptotically. This section bounds remaining movement with quantities available at the current box instead: the residual of one exact round, and a sensitivity bound valid on a region that contains every later box. With a matrix sensitivity bound, the result is an a posteriori estimate of the kind given by vector-valued contraction principles [24]. The OBBT-specific points are the following. The order structure of OBBT removes the need for a spectral-radius condition. The region on which the sensitivity bound must hold can be obtained from a protected box. Only some kinds of observed movement are valid residual information. Finally, the history of an iteration cannot replace a verified sensitivity bound.

7.1 Endpoint maps and invariant regions

Represent a box B by its endpoint vector $p(B) = (-\ell, u) \in \mathbb{R}^{2n}$, as in Section 3, and write $B(p)$ for the box with endpoint vector p . Then $p \leq q$ componentwise exactly when $B(p) \subseteq B(q)$, and shrinking a box decreases p . Fix a cutoff U and let $F(p) = p(T_U(B(p)))$, defined when $K_U(B(p)) \neq \emptyset$. Then $F(p) \leq p$, and monotonicity makes F order preserving: $p \leq q$ implies $F(p) \leq F(q)$. Fixed points of F are the endpoint vectors of fixed boxes. Jacobi iteration is $p_{k+1} = F(p_k)$, a nonincreasing sequence. The *residual* at p is $p - F(p) \geq 0$, the movement of the next exact round.

A set $\mathcal{D}$ of endpoint vectors is *invariant* if F is defined on $\mathcal{D}$ and $F(\mathcal{D}) \subseteq \mathcal{D}$. A sensitivity bound is useful only on an invariant region containing the starting point, because every later iterate must stay where the bound has been verified. Protected boxes supply such regions. For vectors $p' \leq p$, write $[p', p] = \{q : p' \leq q \leq p\}$.

Lemma 7.1 (Invariant order intervals). *Let P be protected with threshold at most U , with endpoint vector $p_P \leq p_0$. Then $[p_P, p_0]$ is invariant.*

Proof. For $q \in [p_P, p_0]$, the box $B(q)$ contains P , so $K_U(B(q)) \supseteq K_U(P) \neq \emptyset$ and $F(q)$ is defined. Order preservation and $F(p_P) = p_P$, which is Theorem 5.7(a), give $p_P \leq F(q) \leq q \leq p_0$. $\square$

If an incumbent $\hat{x}$ with $f(\hat{x}) \leq U$ lies in the current box $B(p_0)$, the protected singleton $P = \{\hat{x}\}$ (Corollary 5.14) gives the region of boxes inside $B(p_0)$ that contain $\hat{x}$. At a node of a search tree the incumbent can lie outside the node box; it then supplies no such region, and another invariant region must be found.

7.2 A matrix majorant for the remaining movement

Theorem 7.2 (Remaining movement). *Let $\mathcal{D}$ be invariant and $p_0 \in \mathcal{D}$. Suppose a fixed matrix $M \geq 0$ satisfies*

$$F(p) - F(q) \leq M(p - q) \quad (p, q \in \mathcal{D}, q \leq p). \quad (20)$$

Let $d = p_0 - F(p_0)$, let $\bar{d} \geq d$, and suppose a vector $e \geq 0$ satisfies

$$\bar{d} + Me \leq e. \quad (21)$$

Then p_k converges to a finite limit p_∞ , and for every $m \geq 0$,

$$0 \leq p_m - p_\infty \leq M^m e. \quad (22)$$

In particular, $p_0 - p_\infty \leq e$ and $p_1 - p_\infty \leq Me \leq e - \bar{d}$.

Proof. Let $\Delta_k = p_k - p_{k+1} \geq 0$. Both p_k and p_{k+1} lie in $\mathcal{D}$, and $p_{k+1} \leq p_k$, so (20) gives $\Delta_{k+1} = F(p_k) - F(p_{k+1}) \leq M\Delta_k$. Since $M \geq 0$ preserves componentwise inequalities, induction gives $\Delta_k \leq M^k d \leq M^k \bar{d}$. Next we show by induction on $N \geq 0$ that

$$\sum_{j=0}^{N-1} M^j \bar{d} + M^N e \leq e.$$

For $N = 0$ this reads $e \leq e$. If it holds for N , multiplying by M and adding $\bar{d}$ gives $\sum_{j=0}^N M^j \bar{d} + M^{N+1} e \leq \bar{d} + Me \leq e$ by (21). Since $M^N e \geq 0$, every partial sum $\sum_{j < N} M^j \bar{d}$ is at most e . For $N > m$,

$$0 \leq p_m - p_N = \sum_{k=m}^{N-1} \Delta_k \leq M^m \sum_{j=0}^{N-m-1} M^j \bar{d} \leq M^m e.$$

The sequence (p_k) is nonincreasing and bounded below by $p_0 - e$, so it converges, and letting $N \rightarrow \infty$ proves (22). For $m = 1$, inequality (21) gives $Me \leq e - \bar{d}$. $\square$

Hypothesis (20) compares only ordered pairs; a two-sided bound $|F(p) - F(q)| \leq M|p - q|$ implies it. Convexity of $\mathcal{D}$ is not needed, although it is useful when (20) is derived by integrating derivative bounds along segments, as in Section 8. If the limit lies in $\mathcal{D}$, for example when $\mathcal{D}$ is closed, it is a fixed point: $0 \leq F(p_k) - F(p_\infty) \leq M(p_k - p_\infty) \rightarrow 0$, so $F(p_\infty) = \lim_k p_{k+1} = p_\infty$.

The after-round bound Me holds for an upper residual $\bar{d}$ as well as for the exact one. It concerns the exact image $p_1 = F(p_0)$, not an arbitrary partially updated state. For $F(s) = s/2$, $p_0 = 1$, and $\bar{d} = d = 1/2$, the vector $e = 1$ satisfies (21) and the exact tail after one round is $1/2 = Me$. If no update has been applied, however, the remaining movement is $1 > e - \bar{d}$.

The theorem also gives a test that can refute a claimed matrix: the exact residuals must satisfy $\Delta_k \leq M^k \bar{d}$. An exact residual violating this inequality shows that (20) is false. Agreement with observed residuals does not prove it.

The next proposition identifies the best majorant obtainable from given M and $\bar{d}$, and gives a cheap explicit one.

Proposition 7.3 (Least majorant). *Let $M \geq 0$ and $r \geq 0$. A finite $e \geq 0$ with $r + Me \leq e$ exists if and only if the series $s = \sum_{k \geq 0} M^k r$ converges. In that case s satisfies $r + Ms = s$ and $s \leq e$ for every such e . If $\rho(M) < 1$, then $s = (I - M)^{-1}r$. If a vector $w > 0$ and a number $q \in [0, 1)$ satisfy $Mw \leq qw$, then $\rho(M) \leq q$, and*

$$e = \frac{t}{1-q} w, \quad t = \max_i \frac{r_i}{w_i},$$

satisfies $r + Me \leq e$.

Proof. If e exists, the proof of Theorem 7.2 bounds every partial sum of the nonnegative series by e , so the series converges and $s \leq e$. Conversely, passing to the limit in $\sum_{k \leq N+1} M^k r = r + M \sum_{k \leq N} M^k r$ gives $s = r + Ms$. If $\rho(M) < 1$, the Neumann series gives $s = (I - M)^{-1}r$. If $Mw \leq qw$ with $w > 0$, the norm $\|y\|_w = \max_i |y_i|/w_i$ satisfies $\|My\|_w \leq q\|y\|_w$ for all y , so $\rho(M) \leq q$. Finally, $r \leq tw$ gives $r + Me \leq tw + \frac{tq}{1-q}w = \frac{t}{1-q}w = e$. $\square$

Thus the strongest bound available from M and $\bar{d}$ is $s = \sum_k M^k \bar{d}$. When M and $\bar{d}$ are rational and $\rho(M) < 1$, it is computed exactly by a rational linear solve. Checking a proposed e requires only the inequality (21), with no eigenvalue computation. When a protected box P is also known, the componentwise minimum of e and $p_0 - p_P$ bounds the remaining movement.

Example 7.4 (A noncontracting direction). For the feasible set $\{0\}$ in $B_0 = [-1, 1]$ with $f \equiv 0$ and $U = 0$, let $\phi_{[\ell, u]}(x) = 0$ for $x \leq u/2$ and $+\infty$ otherwise, on every subbox $[\ell, u]$ of B_0 . This family is valid, monotone because $u/2$ is nondecreasing in u , and lower semicontinuous. For $\ell \leq 0 \leq u$ it gives $K_0([\ell, u]) = [\ell, u/2]$, so in endpoint coordinates $F(p_1, p_2) = (p_1, p_2/2)$ on the invariant region $[0, 1]^2$. Hence (20) holds with $M = \text{diag}(1, 1/2)$, whose spectral radius is 1. From $p_0 = (1, 1)$, the residual is $d = (0, 1/2)$, and $e = (0, 1)$ satisfies (21). The bound is exact: $p_\infty = (1, 0)$. The matrix inequality succeeds because the residual has no component in the noncontracting direction and M propagates none into it.

7.3 Valid residual information

Theorem 7.2 needs an upper bound $\bar{d}$ on the *exact* Jacobi residual. Three sources are valid: the residual of an exact Jacobi round, for example one whose support values are proved by primal-dual certificates (Section 9); the movement of an exact complete sequential round, which ends in a box contained in $T_U(B)$ by Lemma 3.2(b); and the current-round ceilings of a witness pool, which require no support solve at all. Three kinds of observed movement are *not* valid. A selective round reports no movement in unprocessed directions; an interrupted round stops early; and numerically safe OBBT rounds its bounds outward. Each can report less than the exact residual.

Corollary 7.5 (Residual bounds from witnesses). *Let $W \subseteq K_U(B(p_0))$ be nonempty and finite, and let $\bar{d}(W)$ collect its current-round ceilings: $\min_{x \in W} x_i - \ell_i$ in the lower components and $u_i - \max_{x \in W} x_i$ in the upper components. Then $d \leq \bar{d}(W)$, and Theorem 7.2 applies with $\bar{d} = \bar{d}(W)$.*

Proof. This is Proposition 5.3 written in endpoint coordinates. $\square$

A solver with a witness pool and a verified comparison matrix on an invariant region can therefore bound all future movement of exact Jacobi OBBT without solving a support problem. The pool may come from mixing or from a cached frontier at a new cutoff (Propositions 6.3 and 6.5). The bounds also apply to complete sequential rounds S_k started at $B(p_0)$. By Lemma 3.2, $S_k \subseteq B(p_k)$ and the S_k have the same limit $B(p_\infty)$, so after m sequential rounds the remaining movement is at most $p(S_m) - p_\infty \leq p_m - p_\infty \leq M^m e$.

7.4 A rebuilt relaxation with a verified comparison constant

The following example has McCormick coefficients that change in every round. The theorem needs no fixed constraint matrix, only a comparison constant valid over the invariant region.

Example 7.6 (Heron's iteration from a square relaxation). Minimize x^2 over $[0, u_0]$ with the square relaxation of the reference family and cutoff $U = r^2$, where $0 < r \leq u_0$. On $[0, u]$ with $u \geq r$, the rows are $s \geq 0$, $s \geq 2ux - u^2$ and $s \leq ux$, together with $s \leq r^2$. The upper tangent and the cutoff give $x \leq G(u) = (u^2 + r^2)/(2u)$. The point $(G(u), r^2)$ attains this bound: $r \leq G(u) \leq u$ because $u^2 + r^2 \geq 2ur$ and $r \leq u$; the tangent holds with equality; and $r^2 \leq uG(u) = (u^2 + r^2)/2$. The point $(0, 0)$ keeps the lower endpoint at 0. The upper endpoint therefore follows Heron's iteration $u_{k+1} = G(u_k)$, which is Newton's method for $u^2 = r^2$. Since

$$G(u) - r = \frac{(u - r)^2}{2u} \quad (u \geq r),$$

the errors $u_k - r$ are nonnegative and nonincreasing, they tend to 0, and $(u_{k+1} - r)/(u_k - r)^2 \rightarrow 1/(2r)$ while $u_k > r$: the convergence is quadratic.

The box $[0, r]$ is protected at cutoff r^2 by the lifted witnesses $(0, 0)$ and (r, r^2) ; on $[0, r]$ the tangent at r and the secant force $s = r^2$ at $x = r$. Lemma 7.1 gives the invariant region $\{(0, u) : r \leq u \leq u_0\}$. There $G'(u) = (1 - r^2/u^2)/2 \in [0, q]$ with $q = (1 - r^2/u_0^2)/2 < 1/2$, so the mean value theorem gives (20) with $M = \text{diag}(0, q)$. For $r = 1$ and $u_0 = 2$, the residual is $d = (0, 3/4)$ and $q = 3/8$, so $e = (I - M)^{-1}d = (0, 6/5)$. The actual total movement is 1. After the first round, $Me = (0, 9/20)$, while the actual remaining movement is $1/4$. The exact iterates are $5/4, 41/40, 3281/3280$; the linear bound is conservative because the convergence is quadratic.

In this example the protected box $[0, r]$ equals the limit and gives the exact remaining movement $u_0 - r$ directly. The residual certificate is useful in the opposite situation: when no inner box close to the limit is known, but a comparison constant can be verified on a region between a known inner box, such as an incumbent singleton, and the current box.

7.5 Cutoff decreases

When the incumbent improves, an old residual certificate does not automatically transfer to the new cutoff, because both the map and its residual change. A new certificate needs a new residual bound, which witnesses supply without support solves.

Corollary 7.7 (Remaining movement after a cutoff decrease). *Let p be the endpoint vector of the current box, let the cutoff decrease to V , and let F_V be the endpoint map at cutoff V . Let $\mathcal{D} \ni p$ be invariant for F_V , and let M satisfy (20) for F_V on $\mathcal{D}$. If $\bar{d}$ bounds $p - F_V(p)$ from above, for example as the ceilings at cutoff V of a pool obtained by mixing or from a cached frontier, and $\bar{d} + Me \leq e$, then exact Jacobi OBBT at cutoff V moves the endpoints of the current box by at most e in total.*

Proof. Apply Theorem 7.2 to F_V . □

If the new incumbent lies in the current box, it supplies an invariant region through Lemma 7.1. The corollary follows one specific iteration and needs no spectral condition. A comparison between fixed points at two cutoffs is a different statement, and the following proof of it uses a spectral condition.

Proposition 7.8 (Shift of a fixed point). *Let $V \leq U$. Let p_U be a fixed point of F_U and $p_V \leq p_U$ a fixed point of F_V , both in a region $\mathcal{D}$ on which F_V satisfies (20) with $\rho(M) < 1$. If $h \geq 0$ satisfies $F_U(p_U) - F_V(p_U) \leq h(U - V)$, then*

$$0 \leq p_U - p_V \leq (I - M)^{-1}h(U - V).$$

Proof. Let $x = p_U - p_V \geq 0$ and $c = h(U - V)$. Then $x = (F_U(p_U) - F_V(p_U)) + (F_V(p_U) - F_V(p_V)) \leq c + Mx$. Iterating gives $x \leq \sum_{j < N} M^j c + M^N x$ for every N , and $M^N \rightarrow 0$. □

The sensitivity bound h is needed only at the old fixed point, not uniformly over a region. The condition $\rho(M) < 1$ cannot simply be dropped from this proposition. For $F_U = F_V = F$ with $F(s, t) = (s, t/2)$, $M = \text{diag}(1, 1/2)$ and $h = 0$, both $(1, 0)$ and $(0, 0)$ are fixed points, ordered as required, yet they differ.

7.6 What observed progress cannot certify

An exact observed residual is valid input for Theorem 7.2. The comparison matrix is different: neither it nor any nontrivial bound on the remaining movement can be inferred from a finite strictly decreasing trajectory alone.

Example 7.9 (An observed ratio is not a comparison constant). Let $F(s) = 3s/4$ for $0 \leq s \leq 1/2$ and $F(s) = s/4 + 1/4$ for $1/2 \leq s \leq 1$. The map is continuous and nondecreasing with $0 \leq F(s) \leq s$, and it is the upper-endpoint map of a valid monotone family constructed as in Proposition 7.10 below. From $s_0 = 1$ the iterates are $s_1 = 1/2$ and $s_2 = 3/8$, with observed decrement ratio $1/4$. A geometric extrapolation predicts remaining movement $(1/8)/(1 - 1/4) = 1/6$ from s_1 . The iterates below $1/2$ follow the slope $3/4$ and converge to 0, so the actual remaining movement is $1/2$. With the valid constant $3/4$ on the invariant interval $[0, 1]$, Theorem 7.2 gives the correct bounds 2 from s_0 and $1/2$ from s_1 ; the latter is exact.

The example is one instance of a general fact.

Proposition 7.10 (Histories do not determine the remaining movement). *Let $s_0 > s_1 > \dots > s_N > 0$ with $N \geq 1$. For the feasible set $X = \{0\}$ in $B_0 = [0, s_0]$, with $f \equiv 0$ and $U = 0$, there are two valid monotone families with lower semicontinuous projected objectives that satisfy the closedness condition (6), whose Jacobi iterations from B_0 both produce the boxes $[0, s_k]$ for $k \leq N$. In the first, every later box equals $[0, s_N]$; in the second, the boxes converge to $\{0\}$.*

Proof. For a continuous nondecreasing function G on $[0, s_0]$ with $0 \leq G(s) \leq s$, set $\phi_{[\ell, u]}(x) = 0$ if $x \leq G(u)$ and $+\infty$ otherwise, for boxes $[\ell, u] \subseteq B_0$, as in Example 3.8 but with continuous G . Validity holds because a box containing 0 has $\ell = 0$ and $G(u) \geq 0$. Monotonicity holds because G is nondecreasing, and each ϕ_B is the indicator of a closed set. For (6), let boxes C_k decrease to C with upper endpoints $u_k \downarrow u_\infty$, and let $x_k \in K_0(C_k)$ converge to x . Then $x \in C$, and $x_k \leq G(u_k)$ gives $x \leq G(u_\infty)$ by continuity, so $x \in K_0(C)$. Since $K_0([0, u]) = [0, G(u)]$, the iteration is $u_{k+1} = G(u_k)$.

Let G_1 be the piecewise-linear interpolant of the points (s_k, s_{k+1}) for $0 \leq k < N$ together with $(0, 0)$ and (s_N, s_N) , and let G_2 be the interpolant of the same points with (s_N, s_N) replaced by $(s_N, s_N/2)$. At the increasing abscissae $0 < s_N < s_{N-1} < \dots < s_0$ the prescribed values are nondecreasing and lie between 0 and the abscissa, so both functions are continuous and nondecreasing with $0 \leq G(s) \leq s$. Both map s_k to s_{k+1} for $k < N$. On $[0, s_N]$, $G_1(s) = s$ and $G_2(s) = s/2$. $\square$

After the common history, the remaining movement is 0 in one family and s_N , the largest value permitted by nesting, in the other. An upper bound computed from $s_0, \dots, s_N$ alone is therefore valid only if it is at least s_N , the bound that nesting already gives, and no such rule can certify a stall. This holds whether the observed ratios are close to one or small; Section 7.7 gives explicit histories of both kinds. The proposition concerns strictly decreasing histories. An exact round that returns its input box, for an unchanged relaxation and cutoff, does prove a fixed box, and nesting always bounds future movement by the current box. The families are not McCormick relaxations. They show that validity, monotonicity, closedness, and a finite history together do not determine the remaining benefit; a protected box or a verified comparison matrix must supply the missing information.

7.7 Explicit misleading histories

Proposition 7.10 shows that every history $[0, s_0], \dots, [0, s_N]$ with $s_0 > \dots > s_N > 0$ is compatible both with a stall and with convergence. The two examples below make the two misleading patterns explicit. Both use the construction in the proof of that proposition: for the feasible set $\{0\}$ in $B_0 = [0, 1]$, with $f \equiv 0$ and $U = 0$, a continuous nondecreasing map G with $0 \leq G(s) \leq s$ defines the valid monotone family $\phi_{[\ell, s]}(x) = 0$ for $x \leq G(s)$ and $+\infty$ otherwise, whose Jacobi iterates from $[0, 1]$ are $[0, s_k]$ with $s_{k+1} = G(s_k)$.

Example 7.11 (Ratios close to one before convergence). Fix an integer $N \geq 2$, and put $\delta = 1/(2N)$ and $\sigma = 1/2 - \delta$. Let

$$G_{\text{stall}}(s) = \min\{s, \max\{\sigma, s - \delta\}\}, \quad G_{\text{shrink}}(s) = \begin{cases} s/4, & 0 \leq s \leq \sigma, \\ \sigma/4 + 3\sigma(s - \sigma)/(4\delta), & \sigma \leq s \leq 1/2, \\ s - \delta, & 1/2 \leq s \leq 1. \end{cases}$$

G_{stall} equals s on $[0, \sigma]$, σ on $[\sigma, 1/2]$, and $s - \delta$ on $[1/2, 1]$; the pieces of G_{shrink} take the values $\sigma/4$ at σ and σ at $1/2$, matching their neighbors. Both maps are therefore continuous with nonnegative slopes, and $0 \leq G(s) \leq s$ holds at the endpoints of each affine piece and hence on it. Both trajectories from 1 subtract δ until $s_N = 1/2$ and reach $s_{N+1} = \sigma$. Every observed ratio $s_{k+1}/s_k = 1 - \delta/s_k$, $k \leq N$, is at least $1 - 1/N$. Afterwards G_{stall} keeps σ , while G_{shrink} gives $s_{N+1+j} = 4^{-j}\sigma \rightarrow 0$. Arbitrarily many ratios close to one therefore do not indicate a stall. Both maps have slope 1 on $[1/2, 1]$, which every invariant interval containing the trajectory includes. Any valid scalar comparison constant there is at least 1, so the supersolution inequality (21) has no finite solution for a positive residual, so Theorem 7.2 provides no finite tail bound here.

Example 7.12 (A small ratio before a large movement). Define G_{stall} and G_{shrink} by linear interpolation of the knots

$$\text{stall: } (0, 0), \left(\frac{899}{1000}, \frac{899}{1000}\right), \left(\frac{9}{10}, \frac{899}{1000}\right), \left(1, \frac{9}{10}\right); \quad \text{shrink: } (0, 0), \left(\frac{1}{10}, 0\right), \left(\frac{899}{1000}, \frac{1}{10}\right), \left(\frac{9}{10}, \frac{899}{1000}\right), \left(1, \frac{9}{10}\right).$$

The knot values are nondecreasing and lie between 0 and the abscissa, so both maps are admissible. Both trajectories from 1 begin $1, 9/10, 899/1000$. The observed decrements $1/10$ and $1/1000$ have ratio $1/100$, and geometric extrapolation predicts movement of at most $(1/1000)/(1 - 1/100) = 1/990$ after the first round. The stalling map stays at $899/1000$; the shrinking map moves on to $1/10$ and then 0, a movement of $9/10$ after the first round.

Example 8.15 shows the second pattern, less extremely, in a McCormick relaxation whose active rows change. These families do not contradict exact certificates: nesting always bounds the remaining movement by the current box, and an exact round that returns its input box for an unchanged relaxation and cutoff proves a fixed box.

7.8 Finding a comparison matrix versus checking a majorant

Checking (21) for rational $\bar{d}$, M and e takes $O(\text{nnz}(M) + n)$ exact operations, and a rational certificate $Mw \leq qw$ for $\rho(M) < 1$ costs the same. Neither check establishes (20). That hypothesis requires a bound valid over an invariant region containing every later box, and hence over every active regime that the support problems can enter. In Example 7.6 it follows from a closed-form map. For relaxations given by linear programs, Section 8 derives it from an exact cover of the region by basis regions. A derivative at one point, a ratio of observed residuals, or a bound valid only on the current basis region does not suffice. This paper gives no general inexpensive construction of M for rebuilt relaxations.

8 Coupled constraints, parametric supports, and feasible repair

The certificates of Sections 5–7 treat the relaxation construction abstractly. In a solver, each directional OBBT problem is a linear program whose rows couple original and auxiliary variables, and these programs are solved again after the incumbent improves or the box shrinks. This section answers three questions. What does information stored from earlier solves still prove after such a change? When can the comparison inequality (20) required by Theorem 7.2 be established although the optimal basis changes from round to round? How do nonlinear constraints enter a rate bound?

The tools are linear programming duality, right-hand-side parametric linear programming [7], and error bounds for constraint systems. None of these is new. The results below make precise which conclusions survive a change of the cutoff, a change of the box with frozen rows, and a change of the box with rebuilt McCormick rows, and which conclusions need a new verification. Three kinds of conclusion must be kept apart throughout. A dual vector bounds a support from above and therefore guarantees a reduction. A primal witness bounds a support from below and therefore limits the possible reduction. Only a verified basis region gives the exact value.

8.1 Directional support problems

Sections 8.1–8.5 consider lifted relaxations given by linear programs. On a box $B = [\ell, u]$ the relaxation is a polytope $R(B) \subseteq \mathbb{R}^n \times \mathbb{R}^m$ of lifted points $z = (x, y)$, where x are the original and y the auxiliary variables, and the relaxed objective v is affine. Write $d = n + m$. The rows of $R(B)$ include the box rows $-x_i \leq -\ell_i$ and $x_i \leq u_i$. For a cutoff U , put $R_U(B) = \{z \in R(B) : v(z) \leq U\}$.

As noted in Section 3, compactness of $R(B)$ makes $K_U(B)$ the projection of $R_U(B)$ onto x , so the endpoints of $T_U(B)$ are optimal values of $2n$ linear programs. Sections 8.6 and 8.7 return to the general relaxations of Section 3.

Write $p = (-\ell, u) \in \mathbb{R}^{2n}$ and $B(p)$ for the corresponding box. For $i = 1, \dots, n$ let $c_i = -e_i$, and for $i = n + 1, \dots, 2n$ let $c_i = e_{i-n}$, extended by zeros on the auxiliary coordinates. The exact Jacobi endpoint map is

$$F_U(p)_i = \max\{c_i^T z : z \in R_U(B(p))\}, \quad i = 1, \dots, 2n, \quad (23)$$

so that $F_U(p) = (-\ell', u')$ when $T_U(B(p)) = [\ell', u']$. The box rows give $F_U(p) \leq p$. When the whole construction is valid and monotone, as assumed in Section 3, Lemma 3.1 gives $F_U(p) \leq F_U(q)$ whenever $p \leq q$. All vector inequalities and absolute values are componentwise. Statements about F_U concern Jacobi rounds; a sequential round rebuilds the relaxation between directions.

Rows of three kinds react differently when the cutoff or the box changes.

1. *Fixed rows* have coefficients and right-hand sides that do not depend on B or U , such as original linear constraints and globally valid cuts.
2. *Box rows and the cutoff row* have fixed coefficients and right-hand sides that are affine in (p, U) .
3. *Rebuilt rows* are recomputed from the current box. For an auxiliary variable y_k standing for a product $x_i x_j$ ($i = j$ allowed), the McCormick rows [27] are

$$\begin{aligned} y_k &\geq \beta x_i + \alpha x_j - \alpha\beta \quad \text{for } (\alpha, \beta) \in \{(\ell_i, \ell_j), (u_i, u_j)\}, \\ y_k &\leq \beta x_i + \alpha x_j - \alpha\beta \quad \text{for } (\alpha, \beta) \in \{(\ell_i, u_j), (u_i, \ell_j)\}. \end{aligned} \quad (24)$$

Their coefficients on x are box endpoints, their right-hand sides are products of endpoints, and the coefficient of y_k is ± 1 . For $i = j$ the rows are the two endpoint tangents and the secant of x_i^2 .

When only the cutoff changes, the rebuilt rows do not change either, and the support problems form a linear program with a fixed matrix and a right-hand side that is affine in U . When the box changes, rebuilt rows change their coefficients. Sections 8.2 and 8.3 treat the fixed-matrix case, and Section 8.4 treats rebuilt rows.

Consider therefore the family

$$R(\theta) = \{z \in \mathbb{R}^d : Az \leq b + E\theta\}, \quad h_c(\theta) = \max\{c^T z : z \in R(\theta)\}, \quad \theta \in \mathbb{R}^k, \quad (25)$$

with $A \in \mathbb{R}^{m_A \times d}$, $b \in \mathbb{R}^{m_A}$, $E \in \mathbb{R}^{m_A \times k}$, and $h_c(\theta) = -\infty$ if $R(\theta)$ is empty. The parameter θ may contain the cutoff, box endpoints that enter only through box rows, or both. For certificates, all data are rational and every inequality is checked in exact arithmetic. A basis or dual vector returned by a floating-point solver is a proposal to be checked, not a proof. An exact certificate concerns the supplied linear program. Validity of its rows for the original problem and their node-local scope are separate premises.

8.2 Dual envelopes and exact basis regions

Proposition 8.1 (Dual envelope). *Let $\Pi_c = \{\pi \in \mathbb{R}^{m_A} : \pi \geq 0, A^T \pi = c\}$.*

(a) *For every $\pi \in \Pi_c$ and every θ ,*

$$h_c(\theta) \leq \pi^T (b + E\theta). \quad (26)$$

Consequently, for every finite nonempty $Y \subseteq \Pi_c$, $h_c(\theta) \leq \min_{\pi \in Y} \pi^T (b + E\theta)$.

- (b) Suppose $\Pi_c \neq \emptyset$, and let $\Theta = \{\theta : R(\theta) \neq \emptyset\}$. Then Θ is a polyhedron, and for every $\theta \in \Theta$, $h_c(\theta)$ equals the minimum of $\pi^T(b + E\theta)$ over the finitely many vertices π of Π_c . In particular, h_c is finite, concave, piecewise affine, and continuous on Θ .

Proof. (a) If $R(\theta)$ is empty there is nothing to prove. For $z \in R(\theta)$, $c^T z = \pi^T A z \leq \pi^T(b + E\theta)$ because $\pi \geq 0$. Maximize over z .

(b) Θ is the projection of the polyhedron $\{(z, \theta) : Az - E\theta \leq b\}$ and is therefore a polyhedron. For $\theta \in \Theta$ the primal problem is feasible and its dual $\min\{\pi^T(b + E\theta) : \pi \in \Pi_c\}$ is feasible, so linear programming duality gives equal finite optimal values, both attained. The polyhedron Π_c lies in the nonnegative orthant, contains no line, and therefore has a vertex. A linear function that is bounded below on such a polyhedron attains its minimum at a vertex, and there are finitely many vertices. A minimum of finitely many affine functions is concave, piecewise affine, and continuous. $\square$

Part (a) uses neither optimality nor primal feasibility nor basis information at the new parameter. A stored dual vector therefore remains a valid upper bound however the optimal basis changes. In OBBT, an upper bound on h_{c_i} that lies below the current endpoint p_i is itself a valid tightened endpoint and requires no new solve. Storing dual information from OBBT solves and propagating it after later bound and cutoff changes is the principle of the Lagrangian variable bounds of Gleixner et al. [15]. The fixed matrix is the premise that makes (26) valid at every parameter; Section 8.4 shows that it fails for rebuilt rows.

Corollary 8.2 (Cutoff multiplier). *Suppose only the cutoff row depends on the parameter: $\theta = U$ and E is the unit vector of the cutoff row. Let $\pi \in \Pi_c$ be optimal at U_0 , and let $\eta \geq 0$ be its cutoff multiplier. For every U with $R(U) \neq \emptyset$,*

$$h_c(U) \leq h_c(U_0) - \eta(U_0 - U). \quad (27)$$

If, in addition, $U \leq U_0$ and $\pi' \in \Pi_c$ is optimal at U with cutoff multiplier η' , then

$$\eta(U_0 - U) \leq h_c(U_0) - h_c(U) \leq \eta'(U_0 - U).$$

Proof. Optimality gives $\pi^T(b + EU_0) = h_c(U_0)$, so $\pi^T(b + EU) = h_c(U_0) - \eta(U_0 - U)$, and Proposition 8.1(a) gives (27). Applying the same proposition at U_0 with π' gives $h_c(U_0) \leq \pi'^T(b + EU_0) = h_c(U) + \eta'(U_0 - U)$. $\square$

For a lower cutoff, $\eta(U_0 - U)$ is therefore a *guaranteed* reduction of the support, not an upper bound on the possible reduction. A zero multiplier guarantees nothing; it does not show that a later improvement of the incumbent is useless. The right-hand inequality uses a multiplier at the new cutoff, which is unknown until that problem is solved; it describes the reduction by the concavity of h_c but does not predict it. If optimality at U_0 has not been verified, the affine bound remains valid, but subtracting it from a numerical value of $h_c(U_0)$ does not give a certified reduction.

Proposition 8.3 (Exact basis region). *Let I be a set of d row indices with A_I nonsingular. Put*

$$z_I(\theta) = A_I^{-1}(b_I + E_I\theta), \quad \pi_I = A_I^{-T}c,$$

and extend π_I by zeros to a vector $\pi \in \mathbb{R}^{m_A}$. If $\pi_I \geq 0$, then on the closed polyhedron

$$C_I = \{\theta : Az_I(\theta) \leq b + E\theta\}$$

the point $z_I(\theta)$ is optimal and

$$h_c(\theta) = c^T z_I(\theta) = \pi^T(b + E\theta) \quad (\theta \in C_I). \quad (28)$$

Thus h_c coincides on C_I with an affine function whose gradient is $E^T \pi$. Conversely, if every $R(\theta)$ with θ in a set $\mathcal{P}$ is a nonempty polytope, then the regions C_I of all bases I with $A_I^{-T}c \geq 0$ cover $\mathcal{P}$.

Proof. For $\theta \in C_I$ the point $z_I(\theta)$ is feasible by definition. The extended vector satisfies $\pi \geq 0$ and $A^T \pi = A_I^T \pi_I = c$, and the rows in I are tight, so $c^T z_I(\theta) = \pi_I^T A_I z_I(\theta) = \pi^T (b + E\theta)$. Proposition 8.1(a) gives $h_c(\theta) \leq \pi^T (b + E\theta)$, and feasibility gives the reverse inequality.

For the converse, fix $\theta \in \mathcal{P}$. A linear objective attains its maximum over a nonempty polytope at a vertex z^* , and the rows active at z^* span $\mathbb{R}^d$. Optimality gives $c = \sum_{j \in J} \pi_j a_j$ with $\pi \geq 0$ and J a set of active rows, where a_j^T is row j of A ; choose such a representation with $|J|$ minimal. The rows in J are linearly independent: a linear relation among them would allow moving the coefficients along it until one becomes zero while the others stay nonnegative. Extend J by active rows to a set I of d linearly independent active rows and set the added coefficients to zero. Then A_I is nonsingular, $A_I^{-T} c$ is the chosen nonnegative coefficient vector, $z_I(\theta) = z^*$, and $\theta \in C_I$. $\square$

The first part is the classical description of critical regions in right-hand-side parametric linear programming, where optimizers and optimal values are affine on each region [7]. Zero multipliers, ties between bases, and lower-dimensional regions are allowed. At a boundary shared by two regions the two affine formulas agree, but h_c need not be differentiable there; the gradient refers to one affine piece. Rejecting a singular proposed basis rejects that proposal, not the relaxation. The polytope premise in the converse is needed. If s is an original variable and t a free auxiliary variable, the set $\{(s, t) : s = 0\}$ has attained supports $\max s = \max(-s) = 0$, but its two rows have rank one, so no basis of two rows exists. The polytope premise holds whenever every original and auxiliary variable has finite bounds, as in the McCormick relaxations considered here.

A proposed region is verified as follows. If a parameter polytope is the convex hull of finitely many vertices, it is contained in C_I if and only if every vertex satisfies the residual inequalities defining C_I , because C_I is convex. For a scalar cutoff, write $z_I(U) = z^0 + U z^1$ with $z^0 = A_I^{-1} b_I$ and $z^1 = A_I^{-1} E_I$. Row j becomes

$$(a_j^T z^1 - E_j) U \leq b_j - a_j^T z^0, \quad (29)$$

one affine inequality in U . Intersecting them gives the exact basis interval; this is classical right-hand-side ranging. On that interval the support moves exactly by η per unit of cutoff, where $\eta = c^T z^1$ is the cutoff multiplier of π . Outside it the dual expression remains an upper bound, but the primal formula need not be feasible. All nonbasic residuals must be checked in the same exact arithmetic as the basis solve. Exact arithmetic on the basis alone does not certify a point in a model whose other rows are evaluated in floating point.

Example 8.4 (A zero multiplier before a useful tightening). Minimize x_2 over $0 \leq x_1 \leq 3$, $0 \leq x_2 \leq 4$, and

$$x_1 - x_2 \leq 1, \quad 4x_1 - x_2 \leq 7, \quad x_1 \leq 5/2,$$

with the cutoff $x_2 \leq U$, $0 \leq U \leq 4$. There are no auxiliary variables. Maximizing x_1 gives

$$h(U) = \min\{1 + U, 7/4 + U/4, 5/2\} = \begin{cases} 1 + U, & 0 \leq U \leq 1, \\ 7/4 + U/4, & 1 \leq U \leq 3, \\ 5/2, & 3 \leq U \leq 4. \end{cases} \quad (30)$$

On $[0, 1]$ the basis consists of $x_1 - x_2 \leq 1$ and $x_2 \leq U$, with primal point $(1 + U, U)$ and multipliers $(1, 1)$; the row $4x_1 - x_2 \leq 7$ becomes $4 + 3U \leq 7$, so (29) gives exactly $[0, 1]$. On $[1, 3]$ the basis consists of $4x_1 - x_2 \leq 7$ and $x_2 \leq U$, with point $(7/4 + U/4, U)$ and multipliers $(1/4, 1/4)$; the row $x_1 - x_2 \leq 1$ requires $U \geq 1$ and the row $x_1 \leq 5/2$ requires $U \leq 3$. On $[3, 4]$ the basis consists of $x_1 \leq 5/2$ and $x_2 \leq U$, with point $(5/2, U)$ and multipliers $(1, 0)$; the row $4x_1 - x_2 \leq 7$ requires $U \geq 3$, and $x_2 \leq 4$ requires $U \leq 4$. The three verified intervals cover $[0, 4]$, which proves (30), and

the cutoff slopes are 1, 1/4, and 0. The three dual vectors give the affine functions $1 + U$, $7/4 + U/4$, and $5/2$; their minimum equals h on $[0, 4]$, an instance of Proposition 8.1(b).

At $U_0 = 4$ the cutoff multiplier is zero, yet lowering the cutoff to $U = 2$ reduces the support from $5/2$ to $9/4$. At $U_0 = 2$ the old dual remains valid at $U = 1/2$ and gives the bound $7/4 + (1/2)/4 = 15/8$. The extrapolated primal point $(15/8, 1/2)$ is infeasible, since $x_1 - x_2 = 11/8 > 1$, and the actual support is $3/2$. The old multiplier guarantees a reduction of $3/8$, while the actual reduction is $3/4$.

The example is chosen to make these distinctions visible without numerical ambiguity; it is not a difficult instance.

8.3 Reconsidering a support after the incumbent improves

When the incumbent improves from U_0 to $U < U_0$ at a fixed box, every row except the cutoff is unchanged and $R(U) \subseteq R(U_0)$. Stored primal points and stored dual vectors then bracket each new support.

Proposition 8.5 (Primal–dual bracket). *Fix the box and all rows except the cutoff, and write $R(U)$ for the relaxation with cutoff U . Let $c = c_i$ be a signed coordinate objective with current endpoint p_i . Let W be a finite set of complete lifted points that satisfy every row except possibly the cutoff, and let $Y \subseteq \Pi_c$ be finite and nonempty. For a new cutoff U , put $W(U) = \{z \in W : v(z) \leq U\}$. If $W(U) \neq \emptyset$, then*

$$\max_{z \in W(U)} c^T z \leq h_c(U) \leq \min\left\{p_i, \min_{\pi \in Y} \pi^T(b + EU)\right\}, \quad (31)$$

and consequently

$$\max\left\{0, p_i - \min_{\pi \in Y} \pi^T(b + EU)\right\} \leq p_i - h_c(U) \leq p_i - \max_{z \in W(U)} c^T z.$$

Proof. Every point of $W(U)$ is feasible for the new support problem, which proves the left inequality. The box row gives $h_c(U) \leq p_i$, and Proposition 8.1(a) gives the dual bound. Subtracting from p_i gives the second display. $\square$

The left end of the second display is the reduction that can be applied immediately, without a solve. The right end limits what a new solve of this one support problem can achieve. If the two ends of (31) meet, the new support is known exactly. The bracket concerns one support problem on one frozen relaxation. It says nothing about the further tightening that rebuilding the relaxation on the new box may produce. Bounds on all later rounds require a protected box (Section 5) or a comparison matrix (Section 8.5). Bounding the benefit of a whole round requires information for every direction, not for one favorable direction.

A stored witness that attains the old support certifies that the support does not change while the cutoff stays at or above its objective value. The least such value is an exact threshold.

Proposition 8.6 (Exact no-change threshold). *Fix the box and all rows except the cutoff, and let $R(U_0)$ be a nonempty polytope. Let $\mathcal{F}_0 = \{z \in R(U_0) : c^T z = h_c(U_0)\}$ be the optimal face and*

$$\underline{U} = \min\{v(z) : z \in \mathcal{F}_0\}.$$

For every $U \leq U_0$ with $R(U) \neq \emptyset$, $h_c(U) = h_c(U_0)$ if and only if $U \geq \underline{U}$.

Proof. If $U \geq \underline{U}$, a minimizer z_0 defining $\underline{U}$ satisfies the new cutoff, so $h_c(U) \geq c^T z_0 = h_c(U_0) \geq h_c(U)$, the last inequality because $R(U) \subseteq R(U_0)$. If $U < \underline{U}$, let z attain $h_c(U)$, which exists because $R(U)$ is a nonempty polytope. If $h_c(U) = h_c(U_0)$, then $z \in \mathcal{F}_0$ and $v(z) \leq U < \underline{U}$, contradicting the definition of $\underline{U}$. Hence $h_c(U) < h_c(U_0)$. $\square$

Computing $\underline{U}$ requires one additional linear program over the optimal face, or a lexicographic step after the support solve. Its cost must be charged like any other solve. A stored optimal witness with a larger objective value certifies the no-change interval only down to its own value; exclusion of such a witness does not show that the support moves, because another optimal point may remain. Exclusion of the least-objective witness does show this. Proposition 8.6 is the single-direction, single-round counterpart of the face thresholds for protected boxes in Section 6.1.

These results sort the directions after an incumbent improvement into three classes. If $U \geq \underline{U}$ for direction i , the new support equals the old one; when the current endpoint p_i already equals that value, for example because the last round applied it, the direction cannot move in this round. If a stored dual vector gives a bound below p_i , that bound can be applied at once. Otherwise the bracket (31) is the only information available, and a budgeted solve is the way to learn more. The classes concern certified bounds on one support. Whether a solve is worth its cost is a separate question.

Example 8.7 (Example 8.4 continued). At $U_0 = 4$ the optimal face is $\{(5/2, x_2) : 3 \leq x_2 \leq 4\}$, because a point with $x_1 = 5/2$ needs $x_2 \geq 3$ by the row $4x_1 - x_2 \leq 7$. Thus $\underline{U} = 3$, and Proposition 8.6 shows that the support is unchanged exactly for $U \in [3, 4]$. Lower the cutoff to $U = 2$. The witness $(5/2, 3)$ is excluded, so the support decreases. With the feasible anchor $(0, 0)$, the convex combination $(2/3)(5/2, 3) + (1/3)(0, 0) = (5/3, 2)$ is feasible at the new cutoff, and the dual vector stored at $U_0 = 4$ gives $5/2$. The bracket is $[5/3, 5/2]$, and the true value $9/4$ lies inside it. A dual vector that is optimal at a cutoff in $(1, 3)$ is unique and would tighten the upper end to $7/4 + U/4 = 9/4$.

8.4 Rebuilt McCormick rows

When the box shrinks and the McCormick rows (24) are rebuilt, the constraint matrix itself depends on p . Write

$$R(p) = \{z : A(p)z \leq b(p)\},$$

where the entries of $A(p)$ are affine in p and those of $b(p)$ are quadratic in p . For products of original variables the coefficients of the auxiliary variables are constant. This subsection states exactly what remains valid. Lifted monotonicity, $R(B') \subseteq R(B)$ for $B' \subseteq B$ with a common objective (Section 3), holds for the reference family of McCormick rows and fixed linear rows (Lemma 5.1).

Lemma 8.8 (Frozen rows). *Assume lifted monotonicity. Fix p_0 , let $A_0 z \leq b_0$ be the rows of $R(B(p_0))$, and for $p \leq p_0$ define the frozen relaxation*

$$\tilde{R}(p) = \{z : A_0 z \leq b_0, x \in B(p)\}.$$

- (a) $R(B(p)) \subseteq \tilde{R}(p)$. *The frozen family has a fixed matrix and a right-hand side affine in p , so every dual envelope of Proposition 8.1 for the frozen family is a valid upper bound for the rebuilt supports.*
- (b) *Let $p_1 = F_U(p_0)$, and let $\tilde{F}_U$ be the Jacobi map of the frozen family at the same cutoff. Then $\tilde{F}_U(p) = p_1$ for every p with $p_1 \leq p \leq p_0$.*

Proof. (a) Lifted monotonicity gives $R(B(p)) \subseteq R(B(p_0))$, and the box rows of $R(B(p))$ give $x \in B(p)$. (b) The frozen cutoff set is $R_U(B(p_0)) \cap \{x \in B(p)\}$. Every point of $R_U(B(p_0))$ has $x \in T_U(B(p_0)) = B(p_1) \subseteq B(p)$, so this set equals $R_U(B(p_0))$, whose signed supports are p_1 . $\square$

Part (a) is why dual information derived from globally valid rows can be reused at descendant nodes. Part (b) shows the price: frozen-row OBBT stops after one round, and its comparison matrix on $[p_1, p_0]$ is zero. Any progress of iterated OBBT comes from rebuilding. A zero comparison matrix computed from frozen rows must not be used in (20) for the rebuilt map; Theorem 7.2 would then assert that no motion remains after the first round. Example 8.15 below gives a case where this assertion is false.

Combining old multipliers with the *rebuilt* rows is a different shortcut, and it is invalid because in general $A(p)^T \pi \neq c$.

Example 8.9 (Reusing a dual vector after the box shrinks). Take the square relaxation of Example 7.6 with $r = 1$: on $B = [0, u]$, $u \geq 1$, the rows are $s \geq 0$, $s \geq 2ux - u^2$, and $s \leq ux$, with the cutoff $s \leq 1$, and the upper support is $G(u) = (u^2 + 1)/(2u)$. At $u_0 = 2$ the optimal basis consists of the tangent row $4x - s \leq 4$ and the cutoff, with exact multipliers $\pi = (1/4, 1/4)$, and $G(2) = 5/4$. Rebuild on $[0, 5/4]$.

- Old multipliers with rebuilt right-hand sides give $\pi^T b(5/4) = (1/4)(25/16) + 1/4 = 41/64$. This is not a valid bound. The point $(x, s) = (1, 1)$ satisfies every rebuilt row and the cutoff, and $x = 1$ is an original feasible point with objective $1 = U$. The vector π is not dual feasible for the rebuilt matrix, because $2 \cdot (5/4) \cdot (1/4) \neq 1$.
- The frozen rows of $[0, 2]$ give the valid bound $5/4$ and no tightening, as Lemma 8.8(b) predicts.
- The corrected bound of Proposition 8.10 below gives $71/64$.
- The exact rebuilt support is $G(5/4) = 41/40$.

Proposition 8.10 (A stored dual vector after a rebuild). *Let $\pi \geq 0$, let p be a parameter with $R(p)$ nonempty, and suppose every $z \in R(p)$ satisfies known bounds $z^L \leq z \leq z^U$. Put $q = c - A(p)^T \pi$. Then*

$$h_c(p) \leq \pi^T b(p) + \sum_j \max\{q_j z_j^L, q_j z_j^U\}, \quad (32)$$

where terms with $q_j = 0$ are zero even if the corresponding bound is infinite. If π is dual feasible for an earlier parameter p_0 , that is, $A(p_0)^T \pi = c$, and the change from $A(p_0)$ to $A(p)$ affects only columns of original variables, then $q = (A(p_0) - A(p))^T \pi$ vanishes on the auxiliary columns, and only the current box bounds of x enter the sum.

Proof. For $z \in R(p)$, since $\pi \geq 0$,

$$c^T z = \pi^T A(p)z + q^T z \leq \pi^T b(p) + q^T z \leq \pi^T b(p) + \sum_j \max\{q_j z_j^L, q_j z_j^U\}.$$

Maximize over z . The last statement follows from $c = A(p_0)^T \pi$. $\square$

In Example 8.9, $q = (1 - 2 \cdot (5/4)/4, 0) = (3/8, 0)$ and $x \leq 5/4$, so (32) gives $41/64 + (3/8)(5/4) = 71/64$. The bound needs no solve and uses the rebuilt rows. As a function of the upper endpoint $u \in [1, 2]$, the corrected bound with the multipliers from $u_0 = 2$ is $u - u^2/4 + 1/4$; for $u > 2$ the residual changes sign and the correction uses the lower bound 0 of x instead. Applying it repeatedly with these same multipliers produces valid endpoints $u_{k+1} = u_k - u_k^2/4 + 1/4$, which satisfy $u_{k+1} - 1 = (u_k - 1)(3 - u_k)/4$ and converge to the exact limit 1 at a linear rate of at most $1/2$. Exact OBBT converges to the same limit quadratically. This is a property of the example; in

general the correction term grows with the change of the coefficients, and the bound is useful only for moderate changes of the box.

Inequality (32) is the bound of Proposition B.2, written for maximization and applied to an exact dual vector of a different matrix. For products of original variables, the auxiliary columns in (24) have fixed coefficients ± 1 , so the correction needs only the current box. For composite factorable relaxations, products involving auxiliary variables have box-dependent auxiliary coefficients, and valid bounds on those auxiliary variables are also needed.

Exact supports with rebuilt rows are again given by basis certificates, now with parameter-dependent matrices.

Proposition 8.11 (Basis certificate with rebuilt rows). *Let $A(\cdot)$ and $b(\cdot)$ be continuously differentiable on an open set $O \subseteq \mathbb{R}^k$, let I be a set of d row indices, and let $\Pi \subseteq O$ be closed with $A_I(\theta)$ nonsingular for every $\theta \in \Pi$. Put*

$$z_I(\theta) = A_I(\theta)^{-1}b_I(\theta), \quad \pi_I(\theta) = A_I(\theta)^{-T}c,$$

and extend $\pi_I(\theta)$ by zeros to $\pi(\theta) \in \mathbb{R}^{m_A}$. If $\pi_I(\theta) \geq 0$ and $A(\theta)z_I(\theta) \leq b(\theta)$ for every $\theta \in \Pi$, then:

- (a) $h_c(\theta) = c^T z_I(\theta) = \pi(\theta)^T b(\theta)$ for every $\theta \in \Pi$;
- (b) $h_I(\theta) = c^T z_I(\theta)$ is continuously differentiable on an open set containing Π , and

$$\partial_j h_I(\theta) = \pi(\theta)^T (\partial_j b(\theta) - \partial_j A(\theta) z_I(\theta)). \quad (33)$$

Proof. (a) At each fixed $\theta \in \Pi$ this is Proposition 8.3 for the matrix $A(\theta)$ and the right-hand side $b(\theta)$. (b) The set where $\det A_I \neq 0$ is open and contains Π . On it, A_I^{-1} is continuously differentiable by Cramer's rule, and so are z_I and h_I . Differentiating $A_I z_I = b_I$ gives $A_I \partial_j z_I = \partial_j b_I - (\partial_j A_I) z_I$, and therefore $\partial_j h_I = c^T A_I^{-1} (\partial_j b_I - (\partial_j A_I) z_I) = \pi_I^T (\partial_j b_I - (\partial_j A_I) z_I)$. Rows outside I have zero multipliers. $\square$

Formula (33) is the derivative of the Lagrangian $c^T z + \pi^T (b(\theta) - A(\theta)z)$ with respect to θ at the primal–dual pair. For a fixed matrix it reduces to $E^T \pi$, the derivative of the certified affine value in (28); at a boundary shared by two regions the support itself need not be differentiable. For the McCormick rows (24) it is explicit. Let $\sigma = b - Az$ denote the slack of a row. For an underestimator row with corner (α, β) ,

$$\partial \sigma / \partial \alpha = \beta - x_j, \quad \partial \sigma / \partial \beta = \alpha - x_i,$$

and for an overestimator row the signs are reversed. For a square, both corner entries are the same endpoint and the two contributions add; the tangent at an endpoint e gives $\partial \sigma / \partial e = 2(e - x)$, and the secant gives $\partial \sigma / \partial \ell = x - u$ and $\partial \sigma / \partial u = x - \ell$. Since $\ell_j \leq x_j \leq u_j$, each corner entry contributes at most the row's multiplier times the width of the partner variable, and a box row contributes its multiplier. Because $p = (-\ell, u)$, derivatives with respect to a lower endpoint change sign. On a verified region, the matrix of these partial derivatives of the $2n$ supports is their sensitivity matrix. A comparison matrix must bound the sensitivity matrices of all regions that later rounds can enter.

The maximal region on which a basis is optimal is no longer a polyhedron. It is described by rational inequalities in θ . A certificate therefore verifies the conditions of Proposition 8.11 on a chosen closed polyhedron Π , such as an interval or a box of parameters. In one parameter this is exact univariate algebra. In several parameters it requires bounding rational functions over Π , for example by exact interval arithmetic on a subdivision, which is sufficient but can be conservative.

One special case reduces to a fixed matrix. If the coefficient vector of a rebuilt row changes only by a positive factor, dividing by that factor gives a row with fixed coefficients and a parameter-dependent right-hand side. For example, the tangent inequality $2tx - t^2 \leq 0$ obtained from $x^2 \leq 0$ is equivalent to $x \leq t/2$ for every tangent point $t > 0$. McCormick rows of products do not have this property in general, because the ratio between their x - and y -coefficients changes with the box.

8.5 A comparison matrix across active-set changes

Theorem 7.2 needs the comparison inequality (20) on an invariant region. A derivative computed at the current basis does not supply it, because later rounds can enter regions with larger sensitivities. The next theorem derives a two-sided form of (20) from a finite cover by regions on each of which the support is smooth, as for verified basis regions with a fixed or a rebuilt matrix.

Theorem 8.12 (Comparison matrix from a complete region cover). *Let $\mathcal{D} \subseteq \mathbb{R}^k$ be closed and convex, and let $F : \mathcal{D} \rightarrow \mathbb{R}^{2n}$. For each $i \in \{1, \dots, 2n\}$ let $C_{i,1}, \dots, C_{i,N_i}$ be closed sets whose union contains $\mathcal{D}$, such that every line segment contained in $\mathcal{D}$ meets each $C_{i,s}$ in a finite union of closed intervals, single points allowed. Suppose that for each (i, s) there is a function $h_{i,s}$, continuously differentiable on an open set containing $C_{i,s} \cap \mathcal{D}$, with $F_i = h_{i,s}$ on $C_{i,s} \cap \mathcal{D}$. If $M \geq 0$ satisfies*

$$|\partial_j h_{i,s}(\theta)| \leq M_{ij} \quad (\theta \in C_{i,s} \cap \mathcal{D}, \text{ all } i, s, j),$$

then $|F(p) - F(q)| \leq M|p - q|$ for all $p, q \in \mathcal{D}$. In particular, F satisfies (20) on $\mathcal{D}$.

The proof is in Appendix C.1.

No continuity of F is assumed separately: on each piece of the partition F_i coincides with one smooth function, and consecutive pieces share their endpoint values because they evaluate the same function F_i . For the Jacobi map (23), take $\theta = p$ and, for each direction i , regions on which a basis is verified. With a fixed matrix these are the polyhedra C_I of Proposition 8.3, each meets a segment in one interval, and M must bound $|E^T \pi|$ entrywise for the dual vectors π of all bases used. With rebuilt rows, take closed polyhedra Π verified by Proposition 8.11; then M must bound (33) over each $\Pi \cap \mathcal{D}$. A region may also use any smooth function that agrees with F_i on it; this handles parameters such as degenerate boxes, at which a basis matrix becomes singular but the support is known directly.

Coverage of $\mathcal{D}$ by finitely many polyhedra is a separate obligation. In one parameter, sorted interval endpoints decide it. In general it can be decided by finitely many exact linear programs.

Proposition 8.13 (A finite coverage test). *Let $\mathcal{D} \subset \mathbb{R}^k$ be a nonempty polytope, and let $C^r = \{\theta : H^r \theta \leq g^r\}$, $r = 1, \dots, N_C$, be polyhedra, each given by at least one inequality. For every tuple $J = (j_1, \dots, j_{N_C})$ that selects one inequality j_r of each C^r , let*

$$\tau_J = \max\{t : \theta \in \mathcal{D}, t \leq 1, t \leq H_{j_r}^r \theta - g_{j_r}^r \text{ } (r = 1, \dots, N_C)\}.$$

Then $\mathcal{D} \subseteq \bigcup_r C^r$ if and only if $\tau_J \leq 0$ for every tuple J .

The proof is in Appendix C.2.

An exact dual feasible vector of value at most zero for each tuple certifies coverage, and a feasible point with $t > 0$ exhibits an uncovered parameter. Shared boundaries give $\tau_J = 0$ and are handled correctly. The number of tuples is the product of the numbers of inequalities, so the test is a finite verification, not an efficient discovery procedure. Checking only the vertices of $\mathcal{D}$ is not sufficient: on $[-1, 1]^2$ the regions $\{\theta_1 \leq -1/2\}$ and $\{\theta_1 \geq 1/2\}$ contain all four vertices but miss the strip $|\theta_1| < 1/2$, and their one tuple gives $\tau_J = 1/2$. The test applies to each support direction separately; a complete cover for one direction says nothing about another.

Corollary 8.14 (Input for the residual theorem). *Let P be protected with threshold at most U , with endpoint vector $p_P \leq p_0$. If, for each direction i , finitely many verified regions cover $[p_P, p_0]$ and M satisfies the derivative bounds of Theorem 8.12, then (20) holds on the invariant interval $[p_P, p_0]$, and Theorem 7.2 applies with every $e \geq 0$ that satisfies (21).*

Proof. Lemma 7.1 shows that $[p_P, p_0]$ is invariant. It is closed and convex, so Theorem 8.12 applies. $\square$

The supersolution inequality (21) is then a finite exact check. If verified regions cover a convex set of pairs (p, U) , Theorem 8.12 applied with $\theta = (p, U)$ gives the stronger estimate $|F_U(p) - F_V(q)| \leq M|p - q| + h|U - V|$ on that set. Because the cutoff row has fixed coefficients, h_i is a bound on the cutoff multipliers of the bases used for direction i . Such an h also serves in Proposition 7.8, which needs it only at one fixed point.

Example 8.15 (A basis change along the trajectory). Minimize x^2 subject to $x + x^2 \leq 1$ and $0 \leq x \leq 4$. The minimizer is $x^* = 0$ with $f^* = 0$; take the cutoff $U = 0$. On $B = [0, u]$, relax $s = x^2$ by the McCormick rows $s \geq 0$, $s \geq 2ux - u^2$, $s \leq ux$, and keep the coupling row $x + s \leq 1$ and the cutoff $s \leq 0$. The lower endpoint stays at 0, since $(0, 0)$ is feasible on every box. The cutoff and $s \geq 0$ force $s = 0$, so the tangent row gives $x \leq u/2$ and the coupling row gives $x \leq 1$. Hence the upper support is

$$F(u) = \min\{u/2, 1\}, \quad 0 \leq u \leq 4.$$

For $0 < u \leq 2$ the basis of the tangent row $2ux - s \leq u^2$ and the cutoff has primal point $(u/2, 0)$ and multipliers $(1/(2u), 1/(2u))$; all other rows hold because $u/2 \leq 1$. Formula (33) gives $\partial F/\partial u = (1/(2u))(2u - 2 \cdot u/2) = 1/2$, which agrees with direct differentiation of $u/2$. At $u = 0$ the box is $\{0\}$ and $F(0) = 0$, so the smooth function $u/2$ represents F on the closed region $[0, 2]$. For $2 \leq u \leq 4$ the basis of $x + s \leq 1$ and $-s \leq 0$ has primal point $(1, 0)$ and multipliers $(1, 1)$; the tangent row holds because $2u \leq u^2$. Its rows do not depend on u , so the slope is 0.

The degenerate box $P = \{0\}$ is protected at cutoff 0, with the lifted witness $(0, 0)$, and Corollary 8.14 applies on $\{(0, u) : 0 \leq u \leq 4\}$, which the two regions cover. Every point of this interval has $\ell = 0$, and the lower support is identically 0. Hence $M = \text{diag}(0, 1/2)$ satisfies (20) on it. From $u_0 = 4$, the exact iterates are $4, 1, 1/2, 1/4, \dots$. The first residual is $d = (0, 3)$, and $e = (0, 6)$ satisfies (21). Theorem 7.2 bounds the total motion by 6 and the motion after the first round by $Me = 3$; the actual values are 4 and 1.

The observed displacements 3 and $1/2$ have ratio $1/6$. A geometric extrapolation with that ratio would bound the motion after the first round by $(1/2)/(1 - 1/6) = 3/5$, but the actual motion is 1: the first round used the coupling row, and later rounds use the tangent row with ratio $1/2$. This is a McCormick instance of the failure shown abstractly in Example 7.9 and Proposition 7.10. The frozen rows of $[0, 4]$ give the support 1 for every $u \in [1, 4]$, so their comparison matrix on that interval is zero. Used in Theorem 7.2, it would assert that no motion remains after the first round, whereas the rebuilt iterates continue to 0.

Complete covers are practical when the parameter set is low-dimensional, for example a scalar cutoff or a few endpoints that actually move. The number of basis regions is bounded only by the number of bases, which grows combinatorially with the number of rows, and the coverage test of Proposition 8.13 has a combinatorial number of tuples. Theorem 8.12 is therefore a verifiable sufficient condition, not an inexpensive general method.

8.6 An affine equality changes the rate

Assumption 4.3 requires finite relaxation values on the comparison boxes. A projected relaxation that keeps an equality constraint exactly is infinite off the feasible affine set, so the constrained case needs a restricted tangent construction, as observed in Proposition 4.14. The following example computes the restricted map exactly.

Consider $f(x, y) = (x - y)^2 = x^2 + y^2 - 2xy$ on boxes $[-r, r]^2$. Keep the squares exact and replace xy by its McCormick overestimator $\min\{r^2 + r(x - y), r^2 - r(x - y)\} = r^2 - r|x - y|$. The relaxed objective is

$$\phi_r(x, y) = x^2 + y^2 - 2r^2 + 2r|x - y|.$$

Without further constraints, the points (r, r) and $(-r, -r)$ have $\phi_r = 0$ and are true minimizers, so at cutoff 0 every box face is retained and OBBT stalls.

Proposition 8.16 (Exact map with an affine equality). *Impose the equality $y = -x$ exactly in the relaxation, and let $U \geq 0$. A Jacobi round maps $[-r, r]^2$ to $[-G_U(r), G_U(r)]^2$, where*

$$G_U(r) = \min\{r, \sqrt{2r^2 + U/2} - r\}.$$

Put $a = \sqrt{U}/2$ and $r_{k+1} = G_U(r_k)$.

(a) If $U = 0$, then $r_{k+1} = (\sqrt{2} - 1)r_k$.

(b) If $U > 0$ and $r_0 > a$, then r_k decreases strictly to a , and

$$r_{k+1} - a = \frac{(r_k - a)^2}{\sqrt{2r_k^2 + 2a^2} + r_k + a} \leq \min\left\{\frac{r_k - a}{2}, \frac{(r_k - a)^2}{4a}\right\}.$$

(c) If $U > 0$ and $r_0 \leq a$, then $r_k = r_0$ for all k .

Proof. On the equality put $(x, y) = (t, -t)$. Then $\phi_r(t, -t) = 2t^2 + 4r|t| - 2r^2$, and the cutoff $2|t|^2 + 4r|t| - 2r^2 \leq U$ holds exactly when $|t| \leq \sqrt{2r^2 + U/2} - r$, the nonnegative root. The box adds $|t| \leq r$. Both coordinates range over $[-G_U(r), G_U(r)]$, which proves the formula for the map. For $U = 0$ the root is $(\sqrt{2} - 1)r$.

For $U > 0$ we have $U/2 = 2a^2$. Put $X = 2r^2 + 2a^2$. Since $X - (r + a)^2 = (r - a)^2 \geq 0$, we get $\sqrt{X} \geq r + a$ and

$$\sqrt{X} - r - a = \frac{X - (r + a)^2}{\sqrt{X} + r + a} = \frac{(r - a)^2}{\sqrt{X} + r + a}.$$

The root is smaller than r exactly when $X < 4r^2$, that is, when $r > a$. Hence for $r > a$ the map equals the root, lies in (a, r) , and satisfies the displayed identity. The denominator is at least $2(r + a)$, which is at least both $2(r - a)$ and $4a$; this gives the two bounds, and with them strict decrease to a . If $r \leq a$, the root is at least r , so the box is fixed. $\square$

For $U > 0$ and every starting radius r_0 , the limiting radius is $\min\{r_0, a\}$. This is the radius of the original cutoff set within the initial box: on $y = -x$ the objective is $4x^2$, and $4x^2 \leq U$ exactly when $|x| \leq a$. OBBT therefore converges to the box hull of the original cutoff set. When $r_0 > a$ it does not reach it in finitely many rounds, since $r_{k+1} - a > 0$ whenever $r_k > a$, but the error decreases quadratically, whereas in the unconstrained two-variable example the error at a positive cutoff decreases only linearly (Proposition 4.18(d)). At zero cutoff the contraction factor is $\sqrt{2} - 1$. Eliminating the equality before relaxing gives the convex objective $4x^2$, whose exact relaxation

reaches radius $\min\{r, a\}$ in one round. The rate is a property of the formulation and the relaxation, not of the constrained problem alone.

Proposition 4.14 in Section 4.3 gives the general restricted tangent contraction result. Nonlinear equalities require control of their curvature and of the relaxation remainder, as in the feasible-repair analysis below.

8.7 Nonlinear constraints through a feasible repair

For nonlinear constraints, a width bound can be obtained without guessing which constraints will be active at the next support solutions. Each relaxed point is moved to a nearby original feasible point, and feasible quadratic growth then bounds its distance to the minimizer. This transfers a growth condition on feasible points to infeasible relaxed points through a constraint error bound, as in exact-penalty arguments. The OBBT content is the explicit list of uniform constants and their consequence for the iteration.

Let $S = X \cap B_0$ be the original feasible set, including any integrality, and let $x^* \in S$ be a global minimizer with value f^* . Let $\mathfrak{B}^* \subseteq \mathfrak{B}$ be a set of boxes such that $x^* \in B$ and $\phi_B(x^*) \leq f^*$ for every $B \in \mathfrak{B}^*$, and fix $U = f^* + \epsilon$ with $\epsilon \geq 0$. Let $\nu \geq 0$ be a residual function, defined on the boxes of $\mathfrak{B}^*$ and zero on S , such as the maximum violation of the original constraints. Assume constants $\tau, \beta, \kappa, L \geq 0$, $\mu > 0$, and $\alpha \in (0, 1]$ such that every $B \in \mathfrak{B}^*$ and every $x \in K_U(B)$ satisfy

$$\phi_B(x) \geq f(x) - \tau w(B)^2, \quad \nu(x) \leq \beta w(B)^2, \quad (34)$$

and x has a *feasible repair* $y \in S$ with

$$\|x - y\|_\infty \leq \kappa \nu(x)^\alpha, \quad (35)$$

$$f(y) \leq f(x) + L\|x - y\|_\infty, \quad (36)$$

$$f(y) \geq f^* + \mu\|y - x^*\|_\infty^2. \quad (37)$$

Inequality (35) is a Hölder error bound for the constraint system at x . For a polyhedral feasible set $S = \{x : Gx \leq g\}$ and $\nu(x) = \max_j (G_j x - g_j)_+$, Hoffman's theorem [21] gives such a bound with $\alpha = 1$ and a constant κ that depends only on G . Inequality (36) holds if $\|\nabla f\|_1 \leq L$ on a convex set containing x and y ; only the one-sided estimate on the repair pair is needed. Inequality (37) is feasible quadratic growth. The repair need not lie in B . If every box of $\mathfrak{B}^*$ has width at most $\bar{w}$, every repair lies within $\rho = \bar{w} + \kappa(\beta\bar{w}^2)^\alpha$ of x^* , so it suffices that the Lipschitz and growth estimates hold on that neighborhood. These are assumptions about every relaxed point that satisfies the cutoff, including points that violate the original constraints. A small KKT residual at a support optimizer does not establish them. Without constraints one may take $\nu \equiv 0$ and $y = x$; then (34) and (37) combine to the hypothesis $\phi_B(x) \geq f^* + \mu\|x - x^*\|_\infty^2 - \tau w(B)^2$ of Proposition 4.2, and the next theorem reduces to it.

Theorem 8.17 (Width bound from a feasible repair). *For every $B \in \mathfrak{B}^*$, with $w = w(B)$ and $r = \kappa(\beta w^2)^\alpha$,*

$$w(T_U(B)) \leq 2r + 2\sqrt{\frac{\epsilon + \tau w^2 + Lr}{\mu}}. \quad (38)$$

Proof. Validity at x^* gives $x^* \in K_U(B)$, so the set is nonempty. Let $x \in K_U(B)$ with repair y . By (34), $f(x) \leq \phi_B(x) + \tau w^2 \leq f^* + \epsilon + \tau w^2$ and $\nu(x) \leq \beta w^2$, so (35) gives $\|x - y\|_\infty \leq r$. Then (36) gives $f(y) \leq f^* + \epsilon + \tau w^2 + Lr$, and (37) gives $\|y - x^*\|_\infty \leq \sqrt{(\epsilon + \tau w^2 + Lr)/\mu}$. By the triangle inequality every point of $K_U(B)$ lies within r plus this radius of x^* in every coordinate, so each coordinate range of $T_U(B)$ is at most twice that amount. $\square$

Corollary 8.18 (Linear repair). *Let $\alpha = 1$, put $K = \kappa\beta$ and $\lambda_0 = 2\sqrt{(\tau + LK)/\mu}$. For every $B \in \mathfrak{B}^*$,*

$$w(T_U(B)) \leq (\lambda_0 + 2Kw(B))w(B) + 2\sqrt{\epsilon/\mu}. \quad (39)$$

Suppose $\mathfrak{B}^$ contains every subbox of B_0 that contains x^* and has width at most $\bar{w}$, and $\lambda = \lambda_0 + 2K\bar{w} < 1$. Then Jacobi iterates $B_{k+1} = T_U(B_k)$ starting from such a box B'_0 satisfy*

$$w(B_k) \leq \lambda^k w(B'_0) + 2\sqrt{\epsilon/\mu} \frac{1 - \lambda^k}{1 - \lambda}, \quad \limsup_{k \rightarrow \infty} w(B_k) \leq \frac{2\sqrt{\epsilon/\mu}}{1 - \lambda}. \quad (40)$$

The same bounds hold for any update that replaces B_k by a box containing x^ and contained in $T_U(B_k)$, such as a complete sequential round (Lemma 3.2(b)).*

Proof. With $r = Kw^2$, inequality (38) and $\sqrt{s+t} \leq \sqrt{s} + \sqrt{t}$ give (39). The iterates are nested, contain x^* by validity, and therefore remain in $\mathfrak{B}^*$ with width at most $\bar{w}$; so $w(B_{k+1}) \leq \lambda w(B_k) + 2\sqrt{\epsilon/\mu}$. Induction proves the first bound in (40), and letting $k \rightarrow \infty$ proves the second. The final statement uses only the containment in $T_U(B_k)$. $\square$

At zero cutoff slack the widths contract by at least the factor λ in every round, for every box shape in $\mathfrak{B}^*$, whichever constraints are active at the support solutions. At positive slack the limit bound is an upper bound, not an exact floor. The corollary is a statement of *guaranteed progress*: whenever $w(B_k) > 2\sqrt{\epsilon/\mu}/(1 - \lambda)$, the next round reduces the width by at least $(1 - \lambda)w(B_k) - 2\sqrt{\epsilon/\mu} > 0$. It is therefore complementary to the protected boxes of Section 5 and to Theorem 7.2, which bound the remaining benefit from above. The two are consistent: a protected box $P \in \mathfrak{B}^*$ satisfies $w(P) = w(T_U(P)) \leq \lambda w(P) + 2\sqrt{\epsilon/\mu}$, so its width is at most $2\sqrt{\epsilon/\mu}/(1 - \lambda)$, and at zero slack $P = \{x^*\}$. The constants are analytic premises. They involve x^* and f^* and are not computed by any procedure in Section 9.

For $\alpha < 1$ write $r = \kappa\beta^\alpha w^{2\alpha}$. At zero slack and $L\kappa\beta > 0$, the bound (38) contains the term $2\sqrt{L\kappa\beta^\alpha/\mu} w^\alpha$, whose ratio to w is unbounded as $w \downarrow 0$, so this argument does not certify linear contraction. It does not prove a stall either. A local repair constant $L_B = O(w^{2-2\alpha})$ would make the square-root term $O(w)$, but the term $2r$ is $O(w)$ only if $\alpha \geq 1/2$; for $\alpha < 1/2$ the argument needs a stronger repair estimate.

Example 8.19 (A nonlinear equality with explicit constants). Let

$$S = \{(x, y) : y = x^2\} \cap ([-1/4, 1/4] \times [0, 1/16]), \quad f(x, y) = x^2 + y^2 - xy/16.$$

Both x and y are original variables. For boxes $B = [\ell, u] \times [0, \bar{y}]$ with $-1/4 \leq \ell \leq 0 \leq u \leq 1/4$ and $0 \leq \bar{y} \leq 1/16$, use the box rows, the graph relaxation

$$x^2 \leq y \leq (\ell + u)x - \ell u,$$

and the relaxed objective

$$\phi_B(x, y) = x^2 + y^2 - \frac{1}{16}m_B(x, y), \quad m_B(x, y) = \min\{uy, \ell y + \bar{y}(x - \ell)\},$$

where m_B is the McCormick overestimator of xy on B . Then ϕ_B is convex, as the maximum of two convex quadratics, and the relaxed feasible set is convex.

Minimizer. On S , $f(x, x^2) = x^2(1 - x/16 + x^2) \geq (63/64)x^2$, because $x \leq 1/4$. Since $|x| \leq 1/4$ implies $x^2 \leq |x|$, this equals $(63/64)\|(x, x^2)\|_\infty^2$. Hence $(0, 0)$ is the unique minimizer, $f^* = 0$, and (37) holds on S with $\mu = 63/64$.

Validity and monotonicity. For $x \in [\ell, u]$, $(\ell + u)x - \ell u - x^2 = (x - \ell)(u - x) \geq 0$, so the secant is valid, and $m_B - xy = \min\{(u - x)y, (x - \ell)(\bar{y} - y)\} \geq 0$ on B , so $\phi_B \leq f$ on $S \cap B$. For a nested box $B' = [\ell', u'] \times [0, \bar{y}'] \subseteq B$, the factors $x - \ell' \leq x - \ell$ and $u' - x \leq u - x$ show that the new secant is no higher on B' . Also $u'y \leq uy$, and

$$\ell'y + \bar{y}'(x - \ell') \leq \ell'y + \bar{y}(x - \ell') \leq \ell y + \bar{y}(x - \ell),$$

using $x \geq \ell'$ and $\bar{y}' \leq \bar{y}$ for the first inequality and $(\ell' - \ell)(y - \bar{y}) \leq 0$ for the second. Thus $m_{B'} \leq m_B$ and $\phi_{B'} \geq \phi_B$ on B' , and the relaxed feasible set shrinks: the whole construction is monotone. The origin is relaxed feasible and every relaxed point has $y \geq 0$, so OBBT keeps the lower y -bound at 0, and all iterates stay in this family of boxes and contain the origin.

Residual, repair, and objective error. Let $\nu(x, y) = |y - x^2|$. Every relaxed point has the repair $(x, x^2) \in S$, at distance $\delta = y - x^2$ in the infinity norm, and

$$0 \leq \delta \leq (x - \ell)(u - x) = \frac{(u - \ell)^2}{4} - \left(x - \frac{\ell + u}{2}\right)^2 \leq \frac{1}{4}w(B)^2.$$

So $\beta = 1/4$, $\kappa = 1$, and $\alpha = 1$. With $s_1 = u - x$, $s_2 = x - \ell$, $t_1 = y$, and $t_2 = \bar{y} - y$, all nonnegative on B , the arithmetic-geometric mean inequality gives $m_B - xy = \min\{s_1 t_1, s_2 t_2\} \leq \sqrt{s_1 s_2 t_1 t_2} \leq (u - \ell)\bar{y}/4 \leq w(B)^2/4$, so $f - \phi_B \leq w(B)^2/64$ and $\tau = 1/64$. Finally,

$$f(x, x^2) - f(x, y) = \delta \left(\frac{x}{16} - 2x^2 - \delta \right) = \delta \left(\frac{1}{2048} - 2\left(x - \frac{1}{64}\right)^2 - \delta \right) \leq \frac{\delta}{2048},$$

so (36) holds with $L = 1/2048$.

Rate. With $K = 1/4$, $\lambda_0^2 = 4(\tau + LK)/\mu = 43/672 < (13/48)^2$. Every admitted box has $w(B) \leq 1/2$, so (39) gives, for all admitted boxes,

$$w(T_U(B)) \leq \left(\lambda_0 + \frac{w(B)}{2} \right) w(B) + 2\sqrt{\frac{64\epsilon}{63}},$$

with factor at most $13/48 + 1/4 = 25/48$ for every admitted box and at most $13/48 + 1/16 = 1/3$ once $w(B) \leq 1/8$. At zero cutoff, Corollary 8.18 certifies the contraction factor $25/48$ per round from any admitted starting box, and the factor $1/3$ once the width is at most $1/8$.

The example certifies a uniform rate across all box shapes without choosing an active constraint: at a support solution either the graph epigraph or the secant may be active, and the repair treats every relaxed point alike. The constants are sufficient, not exact rates. The constraint need not be relaxed exactly. If $x^2 \leq y$ is replaced by its endpoint tangents $y \geq 2\ell x - \ell^2$ and $y \geq 2ux - u^2$, the construction remains monotone, because on a smaller interval $[\ell', u']$ the new tangents dominate the old ones: $(2\ell'x - \ell'^2) - (2\ell x - \ell^2) = (\ell' - \ell)(2x - \ell' - \ell) \geq (\ell' - \ell)^2$ for $x \geq \ell'$, and symmetrically at the upper endpoint. Relaxed points may now lie below the graph, but still $|y - x^2| \leq w(B)^2/4$. For $\delta < 0$ the same identity gives $f(x, x^2) - f(x, y) = |\delta|(2x^2 - |\delta| - x/16) \leq (9/64)|\delta|$, so $L = 9/64$, $\lambda_0^2 = 13/63 < 1/4$, and the contraction factor is at most $3/4$ for every admitted box. The objective squares, however, are kept exact. If they are also replaced by endpoint tangents, the generic bound on the error of each square is $w(B)^2/4$; with $\tau \geq 1/4$ the constant λ_0 exceeds 1, so this sufficient estimate certifies no contraction. That does not show that the resulting OBBT operator fails to contract. The tangent-map analysis of Section 4 is the appropriate tool for such finite linear relaxations.

8.8 Summary

Table 2 lists the information discussed in this section, the changes after which it remains valid, and what it proves. Observed progress is not on the list, for the reasons given in Section 7.7.

Table 2: Stored information for constrained support problems and the scope of its conclusions.

Information	Remains valid after	Conclusion	Additional verification
Dual vector, fixed matrix (Prop. 8.1)	any change of the right-hand-side parameter	upper bound on the support; guaranteed reduction	exact dual feasibility
Optimal dual with cutoff multiplier η (Cor. 8.2)	a lower cutoff, same box	reduction at least $\eta(U_0 - U)$	exact optimality at U_0
Basis region (Prop. 8.3)	parameters in the region	exact affine support	all primal residuals on the region
Retained witness (Prop. 8.5)	a cutoff at least its objective, same box and rows	lower bound; limits the possible reduction	recheck all rows after any other change
Least-objective optimal witness (Prop. 8.6)	a lower cutoff, same box	exact no-change threshold $\underline{U}$	one additional solve
Frozen-row dual (Lem. 8.8)	a smaller box with rebuilt rows	valid bound, but no progress after one round	lifted monotonicity
Corrected stored dual (Prop. 8.10)	a smaller box with rebuilt rows	valid upper bound	bounds on columns with changed coefficients
Complete region cover (Thm. 8.12)	all parameters in $\mathcal{D}$	comparison matrix for Theorem 7.2	coverage test, derivative bounds, invariance
Repair constants (Thm. 8.17)	all boxes of $\mathfrak{B}^*$	width recurrence; guaranteed progress	uniform error, repair, and growth bounds

9 Algorithms

The certificates of the preceding sections concern an exact operator. Using them requires an algorithm that computes coordinate supports, decides when to stop, and turns floating-point output into valid bounds. This section describes two implementations with different contracts.

The first is an exact reference procedure for a declared rational relaxation family. It commits an OBBT round only after every endpoint has been proved optimal, and it stops with a fixed box only when the finite protected-box certificate of Section 5 holds. The second is a numerical policy that runs inside a branch-and-bound solver. Each bound it applies is justified by a dual bound evaluated with directed rounding, but its cutoff comes from a numerically accepted incumbent, and its decisions about where to spend auxiliary LPs are heuristic. Table 3 summarizes the difference; the experiments of Section 11 evaluate only the second implementation.

9.1 The reference relaxation family

The exact procedure works with the reference family of Section 5 and rational data. A lifted point is $z = (x, y)$, where y_k stands for a product $x_{i_k} x_{j_k}$ and $i_k = j_k$ is allowed. On a rational box $B = [\ell, u]$ the relaxation $R(B)$ consists of fixed rational rows $Gz \leq g$, the bound rows $\ell \leq x \leq u$, and the four McCormick rows of every product; for a square these are the two endpoint tangents and the secant. The objective is a rational affine function $v(z) = c^\top z + c_0$, and $R_U(B) = \{z \in R(B) : v(z) \leq U\}$. By Lemma 5.1, the family is lifted monotone with a common objective, and each $R(B)$ is a polytope. The infimum in the projected objective (2) is therefore attained whenever the fiber over x is nonempty, so that

$$K_U(B) = \pi(R_U(B)), \quad T_U(B) = \text{box } \pi(R_U(B)), \quad (41)$$

	Exact reference procedure	Numerical solver policy
Relaxation	Fixed rational rows and McCormick product rows, rebuilt on each box	Outward-rounded local LP with a retained tangent pool, built at each callback
Endpoint proof	Exactly verified primal–dual pair	Dual bound with a residual term, evaluated with directed rounding
Cutoff	A given rational number U	A numerically accepted incumbent value, rounded upward
Stopping	Protected fixed box, round budget, or failed proof	Work budgets, triggers, screening, and a two-LP pilot
On failure	Inconclusive; earlier rounds are kept	No bound is proposed; native search continues
Guarantee	Every committed box and the stopping certificate	Each applied bound, conditional on the cutoff

Table 3: The two implementations. Only the numerical policy was evaluated in the computational comparison.

where $\pi(z) = x$. The endpoints of $T_U(B)$ are thus the optimal values of the $2n$ LPs that minimize and maximize each x_i over $R_U(B)$.

Validity for an original problem is a premise that the caller must establish. The McCormick rows supply valid lifts of the products, but the fixed rows and the objective must be valid on the original sublevel set and on the domain concerned. Equality of stored row data does not prove where a row is valid.

9.2 Exact endpoint certificates

Each directional LP of the reference family has the form $\min\{d^\top z : Az \leq b\}$, where the rows $Az \leq b$ list the fixed rows, the bound rows, the McCormick rows, and the cutoff row, and $d = \pm e_i$ is a signed unit vector. The following certificate is LP weak duality.

Lemma 9.1 (Exact LP optimality certificate). *Let $A \in \mathbb{Q}^{p \times N}$, $b \in \mathbb{Q}^p$, and $d \in \mathbb{Q}^N$. Suppose $\bar{z} \in \mathbb{Q}^N$ and $\lambda \in \mathbb{Q}^p$ satisfy*

$$A\bar{z} \leq b, \quad \lambda \leq 0, \quad A^\top \lambda = d, \quad d^\top \bar{z} = b^\top \lambda. \quad (42)$$

Then $\bar{z}$ minimizes $d^\top z$ subject to $Az \leq b$, and the optimal value is $d^\top \bar{z}$.

Proof. Let z satisfy $Az \leq b$. Multiplying the r th inequality by $\lambda_r \leq 0$ reverses it, so $\lambda^\top Az \geq \lambda^\top b$. Hence $d^\top z = \lambda^\top Az \geq b^\top \lambda = d^\top \bar{z}$. Since $\bar{z}$ is itself feasible, it is optimal. $\square$

Every condition in (42) is a finite set of rational inequalities and identities. Checking them uses no tolerance and no solver status. Because the bound rows are explicit rows of A , the multiplier λ also carries their dual information; the certificate therefore does not depend on how a numerical solver treats variable bounds. A floating-point LP solver supplies only a proposal $(\tilde{z}, \tilde{\lambda})$. The reference implementation replaces each proposed floating-point number by the rational number with denominator at most $D = 10^9$ that is closest to the number’s shortest decimal representation, and then checks (42) exactly. Checking floating-point proposals exactly after rational reconstruction is an established technique of exact linear programming [1]. A failed check rejects the proposal. It says nothing about the LP itself.

The lemma needs an optimal solution. If $R_U(B)$ is empty, no certificate of this form exists. The procedure below then stops without a conclusion. It does not construct a Farkas certificate, so it never declares a box infeasible.

9.3 Certified closure

The certified closure procedure takes the family data, a rational box B_0 , a rational cutoff U , a round budget $K \geq 0$, and a flag for an optional objective proposal. It keeps a list of committed rounds and a counter of proposal requests.

1. Set $B \leftarrow B_0$ and $k \leftarrow 0$.
2. While $k < K$:
 - (a) Build the rows of $R_U(B)$.
 - (b) For each $i = 1, \dots, n$, request proposals for $\min\{x_i : z \in R_U(B)\}$ and for $\min\{-x_i : z \in R_U(B)\}$, reconstruct rational values, and check (42). Call the verified optima $z^{(i,-)}$ and $z^{(i,+)}$. If any request returns no proposal, or any check fails, return *inconclusive* with the current box B ; the partial round is discarded.
 - (c) Commit the round: $B' = [\ell', u']$ with $\ell'_i = z_i^{(i,-)}$ and $u'_i = z_i^{(i,+)}$. Set $B \leftarrow B'$ and $k \leftarrow k + 1$.
 - (d) Check exactly whether all $2n$ lifted points $z^{(i,\pm)}$, with every auxiliary coordinate, satisfy every row of $R_U(B')$, the relaxation rebuilt on the new box. If they do, optionally request a proposal for $\min\{v(z) : z \in R_U(B')\}$; if its reconstruction satisfies every row of $R_U(B')$ exactly, add it to the points. Return *fixed* with box B' and the lifted pool $\widehat{W}$ of checked points. Otherwise continue with the next round.
3. Return *unfinished* with the current box B .

Step 2(d) is the test of Proposition 5.17(a), applied to the complete lifted optima of the round, which lie in $R_U(B)$ and whose projections attain all endpoints of B' . The reported cost includes every proposal request, including failed ones, and the elapsed time of all exact checks. The optional objective proposal is never needed for a stopping decision.

Theorem 9.2 (Outcomes of certified closure). *Run the procedure on the reference family with rational data. Write B_k for the box after k committed rounds.*

- (a) Exact rounds. *Every committed round is the exact Jacobi round: $B_{k+1} = T_U(B_k) \subseteq B_k$. Hence every returned box equals $T_U^k(B_0)$, where k is the number of committed rounds.*
- (b) Fixed status. *If the procedure returns fixed with box P and lifted pool $\widehat{W}$, then $\pi(\widehat{W})$ is a witness pool for P with threshold at most $U_{\widehat{W}} = \max_{z \in \widehat{W}} v(z) \leq U$, and P is the greatest fixed box of T_U in B_0 and the limit of the exact Jacobi iterates. Every sequence of relaxation-sound steps that starts from a box $C \supseteq P$ and uses cutoffs at least $U_{\widehat{W}}$ retains P , and the relaxation bound of every box $C \supseteq P$ is at most $\min_{z \in \widehat{W}} v(z)$.*
- (c) Prompt detection. *A committed round whose input box is a fixed box of T_U returns fixed. Consequently every committed round that does not return fixed strictly shrinks the box, and if $T_U(B_m) = B_m$, the budget allows $m + 1$ rounds, and all proposals in these rounds pass their checks, then the procedure returns fixed after at most $m + 1$ rounds.*

- (d) Other statuses. *If the procedure returns unfinished or inconclusive, the returned box is the last committed iterate. No claim is made about the uncommitted round. In particular, a missing proposal does not certify infeasibility.*
- (e) Validity. *If the family is valid for an original problem on B_0 , every returned box contains every original feasible point in B_0 with objective at most U .*

Proof. (a) By Lemma 9.1, $z^{(i,-)}$ minimizes and $z^{(i,+)}$ maximizes x_i over $R_U(B_k)$. By (41), these optimal values are the endpoints of $T_U(B_k)$, so the committed box is $T_U(B_k)$. The bound rows of $R(B_k)$ give $B_{k+1} \subseteq B_k$. Induction over committed rounds gives the second statement; failed rounds change nothing.

(b) Let $P = B_{k+1} = T_U(B_k)$ be the box just committed. The verified optima $z^{(i,\pm)}$ lie in $R_U(B_k)$, and by part (a) their projections attain all endpoints of P . The check in step 2(d) is the hypothesis of Proposition 5.17(a), which therefore makes these projections a witness pool for P . An accepted objective point satisfies every row of $R_U(P)$, including the bound rows, so its projection lies in $K_U(P)$, and adding it leaves the hull equal to P . Every point z of $\widehat{W}$ lies in $R(P)$, so $\phi_P(\pi(z)) \leq v(z)$, which bounds the threshold by $U_{\widehat{W}}$. Theorem 5.7 gives $T_U(P) = P$ and the persistence statement; since the reference family is defined on every box, the theorem applies with any box containing C in the role of B_0 . By Lemma 3.6, every Jacobi iterate contains the greatest fixed box P_U in B_0 , so $P_U \subseteq P$, and $P \subseteq P_U$ because P is a fixed box in B_0 . All later Jacobi iterates equal P . The bound on the relaxation value is Corollary 5.12, applied to the points of $\widehat{W} \subseteq R(P)$.

(c) If $T_U(B_k) = B_k$, the round commits $B_{k+1} = B_k$. The rows of $R_U(B_{k+1})$ are then exactly those of $R_U(B_k)$, because the construction depends only on the box and the fixed data, and the verified optima belong to $R_U(B_k)$. The check therefore passes; this is Proposition 5.17(b). If a committed round does not return *fixed*, its input box is therefore not fixed, and $B_{k+1} = T_U(B_k) \neq B_k$. For the last statement, either the check succeeds earlier, or round $m+1$ starts at the fixed box B_m .

(d) This follows from (a): a round changes the box only when it is committed.

(e) By Lemma 3.1(a), each original feasible point $x \in B_k$ with objective at most U lies in $T_U(B_k) = B_{k+1}$. Induction from B_0 proves the claim. $\square$

Part (b) states what the stopping certificate guarantees: the returned box is the greatest fixed box and the limit of exact Jacobi iteration at cutoff U , and no order of exact or conservative coordinate updates in this family, at cutoffs at least $U_{\widehat{W}}$, can remove any part of it. Because $U_{\widehat{W}}$ can lie below U , the guarantee survives incumbent improvements down to that value. The relaxation-value ceiling bounds the possible improvement of this relaxation's lower bound; it does not bound the gain from cuts, integrality reasoning, or a different relaxation. Part (c) shows that the check after a round is at most one round late. It can be late: the verified optima of a round can violate a row rebuilt on the new box although that box is fixed, as in Example 5.18, and only the next round, which starts at the fixed box, produces points that pass.

Every run stops, because the budget K is finite, but no finite budget guarantees the status *fixed*. That status requires exact iterates that reach a fixed box, and the square examples below never do. Rational reconstruction from double-precision proposals can also fail on well-behaved LPs once the exact iterates need many digits. The procedure then reports *inconclusive* and keeps the last committed box. An exact rational LP solver could replace the floating-point proposals without changing the checks. It would remove reconstruction failures, but not the growth of the iterates: in Heron's iteration below, the number of digits roughly doubles in every round.

Table 4 records the archived outcomes of the reference implementation on small fixtures. They illustrate the three statuses; they are correctness checks, not performance evidence. The first fixture is Example 5.13: the cube is fixed at cutoff 0, so the first round starts at a fixed box, and the

objective proposal certifies the ceiling -3 , the relaxation bound, while the original optimum is 0. The linear fixture has the rows $x_1 + x_2 \geq 1$ and $x_1 = x_2$ on $[0, 2]^2$. In the square fixtures, the cutoff 0 halves the upper endpoint in every round, as in Example 5.4, and never reaches the fixed box $\{0\}$. With cutoff 1 the upper endpoint follows Heron’s iteration of Example 7.6, 2, $5/4$, $41/40$, $3281/3280$; the next iterate is $21523361/21523360$, and the denominators roughly square in every round. In the fourth round, the reconstructed proposal for the upper direction violated a row exactly, and the procedure returned *inconclusive* with the third box.

Fixture	Committed rounds	Proposals	Status	Returned box
Three-variable stall, cutoff 0	1	7	fixed	$[-1, 1]^3$, ceiling -3
Linear coupling	1	4	fixed	$[1/2, 2]^2$
Interval with negative bounds	1	2	fixed	$[-3, -1]$
Square, cutoff 0, budget 5	5	10	unfinished	$[0, 1/32]$
Square, cutoff 1, budget 3	3	6	unfinished	$[0, 3281/3280]$
Square, cutoff 1, budget 12	3	8	inconclusive	$[0, 3281/3280]$

Table 4: Archived runs of the certified closure procedure. The square fixtures start at $[0, 1]$ (cutoff 0) and $[0, 2]$ (cutoff 1). The stall count includes one objective proposal. Further archived cases inject an invalid dual vector, a missing proposal after one committed round, and an invalid objective proposal; they return *inconclusive* with no committed round, *inconclusive* with the first committed box, and *fixed* with the certificate unchanged, respectively. A zero round budget returns *unfinished* without a proposal.

9.4 A numerical policy inside branch and bound

The second implementation adds a propagator to SCIP for quadratically constrained problems with quadratic objectives. It leaves SCIP’s own propagation, including its LP-based OBBT [15], unchanged. At selected nodes it builds a separate LP relaxation of the current node box, solves some directional LPs with a floating-point solver, and proposes valid local bounds. The time that the propagator records for this work is its *propagator time*. The propagator never uses an LP status to prune a node.

9.4.1 Local relaxation

Let $B = [\ell, u]$ be the current local box of the original variables, read from SCIP and intersected with the original bounds; every bound must be finite, otherwise the callback records an unsupported box and returns. The product terms are the distinct pairs (i, j) , $i \leq j$, that occur in the objective or a row. The lifted LP has one variable y_{ij} per term and these rows:

- the four McCormick rows (14) for each $i \neq j$;
- for each square, tangent rows $y_{ii} \geq 2px_i - p^2$ at five equally spaced points p of $[\ell_i, u_i]$ and at retained points from earlier callbacks, and the secant $y_{ii} \leq (\ell_i + u_i)x_i - \ell_i u_i$;
- every original row, with each product replaced by its lifted variable;
- if a cutoff U_c is available, the objective in minimization form, with products replaced, bounded by U_c .

The variable bounds are $x \in B$ and, for each term, an interval containing every product value on B : the four endpoint products for $i \neq j$, and $[\min_{s \in [\ell_i, u_i]} s^2, \max\{\ell_i^2, u_i^2\}]$ for a square. These intervals are computed in floating point and widened by one unit in the last place in each direction. A tangent is valid for the square everywhere, so a retained point outside the current interval still yields a valid row. At every callback the five points of the current interval are offered to a pool shared by all nodes, which keeps the first 25 distinct points of each square and accepts no more.

The relaxation is frozen during a callback. Bounds applied in that callback change SCIP's domains but not the auxiliary LP, so a callback performs part of a frozen round; the next callback rebuilds. The relaxations of different callbacks need not form a monotone family, for two reasons. First, the five tangent points move with the box, and such box-relative points violate monotonicity on their own: for $f(x) = x^2$ these rows alone give $T_{1/4}([-1, 1]) = [-1/2, 1/2]$ but $T_{1/4}([-1/2, 1]) = [-1/2, 41/80]$ (Section 3). The pool restores the tangents of earlier boxes only while it has room. Second, tangents that the pool gains later are rows that were absent when lifted points were checked on an earlier box, and they can cut those points. The future-round results therefore do not apply to this sequence of relaxations.

9.4.2 Valid bounds from floating-point data

The model data are stored in binary floating point. The implementation treats each stored number as the exact dyadic rational it represents, constructs every new row exactly, and then rounds its coefficients to floating point with a correction of the right-hand side.

Each candidate uses the dual bound of Proposition B.2, evaluated with directed rounding as justified by Lemma B.3, for an LP whose rows are rounded outward using Proposition B.1. Appendix B gives the formulas and proofs. The argument assumes IEEE binary64 arithmetic with gradual underflow and with flush-to-zero and denormals-are-zero modes disabled. It proves validity for the stored binary model on the local box, without asserting equivalence to a decimal or symbolic model before import.

For a lower direction the solver minimizes x_i and the candidate bound is $\tilde{\mu}$; for an upper direction it minimizes $-x_i$ and the candidate is $-\tilde{\mu}$. Each candidate is moved outward by a further margin $10^{-7} \max\{1, |\tilde{\mu}|\}$ and rounded outward. A candidate is proposed to SCIP only if it lies in the current local domain, improves the current bound by more than $\max\{10^{-9}, 10^{-4}(u_i - \ell_i)\}$, and, for an integer variable, remains inside the domain after rounding to the next integer. A candidate that would empty the domain is discarded rather than used to prune the node. Accepted bounds are local to the node; SCIP restores them on backtracking. LP errors, time limits, infeasibility reports, and unbounded statuses produce no candidate.

Proposition 9.3 (Conditional validity of applied bounds). *Let $x^* \in B$ satisfy the stored original rows, bounds, and integrality restrictions. If the callback uses a cutoff U_c , assume also that the stored objective, in minimization form, satisfies $f(x^*) \leq U_c$. Then x^* satisfies every bound that the callback applies.*

Proof. The exact lift of x^* satisfies the McCormick, tangent, and secant rows, which are valid inequalities for the products on B , the product intervals, the lifted original rows, and the cutoff row. By Proposition B.1 it satisfies every rounded row, so it is feasible for the stored LP. Hence $x_i^* \geq \mu \geq \tilde{\mu}$ for a lower direction and $x_i^* \leq -\tilde{\mu}$ for an upper direction. The outward margin weakens each candidate further. SCIP rounds an integer bound to the next integer, which an integer x_i^* still satisfies. $\square$

The cutoff is the weak point of this guarantee. It is computed from SCIP's best solution $\hat{x}$ only if $\hat{x}$ passes a numerical check of all original rows, bounds, and integrality with absolute tolerance

10^{-6} . Its objective is then evaluated exactly in rational arithmetic, rounded upward, and increased by $10^{-6} \max\{1, |f(\hat{x})|\}$. Exact evaluation prevents cancellation; for example, $10^{16}x^2 + xy - 10^{16}y^2$ at $x = y = 1$ has value 1, while sequential floating-point summation can return 0. Thus $U_c \geq f(\hat{x})$. If $\hat{x}$ is exactly feasible, then every global minimizer of the stored model satisfies $f \leq U_c$, and Proposition 9.3 shows that no applied bound removes one. If $\hat{x}$ is only approximately feasible, its value may lie below the optimal value, and the conclusion can fail. Without an incumbent no cutoff row is used, and no feasible point of the node is removed. This is the ordinary numerical contract of the solver, made explicit; neither the propagator nor the solve as a whole is an exact certificate.

9.4.3 Scheduling policy

The fixed and adaptive variants share the relaxation, the bound validation, the budgets, and the triggers. They differ in direction order, witness screening, and a pilot rule. Table 5 lists the parameters, which were frozen before the comparison of Section 11. SCIP calls the propagator at every node before the node LP, and it may call it several times at a node.

Parameter	Value
Directional LPs per run	64
Directional LPs per callback at the root / elsewhere	12 / 4
Acting callbacks per run; per node	24; 3
Maximum depth	64
Propagator time allowance	10% of the time limit
Time limit of one directional LP	0.25 s
Incumbent retrigger: decrease of the cutoff by at least	$10^{-2} \max\{1, U_{\text{old}} \}$
Domain retrigger: relative width decrease of some variable	0.10
Minimum improvement of a proposed bound	$\max\{10^{-9}, 10^{-4}(u_i - \ell_i)\}$
Outward margin of a proposed bound	$10^{-7} \max\{1, \bar{\mu} \}$
Screening tolerance (adaptive)	$10^{-7} \max\{1, u_i - \ell_i\}$
Pilot (adaptive): LPs; minimum total normalized gain	2; 10^{-3}
Cached LP points (adaptive); retained tangent points per square	24; 25

Table 5: Frozen parameters of the numerical policy.

Admission. A call does nothing if the run has used all its directional LPs or reached its limit of acting callbacks, defined below, if the propagator time allowance or the solve deadline is exhausted, or if the node depth exceeds the limit. These checks are cheap and are not timed; Section 10 accounts for them separately.

Triggers. After admission the propagator’s timer starts. The callback reads the local box and computes the cutoff, and it continues only if a trigger fires. A propagation call that passes admission and a trigger is an *acting callback*. Only acting callbacks build a relaxation, and the callback limits of Table 5 count them. For each node number the policy records the box and the cutoff of the last acting visit and the number of acting callbacks at the node. A first visit triggers. A later visit triggers only if fewer than three acting callbacks have occurred at the node and either the cutoff decreased by at least $10^{-2} \max\{1, |U_{\text{old}}|\}$ (or became finite), or some variable’s width decreased by at least 10% relative to the recorded box. A visit that triggers nothing returns, but its time is charged. For a monotone relaxation family, a smaller box or a smaller cutoff shrinks the cutoff sets (Lemma 3.1) and can move supports that were optimal before, and the cutoff-dependent floors of Section 4 motivate the incumbent trigger. The policy’s own relaxation is not monotone, as explained above, so this reasoning motivates the two triggers without justifying them. The policy does not revisit directions left unprocessed at an unchanged node, and the thresholds are heuristic.

Directions and limits. Candidate variables are those occurring in a product term whose width exceeds 10^{-9} ; each contributes its lower direction followed by its upper direction. The fixed variant uses variable index order. The adaptive variant orders variables by the score

$$\text{score}_i = \sum_{(i,j)} \omega_{ij}(u_i - \ell_i)(u_j - \ell_j),$$

summed over the product terms containing i , where ω_{ij} is the sum of the absolute coefficients of the term in the objective and all rows. A product of factor widths measures the scale of the term’s McCormick error on the box, so the score favors variables whose relaxation error is large. Each callback may solve up to 12 directional LPs at the root and 4 elsewhere, limited by the LPs remaining in the run. Each LP receives the time limit $\min\{0.25\text{ s}, \text{remaining allowance and solve time}\}$, raised to at least one millisecond.

Screening (adaptive). The adaptive variant keeps the 24 most recent LP solutions from any node. At a callback it first checks each cached point for exact feasibility in the current frozen LP, in rational arithmetic after a fast interval test. A point that passes and lies within $10^{-7} \max\{1, u_i - \ell_i\}$ of an endpoint screens that direction. This is the current-round ceiling of Proposition 5.3, with a tolerance: the screened direction can gain at most that tolerance in this frozen LP. After every solved LP the new solution is checked and used in the same way. The check is repeated at every callback, because a point from another node, a larger box, or a weaker cutoff need not satisfy the current rows.

Pilot (adaptive). After the second LP of a callback, the adaptive variant stops if the gains of the bounds that SCIP accepted in this callback sum to less than 10^{-3} . The gain of a bound is the improvement of the proposed value over the current bound, measured before SCIP rounds integer bounds, divided by the variable’s width at the start of the callback. The accumulated sum never decreases, so this is a single test. Without screening, the first two LPs are the two directions of the highest-ranked variable, and the pilot asks whether that variable can be tightened noticeably. It is a heuristic: it does not bound the gain in unsampled directions or in later rounds.

Timing. A callback stops when its LP limit or its share of the remaining propagator allowance is used up. Construction checks the deadline every 64 rows, and screening checks it between points. Individual construction steps, LP calls, and validations are not interrupted, and the one-millisecond floor of the LP limit can exceed the remaining allowance, so a callback can overrun its deadline; the overrun is charged.

9.4.4 What the policy uses from the theory

The policy uses three proved statements: the outward correction and the residual dual bound, which make each applied bound valid under the cutoff premise; and the current-round ceiling of Proposition 5.3, which makes screening sound for the frozen LP. It does not use protected boxes, cutoff thresholds of witness pools, or residual tail bounds to stop tightening permanently, and it neither searches for nor checks such certificates. Using them would require changes that the policy does not implement. Its relaxations are not monotone, so a box protected for the relaxation of one callback need not remain protected for the next. A protected box also covers only steps at cutoffs at least its threshold, whereas incumbent improvements lower the cutoff during the search. SCIP’s own propagation, branching, and integer rounding shrink the node box by operations other than the supports of the auxiliary LP, so a box protected against the propagator’s operator can still be cut, and its containment would have to be rechecked at every node. Finally, proposing a protected box from support LPs takes a complete round. Every model of the prospective comparison has at least ten variables in product terms, so a complete round over them takes at least 20 LPs, more than the

12 allowed per root callback, unless earlier bounds have fixed some of these variables. The measured comparison is therefore a test of this policy and its costs, not of the all-future certificates.

10 Accounting for optional work

OBBT inside a branch-and-bound solver is optional work: the solver remains correct without it. A budget can limit this work on the execution where the policy runs. The question a user cares about is different: does the solver with the policy finish sooner than the same solver without it? This section separates the two questions. Proposition 10.1 bounds the optional work of one execution. Proposition 10.2 shows that such a bound implies nothing about a comparison with an independent execution of the unchanged solver. Appendix D gives comparison guarantees for constructions that leave the baseline execution unchanged. These are elementary accounting and scheduling facts, and we claim no novelty for them. Their role here is to fix which conclusions the measurements of Section 11 can support.

10.1 What must be charged

The work of a callback-based tightening policy is not the number of auxiliary LPs it solves. For the policy of Section 9.4, a propagation call can involve seven kinds of work:

1. an admission test, evaluated at every propagation call;
2. a trigger evaluation, which reads the local box and validates the incumbent to compute a cutoff;
3. construction of the auxiliary relaxation, with exact row arithmetic and outward rounding;
4. exact feasibility checks of cached points for screening;
5. the directional LP solves;
6. validation of each candidate bound by directed rounding, and its application;
7. bookkeeping.

An LP count measures only item 5. A policy that solves fewer LPs but enters more callbacks can spend more in total, since items 2–4 and 7 are paid per callback. If a callback costs c_0 outside its LP solves and an LP with its validation costs c_1 , a change that saves ΔL LPs but adds ΔC callbacks saves time only if $c_1 \Delta L > c_0 \Delta C$. Section 11.4 measures this trade-off: there the non-LP time per recorded callback was about three times the time of one LP with validation. The implementation charges items 2–7 to a timer for the added propagator that starts after the admission test. Item 1 is not timed. Its cost appears only in the total elapsed time of the run. The following accounting keeps such untimed work as a separate term.

10.2 A ledger for the enhanced execution

Consider one execution of the solver with the policy. Let $N(t)$ and $E(t)$ be the native and optional work charged up to time t ; both are nondecreasing, and $E(0) = 0$. Optional work is spent in operations. An operation is admitted at its start time s only if

$$E(s) < b + \eta N(s), \quad b \geq 0, \eta \geq 0. \quad (43)$$

Operations are serial: an admitted operation finishes before the next one starts. Native work may be charged at any time.

Proposition 10.1 (Ledger). *Suppose every increment of E belongs to an admitted operation and every admitted operation adds at most $\delta \geq 0$ to E . Then, at every time t ,*

$$E(t) \leq b + \eta N(t) + \delta, \quad N(t) + E(t) \leq (1 + \eta)N(t) + b + \delta. \quad (44)$$

If each operation instead has an enforced upper bound r on its charge, and admission requires $E(s) + r \leq b + \eta N(s)$, then $E(t) \leq b + \eta N(t)$ for all t . If the admission tests cost J in total and this cost is not charged to E , the total work is at most $(1 + \eta)N(t) + b + \delta + J$.

Proof. If no operation has started by time t , then $E(t) = 0 \leq b$. Otherwise let s be the start of the last operation admitted at or before t . Because operations are serial and every increment of E belongs to an admitted operation, only this operation charges E during $(s, t]$, so $E(t) \leq E(s) + \delta$. Admission and monotonicity of N give $E(s) < b + \eta N(s) \leq b + \eta N(t)$, which proves the first inequality; adding $N(t)$ proves the second. With reservation, $E(t) \leq E(s) + r \leq b + \eta N(s) \leq b + \eta N(t)$. The last statement adds the uncharged cost J . $\square$

The serial premise is necessary for the stated rule. With $b = 1$, $\eta = 0$, and $\delta = 1$, three concurrent operations can all be admitted while $E = 0$; if each then costs one unit, $E = 3 > b + \delta$. A concurrent implementation must include outstanding reservations in its admission test. Callbacks of a sequential solver are serial.

The constant δ must cover the whole operation, not one of its parts. A time limit on each auxiliary LP does not bound relaxation construction, screening, or validation, and an LP solver may itself return slightly after its limit. The policy of Section 9.4 uses b equal to 10% of the time limit and $\eta = 0$. Each callback receives a deadline equal to the remaining allowance and checks it between construction blocks, cached points, and LPs. Steps between these checks are not interrupted, so the implementation enforces no fixed δ ; it records the actual overrun instead. Section 11.4 reports the largest recorded propagator times. The admission test is evaluated outside the timer and contributes to J ; the proposition gives no bound on J without a further premise on the number of propagation calls.

10.3 A ledger does not compare executions

In Proposition 10.1, N is the native work on the *enhanced* execution. Let N_0 be the work of an independent execution of the unchanged solver on the same input. Tightened bounds change LP solutions, branching scores, heuristics, and node order, so the native part of the enhanced execution follows a different trajectory. The ledger contains no premise relating N to N_0 , and none can be derived from it.

Proposition 10.2 (No comparison from a ledger). *Let $b > 0$ and let g be any real function of two variables. There is a deterministic search procedure and an optional operation that satisfies the reservation form of Proposition 10.1 with $\eta = 0$ and $E = b$, such that $N + E > g(N_0, b)$.*

Proof. Consider an abstract search procedure whose first decision has two legal options, explored in the order of a score. Exploring the first option first completes a proof after one unit of work. Exploring the second option first spends M units before the proof is complete, for instance because its subtree must be exhausted first. The unchanged procedure scores the first option higher, so $N_0 = 1$. The optional operation adds a valid redundant row at cost b . Its cost is reserved at the start: the test $E + b \leq b$ holds at $E = N = 0$. The row changes the scores, the second option is explored first, and $N = M$. Then $N + E = M + b$, which exceeds $g(1, b)$ for large M . $\square$

The example is a model of search, not a constructed solver instance, and it predicts no particular solver’s runtime. It shows which premise is missing from a speedup argument based on a budget: a statement connecting the two trajectories. The same caution applies to the certificates of Sections 5 and 7. A certified ceiling on remaining endpoint movement bounds the benefit of a specified operator in a specified measure. It does not bound remaining search time. A small width reduction can enable a decisive inference, and a large one can leave the relevant relaxation error unchanged.

Guarantees against an unchanged baseline require a construction that preserves its execution trace. Appendix D gives two such constructions: interleaving bounds the work until one solver finishes, and a capped enhanced run followed by a fresh baseline run gives an additive bound on the extra work. Both use an ideal preemptible work model with separate solver states, fixed random tapes, and traces that depend only on charged work. The appendix also states the limit on a first action chosen from indistinguishable histories.

10.4 Consequences for evaluation

The results of this section lead to three requirements, which the comparison of Section 11 follows. First, the cost of a policy must include every item of Section 10.1, and auxiliary LP counts must be reported separately from time. Second, a claim about speed must come from independent complete runs of the solver with and without the policy, charging startup, construction, and validation, and keeping unsolved runs and failures in every summary; restricting a comparison to instances solved by every variant hides lost solves and all differences among unsolved runs. Third, a budget statement and a remaining-benefit certificate are statements about one execution and about an operator, respectively. Neither predicts the effect on search time, so neither can replace the measurement.

11 Computational evidence

This section reports two archived computational studies. Neither was rerun for this paper, and their results are not pooled: they used different solvers, budgets, cohorts, and questions. The first, a precursor study, tested iterated OBBT as an external presolve. Its findings shaped the second, a prospective comparison of the numerical policy of Section 9.4 with native SCIP. All numbers come from the archived records, which the supplementary archive contains. Quantities computed afterwards from the per-run records, rather than by the archived analysis, are identified as such; they are descriptive and were not part of the frozen protocol.

11.1 A precursor study of external iterated OBBT

The precursor study asked whether iterating OBBT before a solve pays off. Its candidates were the 377 MINLPLib instances [8, 44] with quadratic nonlinearities that are not flagged convex, have 5–2000 variables and no semicontinuous or SOS variables, and have a known optimal value. It excluded 38 of them because feasibility-based bound tightening left some variable of a nonlinear term without a finite bound of magnitude at most 10^7 , leaving 339 instances from 85 families. The solvers were Gurobi 13 [18] and SCIP 10 [22], with one thread each, two random seeds, and a 600-second limit on total time. A pipeline first ran the solver at the root node for at most 30 seconds to obtain an incumbent $\hat{f}$, set the cutoff $U = \hat{f} + 10^{-6} \max\{1, |\hat{f}|\}$, ran OBBT on a separate McCormick LP with five-point square tangents and secants, and restarted the solver on the tightened box with the incumbent and cutoff. OBBT used sequential coordinate updates with feasibility filtering. Five stopping rules were compared: one round, five rounds, a rule that stopped after a round without change or at a limit of 50 rounds or 400 seconds, and two adaptive rules. The adaptive rules

computed, over the variables that moved in a round, the ratio ρ of their total normalized width after the round to that before it, and stopped once ρ exceeded 0.5 or 0.8, after a round without change, or after 120 seconds of OBBT; later rounds were restricted to moved variables and their bilinear partners. A control arm repeated the root run and the restart without OBBT, which separates the effect of the box from the effect of restarting with an incumbent. Appendix E gives the tables.

No stopping rule solved more instance–seed pairs than the default solver: OBBT pipelines solved 637–638 of 678 pairs with Gurobi against 640 by default, and 561–571 with SCIP against 575. Every pipeline had a larger shifted geometric mean of total time than the default, by factors of 1.31–1.73 for Gurobi and 1.21–1.32 for SCIP. These ratios, like the node and final-solve ratios below, are the archived summaries; Appendix E explains why two solved counts are corrected here but not in these summaries. On the instances where OBBT ran, the tightened boxes reduced node counts against the control—to 0.65 (Gurobi) and 0.62 (SCIP) of the control with the no-change-or-limit rule—but the time of the final solve was 0.96–1.02 of the control’s, so the savings in the final solve did not offset the OBBT time. In the group of pairs where that rule’s box closed at least half of the gap of the McCormick LP, the node count fell to about one fifth of the control, but the mean control time of this group was only 1.2 seconds (SCIP) and 4.9 seconds (Gurobi). More rounds strengthened the relaxation with diminishing returns. With the known optimal value as cutoff, OBBT closed on average 0.236 of the McCormick LP gap after one round, 0.322 after five, and 0.346 with the no-change-or-limit rule; with the root incumbents that rule closed 0.250 (Gurobi) and 0.287 (SCIP). The median closure was zero for every rule. Each of the 4,573 tightened boxes that belonged to an instance with a reference solution contained that solution, up to a numerical tolerance of $10^{-6}(1 + |x_j|)$ per coordinate.

Two observations from this study shaped the integrated policy. First, an external presolve pays for a second root and a restart, which by themselves made the control slower than the default (shifted geometric mean ratios 1.08 for Gurobi and 1.16 for SCIP). An integrated propagator avoids both. Second, much OBBT work was spent where it could not help: on 109 instances whose nonlinear variables are all integer, the first round changed a bound on only 25, yet these instances consumed more than nine tenths of all first-round OBBT time. This suggested a short pilot before a full round, reconsideration after an incumbent improves, and a test cohort that the default solver does not solve quickly. The prospective comparison implemented these suggestions.

11.2 Design of the prospective comparison

The comparison asks whether adding the propagator of Section 9.4 to SCIP improves solving within a short budget, and whether its adaptive variant reduces the propagator’s cost relative to its fixed-order variant. The protocol, the cohort, the policy parameters, and the analysis were frozen before any comparative outcome was observed.

Arms. The *native* arm runs SCIP with default settings. The *fixed* and *adaptive* arms add the propagator with the parameters of Table 5; they share the relaxation, the bound validation, the budgets, and the triggers, and differ in ranking, screening, and the pilot. All arms use identical SCIP settings and the same model construction: quadratic rows become SCIP nonlinear constraints, a quadratic objective is moved into a constraint on an auxiliary variable, and maximization is converted to minimization. In SCIP 10.0.2 the LP-based OBBT propagator runs at the root node by default, and the separate nonlinear OBBT propagator is disabled. Generalized variable bound propagation is enabled by default [37–39]; these defaults were left unchanged in every arm. The comparison therefore measures additional work on top of native OBBT, not the effect of OBBT itself.

Cohort. Table 6 lists the twenty models. The twelve public models were selected by a rule fixed

in advance from the precursor study’s in-scope instances: 5–300 variables, 1–200 nonlinear variables, at most 500 distinct quadratic terms, and at least five seconds in every historical native SCIP run. Of the 41 eligible names, ordered by the SHA-256 digest of a fixed prefix followed by the name, the first twelve were taken, at most one per historical family. No known optimal value or outcome of the current arms entered the rule. The cohort is a selected set of harder instances, not a random sample. Its family labels still separate closely related applications: four of the twelve are circle- or point-packing models. Three public models contain binary variables, and four have variables without finite bounds in their original formulation. The eight synthetic models, two sizes each of four structures generated with seed 20261003, are stress tests: point packing, which maximizes the smallest squared distance among k points in the unit square; coupled square budgets over $[-1, 1]^n$, with concave diagonal and cyclic bilinear objective terms, two quadratic budget rows, and a two-sided sum row; bilinear cycles, with products $x_i x_{i+1}$ of $[0, 1]$ variables around a cycle and a two-sided sum row; and sparse indefinite quadratic programs over $[0, 1]^n$ with random two-sided linear rows. Four tiny synthetic models of other sizes and another seed were used before freezing to check the interface and callback coverage; they are excluded from every result. The normalized public inputs were compared with the original OSiL files at five test points per model, and every returned public incumbent was also evaluated directly from the original files.

Public model	n	m	nonlin.	hist. (s)	Synthetic model	n	m	nonlin.
bayes2_50	86	77	65	600	packing, $k = 5$	11	10	10
ex5_2_5	32	19	27	600	packing, $k = 7$	15	21	14
gabriel01	215	467	100	371	coupled squares, 12	12	3	12
genpooling_meyer04	118	141	28	600	coupled squares, 24	24	3	24
graphpart_3pm-0344-0344	144	48	144	12	bilinear cycle, 12	24	13	12
kall_circles_c6b	18	54	16	73	bilinear cycle, 20	40	21	20
kall_circlespolygons_c1p13	43	48	17	7	indefinite QP, 20	20	5	20
kall_circlesrectangles_c1r12	49	52	19	15	indefinite QP, 36	36	9	36
pointpack08	17	36	16	25				
powerflow0009r	60	103	57	20				
qp3	100	52	100	600				
wastewater15m2	133	151	24	63				

Table 6: The twenty models of the prospective comparison: variables n , rows m , and variables occurring in quadratic terms. “Hist.” is the shorter of the two native SCIP times in the precursor study, rounded; 600 means that both runs reached the 600-second limit.

Runs and resources. Every model was run with SCIP random seed shifts 0 and 1 in all three arms: 120 runs, 40 per arm. The two seeds are repeated runs of one model, not independent problem families. Each run had a ten-second limit on the solver call, which includes model construction and all propagator work; construction time is subtracted from SCIP’s own limit. A driver started each run in a fresh process, recorded the elapsed time of the whole process—interpreter start, model reading, solving, validation, and output—and enforced a 30-second hard limit. SCIP, its LP solver, and the numerical libraries used one thread. At most two runs executed at a time, in an order shuffled with seed 20261003, on a 36-logical-CPU machine that was also running unrelated jobs; the load averages at the start were 17.76, 15.65, and 14.11. The software was SCIP 10.0.2 through PySCIPOpt 6.2.1 [26], with Python 3.13.11, NumPy 2.5.3, and SciPy 1.18.1, whose HiGHS dual simplex solved the auxiliary LPs [23].

Outcomes. Each returned incumbent was checked against the original bounds, integrality, quadratic rows, and objective of the stored model. A row residual is scaled by the larger of one plus the sum of absolute term values at the point and the absolute right-hand side, and a bound

residual by one plus the absolute coordinate; the incumbent passes if the largest scaled residual and the integrality residual are at most 10^{-6} . The objective is evaluated exactly in rational arithmetic before conversion. This is a numerical check, not an exact certificate. A run counts as solved if SCIP reports optimality, the incumbent passes the check, the process did not fail, the dual bound does not exceed the incumbent value by more than 10^{-6} relative, and the reported and independently evaluated objectives agree to 10^{-6} relative. The primary time summary is the mean PAR-2 score over all 40 runs of an arm: the process time of a solved run and 20 seconds, twice the limit, otherwise. The shifted geometric mean uses the uncapped process times of all runs with a shift of one second. With p the validated incumbent value and d the final dual bound, both in minimization form, the normalized gap is $\max\{0, p - d\} / \max\{1, |p|, |d|\}$; the gap score is this value capped at one, or one if there is no valid incumbent or no finite consistent dual bound. Paired comparisons with native use the same model and seed. A PAR-2 win or loss requires a value below 95% or above 105% of native's, and a gap win or loss requires a difference in normalized gap larger than 10^{-4} , among pairs in which both gaps are defined.

11.3 Outcomes

All 120 runs completed. There were no process failures, invalid incumbents, objective mismatches, or inconsistent primal and dual bounds. All 117 returned incumbents passed the check, with largest absolute residual 9.97×10^{-7} and largest scaled residual 7.21×10^{-7} . Each arm had one run without an incumbent, `genpooling_meyer04` with seed 0. Table 7 gives the main summaries.

Cohort	Arm	Solved	Mean PAR-2 (s)	Shifted mean (s)	Mean gap score
All	Native	14/40	13.855	5.957	0.20351
All	Fixed	14/40	13.988	6.285	0.18438
All	Adaptive	14/40	13.997	6.265	0.18806
Public	Native	2/24	18.964	10.248	0.33408
Public	Fixed	2/24	18.999	10.297	0.30209
Public	Adaptive	2/24	19.026	10.324	0.30798
Synthetic	Native	12/16	6.192	2.384	0.00767
Synthetic	Fixed	12/16	6.471	2.772	0.00782
Synthetic	Adaptive	12/16	6.453	2.733	0.00818

Table 7: Outcomes of the prospective comparison. Times are process times in seconds. The shifted mean is the geometric mean of uncapped process times with a shift of one second. Lower gap scores are better.

Solved runs. Every arm solved the same 14 runs: both seeds of `kall_circlespolygons_c1p13` and 12 of the 16 synthetic runs. Neither added policy gained or lost a solve. Because the solved sets coincide and every unsolved run scores 20 seconds, the differences in mean PAR-2 come entirely from the process times of these 14 runs. In total they took 34.21 seconds for native, 39.52 for fixed, and 39.88 for adaptive, increases of 5.32 and 5.67 seconds. The propagator's recorded time in the same runs was 5.23 seconds (fixed) and 5.72 seconds (adaptive), comparable to these increases. In paired terms, mean PAR-2 rose by 0.133 seconds (fixed) and 0.142 seconds (adaptive); adaptive had two PAR-2 wins and eleven losses, fixed one win and ten losses, all among the solved runs. Both adaptive wins are the two seeds of the bilinear cycle with twelve factors. There both added arms finished at the root node, whereas native SCIP needed seven nodes; the adaptive process times were 0.1 seconds below native's, while the fixed arm matched native's time in one seed. The callback

records of these runs show the incumbent trigger at work: in both arms and both seeds, the first root callback, made before any incumbent existed, accepted no reduction; the callback triggered by the first incumbent accepted six, and the following callback, triggered by the resulting domain change, six more. The driver waited for each worker with Python’s standard polling loop, whose interval grows to 50 milliseconds, and the recorded times of short runs cluster in steps of about 0.05 seconds. The time outside the solver call—interpreter start, reading, validation, output, and the wait—was 0.41–0.59 seconds per run.

Final gaps. Both added policies lowered the mean public gap score, from 0.334 to 0.302 (fixed) and 0.308 (adaptive). The two seeds of `bayes2_50` account for this. There both added arms ended with an incumbent of value 0.520 instead of 6.825 while the dual bound stayed near zero, which changed the gap score from 1 to 0.520. The propagator constructs no incumbents, so this is an indirect effect of a changed search. Without `bayes2_50`, the mean public gap score was 0.274 for native, 0.282 for fixed, and 0.289 for adaptive; these values were computed afterwards from the per-run records. Among the 39 pairs in which both gaps are defined, adaptive had four gap wins and twelve losses, and fixed had five wins and twelve losses; 23 of these pairs are public, with four (adaptive) and five (fixed) wins and ten losses each. In eleven of the twelve losses of each arm, the final dual bound was weaker than native’s. Gaps at a wall-clock limit are sensitive to how much search fits into the limit: on `qp3`, where the propagator could never run, the fixed arm still ended one seed with a different dual bound from native.

11.4 Where the added work went

Table 8 decomposes the propagator’s work, using the per-callback records of the archived runs, which store each callback’s node number and depth. A callback is attributed to the root node, to a nonroot node with a positive number, or to a nonroot node with number zero. The release-tagged SCIP 10.0.2 source initializes new nodes with number zero, assigns positive numbers to search-tree children, and leaves temporary probing nodes at zero [36]. Consistently, in runs that finished at the root, every nonroot callback had number zero. This supports interpreting them as temporary probing nodes, although the logs did not record node types. We report them as nonroot, number-zero callbacks. Because the trigger state is keyed by node number, the policy treated all nonroot nodes with number zero in a run as one node and allowed at most three acting callbacks for them. The time in the row of triggers without a call is the remainder of the propagator time: trigger evaluations that started no acting callback, which are timed but produce no record.

	Fixed				Adaptive			
	Calls	LPs	Accepted	Time (s)	Calls	LPs	Accepted	Time (s)
Root node	100	1150	62	8.224	98	336	62	3.554
Tree nodes	166	662	50	5.485	505	1203	244	13.287
Nonroot, number 0	90	360	0	3.684	90	180	2	2.593
Unsupported box (<code>qp3</code>)	48	0	0	0.068	48	0	0	0.066
Triggers without a call				2.069				3.623
Total	404	2172	112	19.530	741	1719	308	23.123
of which LP solves				12.818				10.393

Table 8: Work of the added propagator over all 40 runs of each arm. “Accepted” counts bound changes that SCIP applied. The time of LP solves includes the dual-bound validation of each solve. The totals are those of the archived analysis; the split was computed afterwards from the per-callback records.

The adaptive variant solved 1,719 directional LPs against 2,172, a reduction of 20.9%, but its propagator time rose from 19.530 to 23.123 seconds, an increase of 18.4%. The decomposition explains the difference. The time spent in LP solves and dual-bound validation fell by 2.43 seconds, while the remaining propagator time rose from 6.71 to 12.73 seconds. This remaining time, including trigger visits that started no acting callback, amounted to about 17 milliseconds per recorded callback in both variants, and an LP with its validation took about 6 milliseconds. The adaptive variant made 337 more callbacks; 574 of its 693 supported callbacks stopped after the two-LP pilot. Its 64-LP budget was therefore rarely binding: 20 of its 38 runs on models other than **qp3** ended at the cap of 24 callbacks, and only 5 of its 40 runs used all 64 LPs. The fixed variant used all 64 LPs in 27 runs and never reached the callback cap outside **qp3**; 349 of its 356 supported callbacks ended at an LP limit. Much of the reduction in LPs is thus a consequence of the callback cap binding before the LP budget, not of evidence that the adaptive variant chose its directions better. The same budgets also kept the recorded work small: the largest propagator total of a run was 1.00501 seconds against an allowance of one second, 7 fixed and 11 adaptive runs exceeded the allowance at all, and the summed propagator time was 6.2% (fixed) and 7.4% (adaptive) of the summed process time of all 40 runs. The share was much larger in some short solved runs, up to about two fifths, and the untimed admission tests are not included.

The adaptive variant obtained more accepted bound changes with fewer LPs: 308 against 112. At the root, both variants obtained 62 accepted changes, the adaptive one with 336 LPs instead of 1150; most of its other changes came from callbacks at tree nodes. Of the 470 bounds the adaptive variant proposed, SCIP applied 308; of the fixed variant’s 200 proposals, it applied 112. SCIP’s tightening interface declines a change that it does not regard as a sufficient improvement; the archive does not record individual reasons. Witness screening removed only two directions, one in each seed of **kall_circlespolygons_c1p13**. The archive does not record the outcome of individual feasibility checks; a check of a floating-point LP solution in exact arithmetic fails whenever the solution violates an active row by any positive amount, which limits how often such points can screen. Nonroot, number-zero callbacks consumed 360 of the fixed variant’s LPs without producing an accepted change. Directional LPs that produced no usable bound numbered 4 (fixed) and 26 (adaptive); their individual statuses were not retained. All 48 unsupported callbacks of each arm were on **qp3**, whose local box remained unbounded; the initially unbounded **powerflow0009r** became eligible after SCIP tightened its bounds. The numerical incumbent check never rejected SCIP’s incumbent, and 124 (fixed) and 120 (adaptive) callbacks ran without a cutoff. The time limit was a stopping criterion rather than an exact deadline: all 78 time-limited runs exceeded ten seconds slightly, the longest solver call took 10.093 seconds, and the longest process 10.606 seconds.

11.5 What the comparison shows

The comparison supports four conclusions, each confined to this cohort, budget, and implementation. Neither added policy changed which runs were solved within ten seconds. The adaptive variant solved fewer auxiliary LPs than the fixed variant but spent more time in the propagator. Neither added policy showed a net speedup: on the runs that every arm solved, both took longer, by amounts comparable to their own recorded propagator time. The improvement in mean public gap comes from one model through a better incumbent; on the other public models both policies had worse mean gap scores than native.

Several factors prevent stronger conclusions in either direction. The cohort has twenty models, twelve of them selected using historical results and eight synthetic, and the two seeds are repetitions rather than independent families. The ten-second budget left 22 of 24 public runs unsolved in every arm, so PAR-2 differences reflect only the 14 solved runs. Most of these are short synthetic runs, in

which a fixed overhead of about half a second is a large share of the process time and the recorded times move in steps of about 0.05 seconds. The machine was shared with unrelated jobs. Under these conditions the small timing differences do not establish a slowdown, and the absence of new solves does not show that longer runs would also see none. The propagator was written in Python and builds its relaxation in exact rational arithmetic at every callback; its per-callback cost is a property of this implementation, and an integrated implementation could have a very different cost structure. The policy was compared as a whole, not by a factorial design over ranking, screening, and the pilot. Finally, the protected-box, cutoff-threshold, and residual certificates of Sections 5–7 were not used by the measured policy, so the comparison says nothing about their computational value.

The negative result matters for three reasons. First, it shows concretely that the number of auxiliary LPs is not a cost measure. Work outside the LP solves took about a third of the propagator time of the fixed variant and more than half of that of the adaptive variant, and a policy that saved LPs by stopping early spent the savings, and more, on additional callbacks. Second, the recorded propagator time was at most 1.00501 seconds per run and amounted to 6.2% and 7.4% of the cohort’s total process time. A better selection rule can directly save at most the time the propagator itself spends. A worthwhile gain must come from the search—fewer nodes, earlier incumbents, or new solves. The search did change in places, as `bayes2_50` and the small bilinear cycles show, but 308 accepted reductions produced no additional solve and no demonstrated overall speedup within ten seconds. The archive records associations between policies and search outcomes; without an ablation it does not isolate the effect of particular reductions. This agrees with the precursor study, in which tightened boxes reduced nodes but the final-solve savings did not offset the cost of obtaining them. Third, it marks the boundary between the two kinds of results in this paper. The certificates bound how much a specified operator can still change a box or a relaxation bound, which can justify skipping work. They do not predict that the work, if done, would save time, and Section 10 shows that no budget statement can supply that prediction. On this evidence, native SCIP remains the appropriate default, and the added policy is an experimental implementation.

12 Discussion

The three limits of Section 3.4 translate into specific information for the decisions that a solver makes about OBBT. We summarize what the results of this paper support, and what they do not.

Starting. A round cannot move the endpoints of a box that is already fixed for the given relaxation and cutoff, although its dual solutions can still be useful, for example for Lagrangian variable bounds [15]. A finite certificate proves fixedness (Section 5) and remains valid at every larger cutoff. In a box that strictly contains the certified box, it no longer proves a stall, but it still bounds the total movement of all later rounds. Two structural situations explain when fixed boxes are present. If a model has several global minimizers, for example by symmetry, the iterates contain their hull and cannot converge to a point (Lemma 3.3), although they can still remove parts of the box outside the hull. If on every face of a box shape some point has negative tangent-model value, its second-order relaxation gap exceeds the objective’s second-order growth there. Boxes of that shape around a minimizer are then fixed at every sufficiently small scale and every cutoff $U \geq f^*$ (Theorem 4.10); for quadratic objectives with McCormick relaxations, condition (a) of Proposition 4.19 gives this at every scale. That condition concerns the relaxation, not the problem: the strongly convex objectives of Example 4.20 stall under termwise McCormick relaxations but not under a relaxation that keeps the convex quadratic whole. Other results guarantee that a

round moves the box: the box threshold of Proposition 6.2, a stored dual bound below the current endpoint (Proposition 8.1 and Corollary 8.2), and the contraction conditions of Section 4. None of them guarantees that the movement shortens the search, and Example D.4 shows that the history of earlier calls cannot identify a first action that is guaranteed to be best. The precursor study of Section 11 illustrates the cost of starting blindly: more than nine tenths of its first-round time went to the 109 instances whose nonlinear variables are all integer, and on only 25 of them did the first round change a bound.

Stopping. A protected box inside the current box bounds the total remaining movement of every endpoint under all later rounds of the specified operator at cutoffs not below its threshold. When it equals the current box, the remaining movement is zero and stopping that operator loses nothing it could have achieved. The local theory shows when such certificates can exist. In the stall regime they exist at every sufficiently small scale. In the strictly contracting regime with a positive cutoff slack and the closedness property (6), the limit is a fixed box of width $O(\sqrt{\epsilon})$, so a certificate exists at the limit, and the iterates reach its neighborhood after a number of rounds that grows only logarithmically in $1/\epsilon$ (Section 4.7). With zero slack, no box of positive width is fixed near the minimizer. An iterate that reaches the minimizer can be certified as a fixed singleton. When the optimal incumbent lies in the current box, its protected singleton (Corollary 5.14) bounds the remaining movement, and the bound is exact if the iterates converge to that point (Example 5.11). Without such an inner box, a residual certificate (Section 7) can justify a tolerance stop. These regimes need not cover every relaxation family (Example 4.12). Small observed changes certify nothing beyond the bound that nesting already gives: the ratio of successive widths can tend to one during slow contraction, near a positive floor, or at a stall, and valid monotone families with identical histories can have different futures (Section 4.7 and Proposition 7.10). The exception is an exact complete round that changes nothing, which proves that the box is fixed. Stopping rules based on observed progress are therefore heuristics, which is how the measured policy uses them.

Repeating after a change. Under the contraction hypotheses, the width of the limit box is at most of order $\sqrt{\epsilon}$ and the remaining relaxation gap at most of order ϵ (Corollaries 4.8 and 4.29); both orders are attained in the two-variable quadratic model (Proposition 4.18). An improved incumbent is therefore the main reason to reconsider OBBT at a box where it has already converged: it can shrink the first limit, the hull of the original sublevel set, and it can remove fixed boxes of the second. A certified stop remains valid while the cutoff stays at or above the face threshold of its pool. Below that threshold the stored proof expires, which is a precise event at which reconsidering becomes justified, though not a proof that it will pay. The certified box itself can remain fixed below the face threshold, down to the box threshold of Proposition 6.2, whose computation requires at most $2n$ face optimizations. For a single frozen round, retained relaxation points and stored dual vectors bracket the possible and the guaranteed change of each support after a cutoff decrease (Section 8). A branching step keeps a certificate valid in every child that contains the certified box and requires a new check in a child that does not. Prior work already reevaluates stored dual inequalities when the incumbent improves; the certificates add a statement about the operator’s remaining movement, not a new trigger mechanism.

Width, certificates, and search cost. The guarantees of this paper bound the movement of a specified tightening operator, or the change of its relaxation bound. They do not bound search time, and the evidence shows that the difference matters. In the prospective comparison, a policy that solved 20.9% fewer auxiliary LPs spent more time in its propagator. The adaptive arm recorded

more calls, each with its own construction and checking cost; neither added policy produced an additional solve or a demonstrated speedup (Section 11). In the precursor study, tightened boxes reduced node counts, with the largest mean reductions in groups whose mean control time was a few seconds, while the savings in the final solve did not offset the OBBT cost. Both findings agree with Section 10: a budget on optional work bounds that work, but no budget statement connects the enhanced search with an unchanged baseline. The value of a stopping certificate is therefore to avoid work that cannot change the box. Whether that saves time depends on how much the certificate costs. Some certificates cost little beyond an exact check, such as the singleton of an exactly feasible incumbent in the current box or the optima of a completed exact round when they pass the rebuilt-row check. If finding a certificate costs about as much as a round, reuse across rounds, cutoffs, or nodes may be needed to recover that cost. The measured policy used no certificates of either kind, and the experiments do not assess them. On the present evidence, native SCIP remains the appropriate default and the added policy is an experimental implementation.

Scope. Every rate and stall statement belongs to a relaxation and a formulation. At zero cutoff slack, the convex quadratic $\sum_i x_i^2$ with $n \geq 2$ stalls when each square is relaxed by the McCormick inequalities of the product $x_i \cdot x_i$, contracts by the factor $1/2$ per round when the valid rows $y_i \geq 0$ are added, and collapses in one round under its exact convex relaxation (Example 4.23); eliminating a linear equation before relaxing changes the rate in the example of Section 8. The results on iterated tightening, rates, and future-round certificates assume monotonicity of the whole relaxation construction. A pool of cuts separated at earlier solutions does not guarantee it, and neither do tangent points that move with the box, as in the measured policy (Section 3). The current-round ceilings, the validated dual bounds, and the results for one fixed LP do not need it. The local theory involves the minimizer and tangent data and is not by itself an activation rule. The exact reference procedure certifies its own rational relaxation family; the numerical policy validates each applied bound with directed rounding, but its cutoff comes from a numerically accepted incumbent, and a complete SCIP solve is not an exact certificate. The prospective comparison uses a small, selected cohort with short time limits on a shared machine and supports only the conclusions stated in Section 11.

Three extensions would change the practical picture. Certificates found cheaply from points that a native solver produces anyway could make the all-future guarantees usable at low cost. Tangent models for lifted relaxations with auxiliary variables in the constraints would extend the local theory to the relaxations that solvers build. A validated model of how reductions of width or of the relaxation gap change the later search would be needed before any certificate of remaining benefit can guide a choice among tightening, cutting, and branching.

Supplementary material

A supplementary archive contains, byte for byte, the source code, raw outcomes, and analysis scripts of both computational studies described in Section 11; the frozen inputs, protocol, and logs of the prospective comparison; the model and solution files retained from the precursor study, for which no hash record from before that study exists; and the exact rational reference checker and closure driver of Section 9 with their saved check outputs. A manifest records the SHA-256 digest and size of every archived file. Solver binaries, licensed software, and the original Python environments are not included.

A Proofs for Section 4

A.1 Proof of Theorem 4.24

Fix the compact set $\mathcal{K}$ of shapes and let $\mathcal{S}_{\mathcal{K}} = \{(d, \xi) : d \in \mathcal{K}, \xi \in D(d)\}$, a compact set. Let $R_{\mathcal{K}} = \max_{d \in \mathcal{K}} \|d\|_{\infty}$, so that $\|\xi\|_{\infty} \leq R_{\mathcal{K}}$ on $\mathcal{S}_{\mathcal{K}}$ and $B_t \subseteq x^* + tR_{\mathcal{K}}[-1, 1]^n$. A quantity depending on (t, d, ξ) is $O(t^j)$ if it is bounded by Ct^j on $\mathcal{S}_{\mathcal{K}}$ for $0 < t \leq t_1$, with C and t_1 depending only on $\mathcal{K}$ and the factorable representation, and it is $o(t^2)$ if its supremum over $\mathcal{S}_{\mathcal{K}}$ divided by t^2 tends to zero. For a factor index a we abbreviate $a = v_a(x_t)$, $a^* = v_a^*$, $a^L = v_a^L(B_t)$, $a^U = v_a^U(B_t)$, and write cv_a, cc_a for the relaxations at x_t on B_t ; the letter b is used in the same way for a second factor. We first collect five lemmas.

Lemma A.1 (Median rule). *Let $\text{cv} \leq v \leq \text{cc}$, $v \in Z = [L, U]$, and $z \in Z$. Then $m = \text{mid}\{\text{cv}, \text{cc}, z\}$ lies in $[\text{cv}, \text{cc}] \cap Z$. If $z = L$, then $m = \max\{\text{cv}, L\}$; if $z = U$, then $m = \min\{\text{cc}, U\}$.*

Proof. Since $\text{cv} \leq \text{cc}$, m is the projection of z onto $[\text{cv}, \text{cc}]$. If $z < \text{cv}$, then $m = \text{cv}$ and $L \leq z < \text{cv} \leq v \leq U$. If $z > \text{cc}$, then $m = \text{cc}$ and $L \leq v \leq \text{cc} < z \leq U$. Otherwise $m = z \in Z$. If $z = L$, then $L \leq v \leq \text{cc}$, and the projection of L onto $[\text{cv}, \text{cc}]$ is $\max\{\text{cv}, L\}$. The case $z = U$ is symmetric. $\square$

Lemma A.2 (Lipschitz envelopes). *If ψ is Λ -Lipschitz on $[\alpha, \beta]$, its convex and concave envelopes on $[\alpha, \beta]$ are Λ -Lipschitz.*

Proof. Let c be the convex envelope. The affine functions $A_{\beta}(y) = \psi(\beta) - \Lambda(\beta - y)$ and $A_{\alpha}(y) = \psi(\alpha) - \Lambda(y - \alpha)$ are minorants of ψ , so $c \geq A_{\beta}$ and $c \geq A_{\alpha}$. Since $c \leq \psi$ and $A_{\beta}(\beta) = \psi(\beta)$, we get $c(\beta) = \psi(\beta)$, and likewise $c(\alpha) = \psi(\alpha)$. For $\alpha \leq y < y' < \beta$, convexity gives

$$\frac{c(y') - c(y)}{y' - y} \leq \frac{c(\beta) - c(y)}{\beta - y} \leq \frac{\psi(\beta) - A_{\beta}(y)}{\beta - y} = \Lambda,$$

and for $\alpha < y < y' \leq \beta$,

$$\frac{c(y') - c(y)}{y' - y} \geq \frac{c(y) - c(\alpha)}{y - \alpha} \geq \frac{A_{\alpha}(y) - \psi(\alpha)}{y - \alpha} = -\Lambda.$$

The remaining cases follow by continuity of c at the endpoints, which holds because $A_{\alpha} \leq c \leq \psi$ near α and $A_{\beta} \leq c \leq \psi$ near β . For the concave envelope apply this to $-\psi$. $\square$

Lemma A.3 (Envelopes of a C^2 function on a short interval). *Let h be twice continuously differentiable on $[a^* - r, a^* + r]$, let $c = h''(a^*)$, and let $\omega(s) = \sup\{|h''(y) - c| : |y - a^*| \leq s\}$, so that $\omega(s) \rightarrow 0$ as $s \downarrow 0$, and $\omega(s) \leq M_2 s$ if h'' is M_2 -Lipschitz. For every interval $Z = [L, U] \subseteq [a^* - s, a^* + s]$ with $s \leq r$, and every $y \in Z$,*

$$\begin{aligned} |h(y) - h_Z^{\text{cv}}(y) - \tfrac{1}{2}c^-(y - L)(U - y)| &\leq \omega(s)s^2, \\ |h_Z^{\text{cc}}(y) - h(y) - \tfrac{1}{2}c^+(y - L)(U - y)| &\leq \omega(s)s^2. \end{aligned}$$

Moreover $h_Z^{\text{cv}} - h'(a^*)\text{id}$ and $h_Z^{\text{cc}} - h'(a^*)\text{id}$ are Lipschitz on Z with constant $s \sup_{|y - a^*| \leq s} |h''(y)|$.

Proof. Let $q(y) = h(a^*) + h'(a^*)(y - a^*) + \frac{c}{2}(y - a^*)^2$ and $e = h - q$. Then $e(a^*) = e'(a^*) = 0$ and $|e''| \leq \omega(s)$ on $[a^* - s, a^* + s]$, so $|e| \leq \eta = \omega(s)s^2/2$ on Z . Envelopes are monotone and commute with adding constants, so $q_Z^{\text{cv}} - \eta \leq h_Z^{\text{cv}} \leq q_Z^{\text{cv}} + \eta$, and therefore $|(h - h_Z^{\text{cv}}) - (q - q_Z^{\text{cv}})| \leq 2\eta$. If $c \geq 0$, q is convex and $q - q_Z^{\text{cv}} = 0$. If $c < 0$, q is concave; its convex envelope on Z is the chord through its endpoint values, because the chord is convex and below q , and every convex minorant

of q lies below the chord. The difference $q - \text{chord}$ is a quadratic with leading coefficient $c/2$ that vanishes at L and U , so it equals $\frac{1}{2}|c|(y-L)(U-y)$. The concave envelope is analogous. For the Lipschitz claim, envelopes commute with adding affine functions, so $h_Z^{\text{cv}} - h'(a^*) \text{id}$ is the convex envelope of $\psi = h - h'(a^*) \text{id}$, and $|\psi'| \leq s \sup|h''|$ on Z ; Lemma A.2 applies. $\square$

Lemma A.4 (Monotone functions). *If $h' > 0$ on $Z = [L, U]$, then L is the unique minimizer of h_Z^{cv} and U is the unique maximizer of h_Z^{cc} on Z . If $h' < 0$ on Z , the roles of L and U are exchanged.*

Proof. Let $\mu = \min_Z h' > 0$. The affine function $h(L) + \mu(y - L)$ is a convex minorant of h on Z , so for $y > L$, $h_Z^{\text{cv}}(y) \geq h(L) + \mu(y - L) > h(L) \geq h_Z^{\text{cv}}(L)$. Similarly $h(U) + \mu(y - U)$ is a concave majorant, which gives the claim about h_Z^{cc} . The case $h' < 0$ is symmetric. $\square$

Lemma A.5 (Sign selections). *Let $|s_t - s^*| \leq Ct$ and $\text{cv} \leq v \leq \text{cc}$ with $\text{cc} - \text{cv} \leq C't^2$. Let $\tilde{v} = \text{cv}$ if $s^* > 0$, $\tilde{v} = \text{cc}$ if $s^* < 0$, and $\tilde{v} = v$ if $s^* = 0$. Then $|\min\{s_t \text{cv}, s_t \text{cc}\} - s_t \tilde{v}| \leq CC't^3$, and the difference is zero if $s^* \neq 0$ and $t < |s^*|/C$. The same holds for the maximum, with cv and cc exchanged in the definition of $\tilde{v}$.*

Proof. The minimum equals $s_t \text{cv}$ if $s_t \geq 0$ and $s_t \text{cc}$ if $s_t \leq 0$. If $s_t = 0$, or if $s^* \neq 0$ and s_t has the sign of s^* , the difference is zero; this holds in particular when $s^* \neq 0$ and $t < |s^*|/C$. In the remaining cases, $s^* = 0$ or s_t and s^* have opposite signs, so $|s_t| \leq |s_t - s^*| \leq Ct$; both the minimum and $s_t \tilde{v}$ lie between $s_t \text{cv}$ and $s_t \text{cc}$, an interval of length $|s_t|(\text{cc} - \text{cv}) \leq CC't^3$. $\square$

Proof of the theorem. By induction over k , each v_k is twice continuously differentiable on a neighborhood N of x^* : sums, scalings, and products preserve this, and $h_k(v_a)$ is twice continuously differentiable near x^* because v_a is continuous and h_k is twice continuously differentiable near $v_a(x^*)$. For t small, $x_t \in N$, and Taylor's theorem gives, uniformly on $\mathcal{S}_K$,

$$v_k(x_t) = v_k^* + t\pi_k(\xi) + O(t^2). \quad (45)$$

We prove (11)–(13) by induction over k ; we refer to them as (I_k) , (R_k) , and (P_k) . By (I_a) , the interval of a univariate argument lies in $[a^* - Ct, a^* + Ct]$, so for small t it lies in the interval where h_k is twice continuously differentiable, and all objects are defined. By (R_a) and boundedness of E_a on the compact set $\mathcal{S}_K$, $\text{cc}_a - \text{cv}_a = O(t^2)$.

Variables and constants. The intervals are exact, and $\text{cv} = \text{cc} = v$, so all three properties hold with $E = 0$.

Sums and scalings. The expansions add. For a sum, $v^L - \text{cv} = (a^L - \text{cv}_a) + (b^L - \text{cv}_b) \leq o(t^2)$. For $c < 0$, $\text{cv} = c \text{cc}_a = ca - t^2|c|E_a^{\text{cc}} + o(t^2)$ and $v^L - \text{cv} = ca^U - c \text{cc}_a = |c|(\text{cc}_a - a^U) \leq o(t^2)$.

Products $v_k = v_a v_b$: intervals. Each endpoint product is $a^i b^j = a^* b^* + t(b^* \pi_a^i + a^* \pi_b^j) + O(t^2)$, where π_a^i ranges over $\underline{\pi}_a, \bar{\pi}_a$ and π_b^j over $\underline{\pi}_b, \bar{\pi}_b$. The minimum and maximum are 1-Lipschitz in the sup norm, and $\min_{i,j} (b^* \pi_a^i + a^* \pi_b^j) = \min_i b^* \pi_a^i + \min_j a^* \pi_b^j$, which is the lower endpoint of the interval in the recursion; the maximum is analogous. This proves (I_k) .

Products: relaxations. By (I_a) and (I_b) , the coefficients b^L, b^U are within $O(t)$ of b^* and a^L, a^U are within $O(t)$ of a^* . Lemma A.5 gives

$$\alpha_1 = b^L \tilde{a} + O(t^3), \quad \beta_1 = b^U \tilde{a} + O(t^3), \quad \alpha_2 = a^L \tilde{b} + O(t^3), \quad \beta_2 = a^U \tilde{b} + O(t^3),$$

with the same $\tilde{a} \in \{\text{cv}_a, \text{cc}_a, a\}$, selected by the sign of b^* , in α_1 and β_1 , and the same $\tilde{b}$, selected by the sign of a^* , in α_2 and β_2 . This is the key point: for small t , b^L and b^U have the sign of b^* when $b^* \neq 0$, and the selection costs only $O(t^3)$ when $b^* = 0$. The identities

$$b^L \alpha + a^L \beta - a^L b^L = \alpha \beta - (\alpha - a^L)(\beta - b^L), \quad b^U \alpha + a^U \beta - a^U b^U = \alpha \beta - (a^U - \alpha)(b^U - \beta)$$

and the fact that the maximum is 1-Lipschitz give

$$cv_k = \tilde{a}\tilde{b} - \min\{(\tilde{a} - a^L)(\tilde{b} - b^L), (a^U - \tilde{a})(b^U - \tilde{b})\} + O(t^3). \quad (46)$$

Since $\tilde{a} - a = O(t^2)$, $\tilde{b} - b = O(t^2)$, $b - b^* = O(t)$, and $a - a^* = O(t)$,

$$\tilde{a}\tilde{b} = ab + b^*(\tilde{a} - a) + a^*(\tilde{b} - b) + O(t^3).$$

By (R_a) , $b^*(\tilde{a} - a) = -t^2((b^*)^+ E_a^{cv} + (b^*)^- E_a^{cc}) + o(t^2)$: the two cases are $\tilde{a} = cv_a$ with $b^* > 0$ and $\tilde{a} = cc_a$ with $b^* < 0$. The term $a^*(\tilde{b} - b)$ is analogous. Next, by (45) and (I_a) , $\tilde{a} - a^L = (a - a^L) + (\tilde{a} - a) = t(\pi_a - \underline{\pi}_a) + O(t^2)$, and likewise for the other three differences, so each product in the minimum of (46) is t^2 times the corresponding product in E_k^{cv} , plus $O(t^3)$. This proves the convex half of (R_k) . The concave half is the same argument with the maximum selections, which select cc_a for $b^* > 0$ and cv_a for $b^* < 0$, and the identities $b^L\alpha + a^U\beta - a^U b^L = \alpha\beta + (a^U - \alpha)(\beta - b^L)$ and $b^U\alpha + a^L\beta - a^L b^U = \alpha\beta + (\alpha - a^L)(b^U - \beta)$.

Products: dominance. The function $M(\alpha, \beta) = \max\{b^L\alpha + a^L\beta - a^L b^L, b^U\alpha + a^U\beta - a^U b^U\}$ is, on $\Pi = [a^L, a^U] \times [b^L, b^U]$, the convex envelope of $\alpha\beta$, so it is at least $\min_{\Pi} \alpha\beta = v_k^L$ there. On $\mathbb{R}^2$ it is Lipschitz in the sup norm with constant $\max\{|a^L| + |b^L|, |a^U| + |b^U|\} = O(1)$. Since $\tilde{a} \in [cv_a, cc_a]$, (P_a) gives $a^L - o(t^2) \leq \tilde{a} \leq a^U + o(t^2)$, and similarly for $\tilde{b}$, so $(\tilde{a}, \tilde{b})$ lies within sup-distance $o(t^2)$ of Π . Together with (46), which states $cv_k = M(\tilde{a}, \tilde{b}) + O(t^3)$, this gives $v_k^L - cv_k \leq o(t^2)$. The concave side is symmetric.

Univariate factors $v_k = h(v_a)$. Let $Z = [a^L, a^U]$, which lies in $[a^* - s_t, a^* + s_t]$ with $s_t = Ct$, and let $h_1 = h'(a^*)$, $h_2 = h''(a^*)$.

For (I_k) : on Z , $h(y) = h(a^*) + h_1(y - a^*) + O(t^2)$, and the minimum over Z of $h_1(y - a^*)$ is $t \min\{h_1 \underline{\pi}_a, h_1 \bar{\pi}_a\} + O(t^2)$.

For (P_k) , which here holds exactly: let $m = \min\{cv_a, cc_a, z^{\min}\}$. By Lemma A.1, $m \in Z$, so $cv_k = h_Z^{cv}(m) \geq \min_Z h_Z^{cv} = \min_Z h = v_k^L$; the convex envelope has the same minimum as h , because the constant $\min_Z h$ is a convex minorant. The concave side is analogous.

For (R_k) write

$$cv_k = h(a) - [h(a) - h_Z^{cv}(a)] + [h_Z^{cv}(m) - h_Z^{cv}(a)].$$

By Lemma A.3, the first bracket equals $\frac{1}{2}h_2^-(a - a^L)(a^U - a) + O(\omega(s_t)s_t^2) = \frac{1}{2}t^2 h_2^- \sigma_a + o(t^2)$. This holds in every case, including $h_2 = 0$ and sign changes of h'' inside Z , because the comparison is with the quadratic model of h and not with the convexity of h on Z . By Lemma A.1, $m, a \in Z$ and $|m - a| \leq cc_a - cv_a = O(t^2)$, so the Lipschitz part of Lemma A.3 gives $h_Z^{cv}(m) - h_Z^{cv}(a) = h_1(m - a) + O(t)O(t^2)$. It remains to evaluate $h_1(m - a)$.

- If $h_1 = 0$, the term is zero. The minimizer $z^{\min}$ may then lie anywhere in Z and may jump as t varies; this does not matter.
- If $h_1 > 0$, then $h' > 0$ on Z for small t , Lemma A.4 gives $z^{\min} = a^L$, and Lemma A.1 gives $m = \max\{cv_a, a^L\}$. Hence $m - a = (cv_a - a) + (a^L - cv_a)^+$, and (P_a) gives $(a^L - cv_a)^+ = o(t^2)$, so $m - a = -t^2 E_a^{cv} + o(t^2)$. Without (P_a) , the clipping of the argument at a^L could change the second-order term; this is the reason for proving (P_k) along with (R_k) .
- If $h_1 < 0$, then $z^{\min} = a^U$, $m = \min\{cc_a, a^U\}$, and $m - a = t^2 E_a^{cc} + o(t^2)$, so $h_1(m - a) = -t^2 |h_1| E_a^{cc} + o(t^2)$.

Adding the three parts gives $cv_k = v_k(x_t) - t^2 E_k^{cv} + o(t^2)$. The concave side uses $z^{\max}$, which is a^U if $h_1 > 0$ and a^L if $h_1 < 0$, and gives E_k^{cc} . This completes the induction.

Remainder order. If each h_k'' is Lipschitz near the relevant point, then $\omega(s) = O(s)$ in Lemma A.3, every remainder in the induction, including those in (P_k) , is $O(t^3)$, and f has a Lipschitz second derivative near x^* , so its Taylor remainder is also $O(t^3)$.

The objective. Since $f = v_m$ is twice continuously differentiable near x^* , $f(x_t) = f(x^*) + t\nabla f(x^*)^T \xi + \frac{1}{2}t^2 \xi^T \nabla^2 f(x^*) \xi + o(t^2)$ uniformly; combine with (R_m) . Continuity of Q follows from continuity of E_m^{cv} .

Clipping. Clipping does not change the intervals, which do not use the relaxations. Clipped relaxations still bracket the factor values, because $v_k^L \leq v_k \leq v_k^U$; hence Lemma A.1 still applies. Every rule is Lipschitz in its relaxation inputs with constants $O(1)$: the product rule has interval endpoints as coefficients, and the univariate rule composes the 1-Lipschitz median with h_Z^{cv} , which is Lipschitz with constant $\max_Z |h'| = O(1)$ by Lemma A.2. By induction, the rule applied to clipped inputs differs from its unclipped value by $o(t^2)$, and the clipping of its output moves it by at most $(v_k^L - cv_k)^+ + o(t^2) = o(t^2)$ by (P_k) . Hence the clipped relaxations have the same expansions. $\square$

A.2 Proofs for minimizers on the boundary

Proof of Lemma 4.26. Let $\mathcal{K} = \{d \in \mathcal{S} : d_F \in \mathcal{K}_F, 0 \leq d_i^+ \leq 1 \text{ for } i \in A\}$, a compact set. Let ω be a modulus of uniform continuity of Q on the compact set $\{(d, \xi) : d \in \mathcal{K}, \xi \in D(d)\}$ with respect to the sup norm, and let $\varrho = \varrho_{\mathcal{K}}$. Choose $\bar{\sigma} \in (0, 1]$ with $\omega(\bar{\sigma}) \leq \delta/4$, and $\bar{t}$ such that $\varrho(t) \leq \delta/4$ and the comparison boxes belong to $\mathfrak{B}$ for $t \leq \bar{t}$. Let $\sigma = \max_{i \in A} d_i^+ \leq \bar{\sigma}$. For $\xi \in D(d)$, the pair $(\iota(d_F), \iota(\xi_F))$ is within sup-distance σ of (d, ξ) , because $d_A^- = 0$ and $0 \leq \xi_A \leq d_A^+$. Hence

$$|Q(d, \xi) - Q^F(d_F, \xi_F)| \leq \omega(\sigma) \leq \delta/4. \quad (47)$$

A point $x = x^* + t\xi \in K_U(B)$ satisfies, by Assumption 4.25,

$$tg^T \xi + t^2 Q(d, \xi) \leq \epsilon + \varrho(t)t^2, \quad g^T \xi = \sum_{i \in A} g_i \xi_i \geq 0. \quad (48)$$

(i) Dropping $g^T \xi \geq 0$ and using (47), $Q^F(d_F, \xi_F) \leq \epsilon/t^2 + \varrho(t) + \omega(\sigma) \leq \delta$, so ξ_F lies in the sublevel set defining $\Phi_\delta^F(d_F)$.

(ii) For $\xi = \iota(\xi_F)$, $g^T \xi = 0$ and $Q(d, \xi) \leq -\delta + \delta/4$, so $\phi_B(x^* + t\xi) \leq f^* + t^2(-3\delta/4 + \delta/4) < f^* \leq U$.

(iii) By (48) and (47), for $i \in A$,

$$g_i \xi_i \leq g^T \xi \leq \frac{\epsilon}{t} + t(-Q(d, \xi) + \varrho(t)) \leq \frac{\epsilon}{t} + t(G^F(d_F) + \omega(\sigma) + \varrho(t)).$$

The lower bound of coordinate i stays at x_i^* , because $x^* \in K_U(B)$ by validity. Hence

$$w_i(T_U(B)) \leq (\epsilon + (G^F(d_F) + \omega(\sigma) + \varrho(t))t^2)/g_i, \quad (49)$$

which gives (iii).

(iv) If $\epsilon + (G^F(d_F) - \delta/2)t^2 < 0$, the asserted bound is negative and holds trivially. Otherwise let ξ_F minimize $Q^F(d_F, \cdot)$ over $D_F(d_F)$, fix $i \in A$ and $0 \leq s \leq d_i^+$, and let ξ have free part ξ_F , component s in coordinate i , and zero in the other active coordinates. Then $\xi \in D(d)$ and, by (47),

$$\phi_B(x^* + t\xi) \leq f^* + tg_i s + t^2(-G^F(d_F) + \delta/4 + \delta/4),$$

which is at most U whenever $g_i s \leq \epsilon/t + t(G^F(d_F) - \delta/2)$. The right side is nonnegative, so $s = 0$ qualifies; taking the largest such $s \leq d_i^+$ gives (iv). $\square$

Proof of Theorem 4.27. Let $M' = \max\{M, 1\}$ and $\mathcal{K}' = \{d \in \mathcal{S} : d_F = u_F, 0 \leq d_i^+ \leq M' \text{ for } i \in A\}$. Let ω be a modulus of uniform continuity of Q on $\{(d, \xi) : d \in \mathcal{K}', \xi \in D(d)\}$, and $\varrho = \varrho_{\mathcal{K}'}$. Choose $\bar{\sigma} \in (0, 1]$ with $\omega(\bar{\sigma}) \leq \delta/4$. Let $C = (G_M + \delta)/(\lambda g_{\min})$, and choose $\bar{t}$ such that, for $t \leq \bar{t}$, $\varrho(t) \leq \delta/4$, the comparison boxes belong to $\mathfrak{B}$, and $C\bar{t} \leq \bar{\sigma}$. The proof of Lemma 4.26 applies on $\mathcal{K}'$ with these constants, including the exact form (49). By Lemma 3.2, every bound below for Jacobi rounds also holds for complete sequential rounds, since the result of such a round is contained in the Jacobi image of the same box.

First round. Let $C_0 = x^* + t_0 D(d^0)$. Then $B_1 = T_U(B_0) \subseteq T_U(C_0)$ by Lemma 3.1, and $B_1 \subseteq B_0$, so the free part of B_1 lies in $x_F^* + t_1 D_F(u_F)$; this is (a) for $k = 1$. For $x = x^* + t_0 \xi \in K_U(C_0)$, (48) holds with $d = d^0$, and $-Q(d^0, \xi) \leq G_M$ by the definition of G_M . As in the proof of Lemma 4.26(iii), $w_i(B_1) \leq (\epsilon + (G_M + \delta/4)t_0^2)/g_i$, the first claim of (b).

Induction. Suppose $N \geq 2$, so that $\epsilon \leq \delta t_1^2/2$. Let $\sigma_k = \max_{i \in A} w_i(B_k)/t_k$. Since $t_1 = t_0$,

$$\sigma_1 \leq \frac{\delta/2 + G_M + \delta/4}{g_{\min}} t_0 \leq C\bar{t} \leq \bar{\sigma}.$$

Suppose, for some $1 \leq k \leq N - 1$, that the free part of B_k lies in $x_F^* + t_k D_F(u_F)$ and $\sigma_k \leq \bar{\sigma}$. Then $B_k \subseteq C_k = x^* + t_k D(d^k)$, where $d_F^k = u_F$ and the active components of d^k are at most σ_k , so $d^k \in \mathcal{K}'$. Since $k < N$, $t_k^2 \geq 2\epsilon/\delta$. Lemma 4.26(i) and Lemma 3.1 place the free part of B_{k+1} in $x_F^* + t_k D_F(\Phi_\delta^F(u_F)) \subseteq x_F^* + t_{k+1} D_F(u_F)$, and (49) gives

$$w_i(B_{k+1}) \leq (\epsilon + (G_0 + \eta_k)t_k^2)/g_i, \quad \eta_k = \omega(\sigma_k) + \varrho(t_k) \leq \delta/2.$$

Using $\epsilon \leq \delta t_k^2/2$ and $G_0 \leq G_M$,

$$\sigma_{k+1} \leq \frac{G_0 + \delta}{\lambda g_{\min}} t_k \leq C\bar{t} \leq \bar{\sigma}.$$

This completes the induction and proves (a) for $k \leq N$ and (b) for $k \leq N - 1$. For $k = N$, the induction hypothesis holds at B_N , and (49) does not require $\epsilon \leq \delta t^2/2$, so the estimate of (b) also holds for $k = N$. Finally $\sigma_1 \leq C t_0$ and $\sigma_k \leq C t_{k-1}$ for $2 \leq k \leq N$, so $\eta_k = \omega(\sigma_k) + \varrho(t_k)$ tends to zero as $t_k \rightarrow 0$, by the continuity and remainder moduli.

Limits. If $\epsilon = 0$, then $N = \infty$; the free parts shrink to x_F^* and the active widths are at most $(G_0 + \eta_k)t_k^2/g_i \rightarrow 0$, so $B_\infty = \{x^*\}$, and $\eta_k \rightarrow 0$ gives the stated asymptotic form. If $\epsilon > 0$, then $N < \infty$ and $B_\infty \subseteq B_N$, whose free part lies in $x_F^* + t_N D_F(u_F)$ with $t_N < \sqrt{2\epsilon/\delta}$. For the active widths, if $N \geq 2$, then

$$w_i(B_\infty) \leq w_i(B_{N+1}) \leq \left(\epsilon + (G_0 + \frac{\delta}{2}) \frac{2\epsilon}{\delta} \right) / g_i = \left(2 + \frac{2G_0}{\delta} \right) \frac{\epsilon}{g_i},$$

and if $N = 1$, then $t_0^2 = t_1^2 < 2\epsilon/\delta$ and

$$w_i(B_\infty) \leq w_i(B_1) \leq \left(\epsilon + (G_M + \frac{\delta}{4}) \frac{2\epsilon}{\delta} \right) / g_i \leq \left(\frac{3}{2} + \frac{2G_M}{\delta} \right) \frac{\epsilon}{g_i}.$$

Both are at most $(2 + 2G_M/\delta)\epsilon/g_i$. The last sentence of the theorem follows from the definition of $r^*(\Phi^F)$. $\square$

B Validation of floating-point bounds

This appendix gives the row correction and dual-bound evaluation used by the numerical propagator of Section 9.4. Each stored floating-point number is treated as the exact dyadic rational it represents. The LP variables have a finite box of bounds.

Proposition B.1 (Outward row correction). *Let $a^\top z \leq \beta$ be valid for a set S contained in the box $Z = [\underline{z}, \bar{z}]$. For any vector $\tilde{a}$, every $z \in S$ satisfies $\tilde{a}^\top z \leq \tilde{\beta}$ whenever*

$$\tilde{\beta} \geq \beta + \sum_j \max\{(\tilde{a}_j - a_j)\underline{z}_j, (\tilde{a}_j - a_j)\bar{z}_j\}. \quad (50)$$

Proof. For $z \in S$, $\tilde{a}^\top z = a^\top z + \sum_j (\tilde{a}_j - a_j)z_j \leq \beta + \sum_j \max_{\underline{z}_j \leq s \leq \bar{z}_j} (\tilde{a}_j - a_j)s$, and a linear function of s attains its maximum over an interval at an endpoint. $\square$

The implementation takes $\tilde{a}_j$ to be the floating-point number nearest to a_j , evaluates the right-hand side of (50) in exact rational arithmetic over the LP's variable bounds, and rounds it upward to a floating-point $\tilde{\beta}$. The coefficients of the original rows, of the McCormick rows, and of the tangents are floating-point numbers already, while the secant slope $\ell_i + u_i$ may need correction; right-hand sides such as $\ell_i \ell_j$ and p^2 stay exact until the final upward rounding. Proposition B.1 applies to each product row with S the set of exact lifts $(x, (x_i x_j))$ of the points $x \in B$, and to each original row and the cutoff row with S the set of exact lifts of the points of B that satisfy that row. Both sets lie in the LP's variable box by construction.

The LP solver returns approximate inequality multipliers. They need not be dual feasible. The following bound uses any nonpositive multipliers and absorbs the dual residual through the variable box. Bounds of this kind, evaluated with directed rounding, are an established way to obtain safe LP bounds from approximate dual solutions [31]; we state the form we use. It is the minimization form of Proposition 8.10, which applies the same idea to a stored dual vector after the rows have changed; here the rows are current and the multipliers inexact.

Proposition B.2 (Dual bound with a residual term). *Consider $\mu = \min\{d^\top z : Az \leq b, z \in Z\}$ with a finite box $Z = [\underline{z}, \bar{z}]$, and $\mu = +\infty$ if the LP is infeasible. For every $y \leq 0$, with $r = d - A^\top y$,*

$$\mu \geq y^\top b + \sum_j \min\{r_j \underline{z}_j, r_j \bar{z}_j\}. \quad (51)$$

The right-hand side is the LP dual function at y , with the multipliers of the variable bounds eliminated; its maximum over $y \leq 0$ equals μ when the LP is feasible.

Proof. If the LP is infeasible, there is nothing to prove. For feasible z , $y \leq 0$ and $Az \leq b$ give $y^\top Az \geq y^\top b$, so $d^\top z = y^\top Az + r^\top z \geq y^\top b + r^\top z$. Minimizing $r^\top z$ separately over each interval gives (51). For the second statement, the LP dual of μ has multipliers $y \leq 0$ for $Az \leq b$ and $\alpha, \gamma \geq 0$ for $z \geq \underline{z}$ and $-z \geq -\bar{z}$, with $d = A^\top y + \alpha - \gamma$ and objective $y^\top b + \underline{z}^\top \alpha - \bar{z}^\top \gamma$. For fixed y , the best choice is $\alpha_j = \max\{r_j, 0\}$ and $\gamma_j = \max\{-r_j, 0\}$, which gives the right-hand side of (51) because $\underline{z}_j \leq \bar{z}_j$. LP strong duality completes the proof. $\square$

The implementation clips the returned multipliers to $y = \min\{\tilde{y}, 0\}$ and evaluates (51) with directed rounding; it rejects multipliers that are not finite. Because the proposition holds for every $y \leq 0$, the validity of the result depends neither on the accuracy of the multipliers nor on the solver's sign convention for them: poor or wrongly signed multipliers only weaken the bound.

Let $\bar{F}$ be the set of binary64 numbers together with $\pm\infty$, and let Ω be its largest finite element. For $t \in \bar{F}$, let $\text{pred}(t)$ and $\text{succ}(t)$ be the next smaller and the next larger element of $\bar{F}$, the IEEE 754 operations `nextDown` and `nextUp`, with $\text{pred}(-\infty) = -\infty$ and $\text{succ}(+\infty) = +\infty$. Thus $\text{pred}(+\infty) = \Omega$ and $\text{succ}(-\infty) = -\Omega$, and both maps are nondecreasing. For an addition, subtraction, or multiplication of elements of $\bar{F}$ whose exact result s is defined in the extended reals, write $\text{fl}(s)$ for the computed result. We assume that rounding is *faithful*: $\text{fl}(s) = s$ if $s \in \bar{F}$, and otherwise $\text{fl}(s)$ is one of the two consecutive elements of $\bar{F}$ that enclose s .

Lemma B.3 (One-step enclosure). *Under faithful rounding, $\text{pred}(\text{fl}(s)) \leq s \leq \text{succ}(\text{fl}(s))$ for every exact result s .*

Proof. If $\text{fl}(s) = s$, the claim holds because $\text{pred}(t) \leq t \leq \text{succ}(t)$ for every $t \in \bar{F}$. Otherwise let $s^- < s < s^+$ be the consecutive elements of $\bar{F}$ that enclose s . If $\text{fl}(s) = s^-$, then $\text{pred}(s^-) \leq s^- < s < s^+ = \text{succ}(s^-)$; the case $\text{fl}(s) = s^+$ is symmetric. $\square$

With gradual underflow, every IEEE 754 rounding direction is faithful, including overflow to $\pm\infty$ or to $\pm\Omega$. The argument therefore also covers a host solver that leaves a directed rounding mode active. Flush-to-zero and denormals-are-zero modes are not faithful, and the argument assumes that they are disabled.

The evaluation computes three enclosures, each preserved by Lemma B.3; the first two are accumulated in one sweep over the rows with $y_r \neq 0$.

1. A lower bound σ of $y^\top b$: starting from $\sigma = 0$, add $\text{pred}(\text{fl}(y_r b_r))$ for every row with $y_r \neq 0$ and apply pred to every rounded sum.
2. An interval $[r_j, \bar{r}_j]$ containing r_j : start from d_j , and for each nonzero a_{rj} with $y_r \neq 0$, let $t = \text{fl}(a_{rj} y_r)$ and update $\underline{r}_j \leftarrow \text{pred}(\text{fl}(\underline{r}_j - \text{succ}(t)))$ and $\bar{r}_j \leftarrow \text{succ}(\text{fl}(\bar{r}_j - \text{pred}(t)))$.
3. The bound $\tilde{\mu}$: starting from σ , add, for each j , pred of the smallest of the four rounded products of $\{\underline{r}_j, \bar{r}_j\}$ and $\{\underline{z}_j, \bar{z}_j\}$, again applying pred to every rounded sum.

Overflow cannot produce an undefined operation in the first two steps. The lower bounds σ and $\underline{r}_j$ start finite and are afterwards results of pred , so they are never $+\infty$, and neither is the term $\text{pred}(\text{fl}(y_r b_r))$ added to σ ; the term $\text{succ}(t)$ subtracted from $\underline{r}_j$ is never $-\infty$. Symmetrically, $\bar{r}_j$ is never $-\infty$, and $\text{pred}(t)$ is never $+\infty$. No sum or difference therefore has the form $\infty - \infty$. An infinite intermediate value is $-\infty$ in a lower bound or $+\infty$ in an upper bound, which remains valid. The third step rejects the computation if any of its products or the final value is not finite; an accepted result thus uses finite $\underline{r}_j$ and $\bar{r}_j$. For each j , let p_j be the smallest of the four exact vertex products. The smallest rounded product is at most $\text{fl}(p_j)$, and pred is nondecreasing, so the term added for j is at most $\text{pred}(\text{fl}(p_j)) \leq p_j$. Since the bilinear function ρs attains its minimum over a rectangle at a vertex, an accepted result satisfies

$$\tilde{\mu} \leq y^\top b + \sum_j \min_{\rho \in [\underline{r}_j, \bar{r}_j], s \in [\underline{z}_j, \bar{z}_j]} \rho s \leq y^\top b + \sum_j \min\{r_j \underline{z}_j, r_j \bar{z}_j\} \leq \mu.$$

These arguments assume IEEE binary64 arithmetic with gradual underflow. They establish a bound for the stored floating-point LP. Together with Proposition B.1, that LP is a relaxation of the exact product graph of the stored model on B . They do not relate the stored binary model to a decimal or symbolic model upstream of its import.

C Proofs for parametric support certificates

C.1 Proof of Theorem 8.12

Proof. Fix i and $p \neq q$ in $\mathcal{D}$, and put $\gamma(t) = q + t(p - q)$ for $t \in [0, 1]$; convexity keeps $\gamma(t)$ in $\mathcal{D}$. The sets $S_s = \{t : \gamma(t) \in C_{i,s}\}$ are finite unions of closed intervals and cover $[0, 1]$. Let $\mathcal{J}$ be the finite collection of the intervals of positive length among their components. The union of $\mathcal{J}$ is closed, and its complement in $[0, 1]$ is relatively open and contained in a finite set, hence empty. Let $0 = t_0 < t_1 < \dots < t_q = 1$ be the endpoints of the intervals in $\mathcal{J}$. The midpoint of (t_{r-1}, t_r)

lies in some $[\alpha, \beta] \in \mathcal{J}$ with $[\alpha, \beta] \subseteq S_s$. Since α and β are among the t 's and none lies strictly between t_{r-1} and t_r , we get $[t_{r-1}, t_r] \subseteq S_s$. On this interval $F_i(\gamma(t)) = h_{i,s}(\gamma(t))$, a continuously differentiable function of t whose derivative $\sum_j \partial_j h_{i,s}(\gamma(t))(p_j - q_j)$ has absolute value at most $\sum_j M_{ij}|p_j - q_j|$. The mean value theorem gives

$$|F_i(\gamma(t_r)) - F_i(\gamma(t_{r-1}))| \leq (t_r - t_{r-1}) \sum_j M_{ij}|p_j - q_j|.$$

Summing over r proves $|F_i(p) - F_i(q)| \leq \sum_j M_{ij}|p_j - q_j|$. For $q \leq p$, the two-sided bound gives $F(p) - F(q) \leq M(p - q)$. $\square$

C.2 Proof of Proposition 8.13

Proof. Each program is feasible, since any $\theta \in \mathcal{D}$ with a sufficiently negative t is feasible, and its optimal value is attained, since it is the maximum over the compact set $\mathcal{D}$ of the continuous function $\min\{1, H_{j,r}^r \theta - g_{j,r}^r\}$. If some $\theta \in \mathcal{D}$ lies in no C^r , choose for each r an inequality of C^r that θ violates strictly. The minimum of these violations and 1 is positive and gives $\tau_J > 0$ for the selected tuple. Conversely, a feasible point with $t > 0$ violates strictly one inequality of every C^r and so lies outside the union. $\square$

D Scheduling guarantees and limits of observation

D.1 Comparisons that keep the baseline unchanged

A guarantee relative to the unchanged solver is available if the unchanged solver runs unchanged. The following statements use an ideal preemptible work model. Two solvers A (the baseline) and H (with the policy) run on the same input with separate states and fixed random tapes. Each solver's sequence of states depends only on the amount of work it has received, not on wall-clock time or on the other solver. Run alone, they finish after T_A and T_H units of work. A scheduler shares one processor; by aggregate time t , solver S has received $a_S(t)$ units of work.

Theorem D.1 (Interleaving). *Let $0 < \alpha < 1$ and $h \geq 0$. Suppose that, as long as neither solver has finished, $a_A(t) \geq (1 - \alpha)t - h$ and $a_H(t) \geq \alpha t - h$. Then the first of the two solvers finishes no later than*

$$\min \left\{ \frac{T_A + h}{1 - \alpha}, \frac{T_H + h}{\alpha} \right\}. \quad (52)$$

Proof. Suppose neither solver has finished by $t_A = (T_A + h)/(1 - \alpha)$. Then A has received at least T_A units of work. Because its sequence of states depends only on the work received, it has reached the state in which it finishes, a contradiction. The same argument applies at $t_H = (T_H + h)/\alpha$. $\square$

Corollary D.2 (Equal shares). *If the scheduler alternates quanta of length q between the two solvers and switching costs nothing, one of the solvers finishes by $2 \min\{T_A, T_H\} + q$.*

Proof. Let A receive the first quantum. During the period $[2kq, 2(k+1)q]$, solver A holds at least $kq + \min\{t - 2kq, q\} \geq t/2$ units of work and H at least $kq + \max\{t - 2kq - q, 0\} \geq t/2 - q/2$. Apply Theorem D.1 with $\alpha = 1/2$ and $h = q/2$: the bound is $\min\{2T_A + q, 2T_H + q\}$. Since $a_A(t) \geq t/2$, the term for A can be sharpened to $2T_A$. $\square$

Proposition D.3 (Serial rescue). *Run H for at most b units of charged work, including its startup. If it has not finished, discard it and run A from its initial state with its original input and random tape. The total work is T_H if $T_H \leq b$, and $b + T_A$ otherwise. If the cap on H can be overrun by at most δ , the second case becomes at most $b + \delta + T_A$.*

Proof. If H finishes within its allowance, the total is its own work. Otherwise its charged work is at most b , or $b + \delta$ with overrun, and A then executes exactly its standalone trace, which costs T_A including startup. $\square$

Both guarantees depend on the baseline trace being unaffected. Shared memory or cache contention, uncharged process startup, decisions that depend on wall-clock time, and communication between the solvers can all violate the premises. In particular, passing an incumbent or a row from H to A changes A 's trajectory, and the bound T_A no longer applies. The statements bound work, not elapsed time on a shared machine. The computational comparison of Section 11 does not deploy either construction; it compares independent full runs.

D.2 History does not identify the best first action

The triggers and the pilot rule of Section 9.4 choose actions from what the policy has observed. The following example shows that such a choice cannot be guaranteed correct from the observations alone.

Example D.4 (Indistinguishable histories). Two actions each cost one unit. In environment 1 only action 1 produces a proof; in environment 2 only action 2 does. Suppose the information available to a selector before its first action is identical in the two environments. A deterministic selector then makes the same first choice in both and chooses wrongly in one of them. If a randomized selector chooses action 1 with probability p , its first choice is correct with probability p in environment 1 and $1 - p$ in environment 2, and $\min\{p, 1 - p\} \leq 1/2$.

The example limits only the first choice under identical information. It does not rule out learning under statistical assumptions, structural tests that distinguish the environments, or good average behavior on a class of instances. Adaptive selection of solver components has been studied under such assumptions, for example with bandit algorithms and online learning in mixed-integer programming [12, 20]. The point for this paper is narrower. The triggers, ranking, and pilot of the measured policy are heuristics; their value can be judged only by complete runs on instances, not by their observed rewards.

E Details of the precursor study

This appendix records the design and the archived results of the external presolve study summarized in Section 11.1. The study is separate from the prospective comparison; its results are not pooled with those of Section 11.

E.1 Design

Reference solutions. For 275 of the 339 instances the reference solution was a MINLP Lib solution file whose objective matched the library's primal bound; for 63 it was the Gurobi solution with default settings, which had to be optimal, match the known value to 10^{-4} , and be feasible to 10^{-5} . One instance had no reference solution.

Relaxation and OBBT. The OBBT relaxation was an LP built with Gurobi, with integrality dropped: the four McCormick inequalities for each distinct bilinear term, five tangents at equally spaced points and the secant for each square (with x_i^2 replaced by x_i for binary x_i), every original row with its quadratic terms replaced by the lifted variables, and the cutoff row. After a bound change the affected rows were rewritten in place and the LP was warm started; presolve was off. An LP reported infeasible produced no tightening; this occurred for two LPs. Each new bound was relaxed by $10^{-6}(1 + |\beta|)$ for a bound β and rounded for integer variables, and it was accepted only if it reduced the width by at least a relative 10^{-4} . A round minimized and maximized every variable occurring in a nonlinear term. The main trajectories used sequential (Gauss–Seidel) updates and skipped a direction for the rest of a round when an earlier LP solution of that round already attained its bound. With sequential updates this filtering is a heuristic: a point from an earlier LP of the round may violate rows rewritten after later updates, so a skipped direction could still have tightened. It never produces an invalid bound. After each round the study recorded ρ , the total normalized width after the round divided by the total before, over the variables whose width shrank by at least 0.1%, where widths are normalized by the box obtained from feasibility-based tightening.

Stopping rules. **r1** and **r5** stop after one and five rounds; **fp** stops after a round in which no bound changed, after 50 rounds, or after 400 seconds. Because of the margins, the filtering, and the improvement threshold, a round without change is a numerical stopping event, not an exact fixed-point certificate. The adaptive rules **ad0.5** and **ad0.8** stop after a round with $\rho > \theta$ for $\theta = 0.5$ or 0.8 , after a round without change, or after 120 seconds of OBBT, and restrict rounds after the first to the variables that moved in the previous round and their bilinear partners. No time limit below 400 seconds applied to the first round, even for the adaptive rules, whose 120-second cap was checked only between rounds.

Pipelines and baselines. A root-node run of the target solver (seed 0, node limit one, 30 seconds) supplied the incumbent and the cutoff $U = \hat{f} + 10^{-6} \max\{1, |\hat{f}|\}$. If this run proved optimality (Gurobi 101 instances, SCIP 66), the pipeline ended there and was charged its time; if it found no incumbent (18 and 20 instances), $U = +\infty$. Otherwise OBBT ran with cutoff U , and the final solve started on the tightened box with the incumbent as a start solution, U as a cutoff or objective limit, and a time limit of 600 seconds minus the root and OBBT time. The control arm was identical but used the box from feasibility-based tightening. The baselines were the default solvers on the original model, and for Gurobi also its built-in aggressive OBBT setting (**OBBT=3**). A diagnostic arm, kept separate, used the known optimal value as cutoff, the box of the **fp** rule, no root run, and no start solution; it was run with seed 0 on the instances that the solver’s root run did not solve. Final solves with identical boxes were shared between rules, each rule being charged its own OBBT time.

Metrics and timing. A run counts as solved if optimality is proved within 600 seconds of total time and the objective is within a relative 10^{-3} of the known value. Times are shifted geometric means with shift one second over instance–seed pairs, using the total time capped at 600 seconds; node ratios are shifted geometric means with shift ten over pairs solved by both arms. The McCormick LP gap closed by a box B is $\min\{1, \max\{0, (L(B) - L_0)/(f^* - L_0)\}\}$, where L is the bound of the OBBT relaxation on B , L_0 its bound on the box from feasibility-based tightening, and f^* the optimal value; it is averaged over instances with $f^* - L_0 > 10^{-6} \max\{1, |f^*|\}$. All runs were wall-clock timed on a machine with 18 cores and 36 hardware threads, with up to 24 concurrent single-thread jobs and load averages of about 23–25.

Correction of two solved counts. Two SCIP runs reported optimality with the value -79.35 instead of -79.75 : **crudeoil_lee1_05** in the control arm and **crudeoil_lee1_09** in the **r5** pipeline, both with seed 1. Both are numerical failures of the final solve; the corresponding boxes contain the reference solution. The solved counts in this appendix exclude them. All time, node, and final-solve summaries below are the archived originals: they use the recorded times of these two runs, 30.8 and

105.0 seconds, and count both runs as solved when selecting pairs solved by both arms. With 678 pairs per arm, the two runs have little weight, but these summaries are not corrected reanalyses.

E.2 Solver-level results

Solver	Arm	Solved	Time (s)	Time ratio	Node ratio	Root+OBBT (s)
Gurobi	default	640	2.37	1.00	1.00	0
Gurobi	OBBT=3	640	2.49	1.05	0.82	0
Gurobi	control	642	2.57	1.08	1.05	0.5
Gurobi	r1	637	3.11	1.31	0.99	14.2
Gurobi	r5	638	3.40	1.44	0.83	17.6
Gurobi	ad0.5	637	3.14	1.33	0.97	14.5
Gurobi	ad0.8	637	3.23	1.37	0.90	15.3
Gurobi	fp	637	4.08	1.73	0.77	30.8
SCIP	default	575	7.18	1.00	1.00	0
SCIP	control	573	8.30	1.16	1.11	3.8
SCIP	r1	570	8.67	1.21	1.05	16.2
SCIP	r5	563	8.84	1.23	0.85	19.7
SCIP	ad0.5	570	8.71	1.21	0.98	16.5
SCIP	ad0.8	571	8.69	1.21	0.88	17.1
SCIP	fp	561	9.50	1.32	0.73	31.4
Gurobi	known-value fp	217/220	9.26/4.50	2.06	0.70	48.5
SCIP	known-value fp	220/221	15.1/12.3	1.23	0.55	43.1

Table 9: Precursor study, all 339 instances and two seeds (678 pairs). Ratios are relative to the default solver; the last column is the mean root and OBBT time per pair. The known-value diagnostic uses seed 0 on 238 (Gurobi) and 273 (SCIP) instances and is compared with the default on the same pairs; its time includes OBBT.

Table 9 gives the solver-level results. No pipeline solved more pairs than the default, and every pipeline was slower in shifted geometric mean. Gurobi’s built-in OBBT reduced nodes to 0.82 at a time ratio of 1.05 without a restart. Even with the known optimal value as cutoff, the **fp** pipeline was slower than the default on the instances it was run on. On the 69 (Gurobi) and 122 (SCIP) instances that the default needed at least ten seconds for or did not solve in some seed, the archived summaries give the pipelines node ratios of 0.99–1.02 and 1.06–1.27 and time ratios of 1.15–1.23 and 1.27–1.51. They solved 97–98 of these pairs against 100 for the default with Gurobi, and 127–137 against 141 with SCIP.

Table 10 isolates the box. The pipelines and the control share the root run, the incumbent, and the cutoff, and differ only in the box. The tightened boxes reduced nodes, but the savings in the final solve did not offset the root and OBBT time in the total-time summaries. Split by the McCormick LP gap closed by the **fp** box, the pairs solved by both arms show where the node reductions occur. With Gurobi, 112 pairs with closure at least 0.5 had node ratio 0.18 and final-solve ratio 0.87, at a mean control time of 4.9 seconds; 98 pairs with positive closure below 0.5 had 0.77 and 0.95 (24 seconds); and 219 pairs without closure had 0.96 and 1.00 (27 seconds). With SCIP the corresponding values were 142 pairs with 0.20 and 0.65 (1.2 seconds), 76 pairs with 0.83 and 1.02 (50 seconds), and 198 pairs with 0.97 and 1.00 (61 seconds). The largest mean node reductions occurred in the groups whose mean control time was only a few seconds.

Solver	Rule	Time ratio	Node ratio	Final-solve ratio	Solved (control)
Gurobi	r1	1.27	0.93	1.02	435 (440)
Gurobi	ad0.8	1.33	0.81	0.98	435 (440)
Gurobi	fp	1.79	0.65	0.98	435 (440)
SCIP	r1	1.05	0.96	1.02	438 (441)
SCIP	ad0.8	1.06	0.77	1.00	439 (441)
SCIP	fp	1.17	0.62	0.96	429 (441)

Table 10: Effect of the tightened box: pipelines against the control on the instances where OBBT ran (238 for Gurobi, 273 for SCIP). The final-solve ratio excludes root and OBBT time and is taken over pairs solved by both arms.

E.3 OBBT-level results

Cutoff	Instances	1	2	3	5	10	fp
Known optimal value	305	0.236	0.274	0.296	0.322	0.338	0.346
Gurobi root incumbent	235	0.147	0.181	0.203	0.231	0.247	0.250
SCIP root incumbent	267	0.176	0.213	0.237	0.260	0.278	0.287

Table 11: Mean McCormick LP gap closed after the given number of sequential rounds and with the rule **fp**; trajectories that stopped earlier keep their final value. Shifted geometric means of cumulative OBBT time were 1.1, 1.4, and 1.1 seconds after one round and 2.2, 2.9, and 2.1 seconds with **fp**.

Table 11 shows diminishing returns from additional rounds. The median closure was zero for every rule. With the known optimal value as cutoff, the first round changed at least one bound on 239 of the 339 instances. Of the 104 **fp** trajectories that ended within their first round, 97 completed that round without a bound change and seven were interrupted by the 400-second cap. A further 22 trajectories changed a bound in the first round and then completed a second round with no accepted change; one more recorded no change in a second round that the cap interrupted. OBBT time was heavy-tailed: the median was 0.04 seconds for one round and 0.23 seconds for **fp**, but 36 instances needed more than 10 seconds for the first round and 14 more than 120 seconds, and 18 **fp** trajectories reached the 400-second cap.

Among the 149 trajectories with at least four rounds in which a variable moved, the median of ρ over rounds 3–10 exceeded 0.95 on 79 and was at most 0.5 on 34. This sample excludes every trajectory that converged quickly, so it does not show that slow contraction is typical. With a cutoff equal to the optimal value up to 10^{-6} , it also cannot separate a stalled limit from slow approach to a cutoff-dependent floor.

The adaptive rule **ad0.8** stopped after the first round on 46–54% of instances, depending on the cutoff source. Restricting later rounds to moved variables and their bilinear partners saved only 3% of the LPs (105,878 against 109,107), because these neighborhoods covered almost all nonlinear variables. Feasibility filtering reduced the LPs of the first round from 87,443 to 25,712 and their time from 7,834 to 5,277 seconds, while the mean closure changed from 0.239 to 0.236. Until the **fp** rule stopped them, simultaneous (Jacobi) rounds used 300,710 LPs and 12,053 seconds, against 289,941 LPs and 12,031 seconds for sequential rounds; both had a median of four rounds and a mean closure of 0.346.

These filtering and Jacobi–sequential comparison totals are taken from the archived report; part

of their calculation code is absent from the saved analysis script.

Every one of the 4,573 tightened boxes that belonged to an instance with a reference solution contained it numerically, with tolerance $10^{-6}(1 + |x_j|)$ in each coordinate. Three SCIP runs failed with an LP solver error and count as unsolved.

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